

15. Interrupt Controller (ICUD)

15.1 Overview

The interrupt controller (ICU) controls various interrupt requests from the peripheral modules and the IRQi pin (i = 0 to 15), and generates an interrupt request to the CPU and a transfer request to the DTC and DMAC.

Table 15.1 lists the ICU specifications, and Figure 15.1 shows a block diagram of the interrupt controller.

Table 15.1 ICU Specifications (1/2)

Item	Description	
Interrupts	Peripheral interrupts	Interrupts from peripheral modules - Interrupt detection method: Edge detection/level detection (fixed for each interrupt source) - Group interrupt: Multiple interrupt sources are grouped together and treated as an interrupt source.*1 - Group IE0 interrupt: Interrupt sources of coprocessors that use ICLK as the operating clock (edge detection) - Group BE0 interrupt: Interrupt sources of peripheral modules that use PCLKB as the operating clock (edge detection) - Group BL0/BL1/BL2 interrupt: Interrupt sources of peripheral modules that use PCLKB as the operating clock (level detection) - Group AL0/AL1 interrupt: Interrupt sources of peripheral modules that use PCLKA as the operating clock (level detection) - Software configurable interrupt B: Any of the interrupt sources for peripheral modules that use PCLKB as the operating clock can be assigned to interrupt vector numbers 128 to 207. - Software configurable interrupt A: Any of the interrupt sources for peripheral modules that use PCLKA as the operating clock can be assigned to interrupt vector numbers 208 to 255.
	External pin interrupt	Interrupt by the input signal to the IRQi pin (i = 0 to 15) - Interrupt detection method: Detection of low level, falling edge, rising edge, rising and falling edges - One of these detection methods can be set for each source. - Digital filter can be used to remove noise.
	Software interrupt	- Interrupt request can be generated by writing to a register. - Two interrupt sources
	Interrupt priority	Priority level can be set with interrupt source priority register r (IPRr) (r = 000 to 255).
	Fast interrupt function	CPU interrupt response time can be reduced. This function can be used for only one interrupt source.
	DTC/DMAC control	Interrupt sources can be used to start the DTC and DMAC.*2
	EXDMAC control	Interrupt selected by software configurable interrupt B source select register 144 or software configurable interrupt A source select register 208 can be used to start EXDMAC0. Interrupt selected by software configurable interrupt B source select register 145 or software configurable interrupt A source select register 209 can be used to start EXDMAC1.
Non-maskable interrupts *3	NMI pin interrupt	Interrupt by the input signal to the NMI pin - Interrupt detection: Falling edge/rising edge - Digital filter can be used to remove noise.
	Oscillation stop detection interrupt *4	This interrupt occurs when the main clock oscillator stop is detected.
	WDT underflow/refresh error interrupt *4	This interrupt occurs when the watchdog timer (WDT) underflows or a refresh error occurs.
	IWDT underflow/refresh error interrupt *4	This interrupt occurs when the independent watchdog timer (IWDT) underflows or a refresh error occurs.
	Voltage monitoring 1 interrupt *4	Interrupt from voltage detection circuit 1 (LVD1)
	Voltage monitoring 2 interrupt *4	Interrupt from voltage detection circuit 2 (LVD2)
	RAM error interrupt *4	This interrupt occurs when a parity check error is detected in the RAM (including the expanded RAM) or an ECC error is detected in the ECCRAM.
	Double-precision floating-point exceptions *4	Exceptions from double-precision floating-point coprocessor

Table 15.1 ICU Specifications (2/2)

Item		Description
Return from low power consumption states	Sleep mode	Exit sleep mode by any interrupt source.
	All-module clock stop mode	Exit all-module clock stop mode by the NMI pin interrupt, external pin interrupt, or peripheral interrupt (voltage monitoring 1, voltage monitoring 2, oscillation stop detection interrupt, USB0 resume, RTC alarm, RTC period, IWDG, software configurable interrupt 146 to 157).
	Software standby mode	Exit software standby mode by the NMI pin interrupt, external pin interrupt, or peripheral interrupt (voltage monitoring 1, voltage monitoring 2, USB0 resume, RTC alarm, RTC period, IWDG).
	Deep software standby mode	Exit deep software standby mode by the NMI pin interrupt, specific external pin interrupt, or peripheral interrupt (voltage monitoring 1, voltage monitoring 2, USB0 resume, RTC alarm, RTC period).

Note 1. Groups to which no interrupt source is assigned are reserved. Also, there is no register corresponding to that group.

Note 2. For the DTC and DMAC triggers, refer to Table 15.5, Interrupt Vector Table.

Note 3. Once non-maskable interrupts are enabled, they cannot be disabled.

Note 4. Each source for these non-maskable interrupts can be used for maskable interrupts. When using for maskable interrupts, do not change the NMIER and EXNMIER register values from the value after reset. To enable the voltage monitoring 1 interrupt, set the LVD1CR1.LVD1IRQSEL bit to 1, and to enable the voltage monitoring 2 interrupt, set the LVD2CR1.LVD2IRQSEL bit to 1.

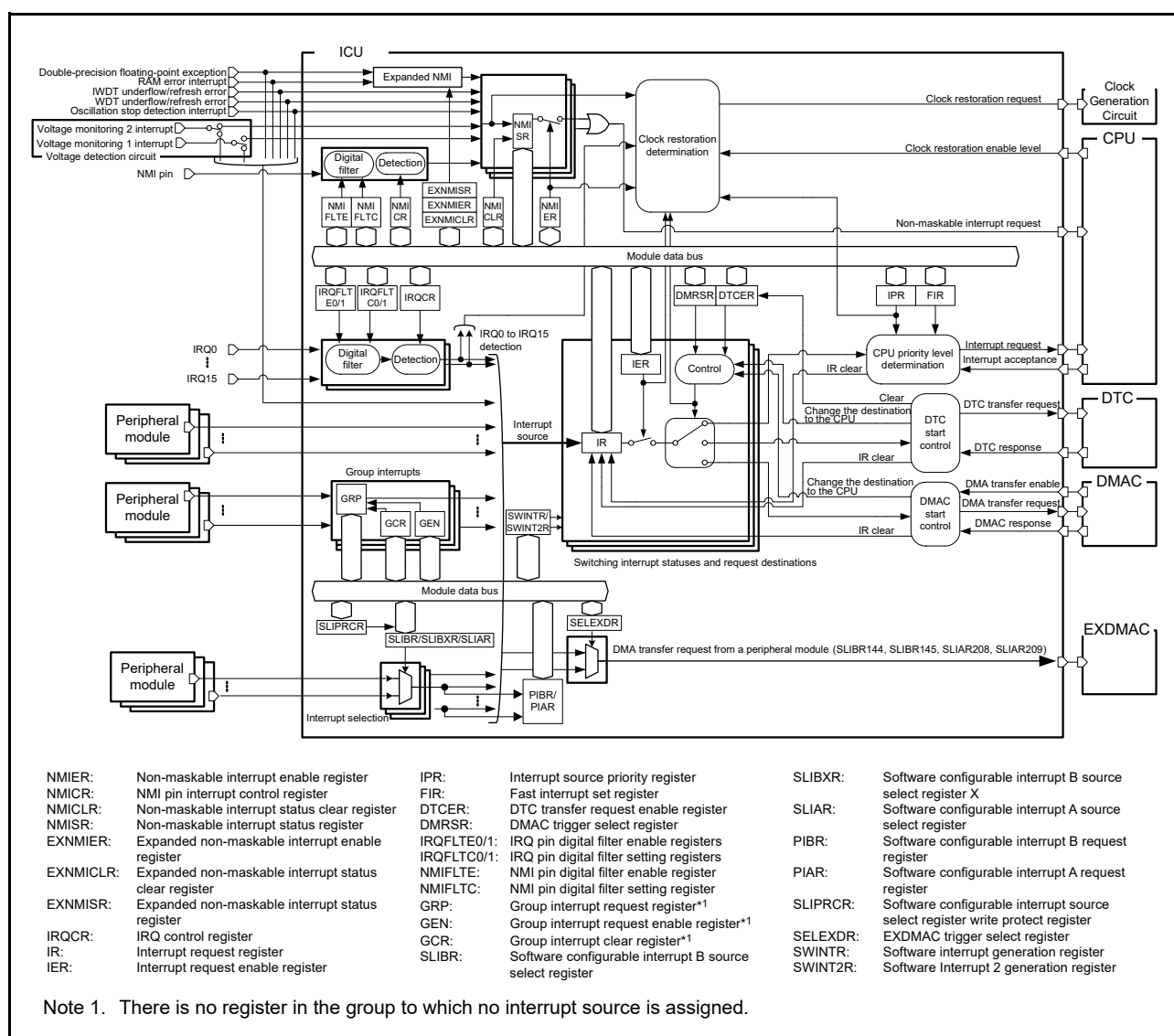


Figure 15.1 Block Diagram of the ICU

Table 15.2 lists I/O pins used for the ICU.

Table 15.2 ICU I/O Pins

Pin Name	I/O	Description
NMI	Input	Non-maskable interrupt request pin
IRQ0 to IRQ15	Input	External interrupt request pins

15.2 Register Descriptions

15.2.1 Interrupt Request Register n (IRn) (n = 016 to 255)

Address(es): ICU.IR016 0008 7010h to ICU.IR255 0008 70FFh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	IR
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	IR	Interrupt Status Flag	0: No interrupt request is generated. 1: An interrupt request is generated.	R/(W) *1
b7 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. For an edge detection interrupt source, only 0 can be written to this bit; do not write 1.
For a level detection interrupt source, neither 0 nor 1 can be written.

The IRn register indicates whether an interrupt request has been generated.

This register is provided for each interrupt vector number, and n matches the interrupt vector number.

For the correspondence between interrupt sources and interrupt vector numbers, refer to Table 15.5, Interrupt Vector Table.

IR Flag (Interrupt Status Flag)

The IR flag is a status flag indicating whether an interrupt request has been generated. This flag becomes 1 when an interrupt request is generated. To detect an interrupt request, set the interrupt enable bit of the peripheral module to enable output of the interrupt request.

An interrupt request can be detected by edge detection or level detection. For interrupts from peripheral modules, the detection method (edge detection or level detection) is determined depending on the source. Refer to Table 15.5, Interrupt Vector Table for details on the detection method for each source. For interrupts from the IRQi pin (i = 0 to 15), edge detection or level detection can be selected by setting the IRQCRi.IRQMD[1:0] bits.

The interrupt status flag for group interrupts is the ISj flag (j = 0 to 31) in the group interrupt request register (GRPIE0, GRPBE0, GRPBL0, GRPBL1, GRPBL2, GRPAL0, GRPAL1). When any of the ISj flags becomes 1, the IRn.IR flag corresponding to each group interrupt becomes 1. Group interrupts are detected by level detection.

Refer to section 15.4.4, Group Interrupts for details on group interrupts.

(1) Edge detection

This flag becomes 1 under the following condition:

- The IR flag becomes 1 when an interrupt request for peripheral interrupts or external pin interrupts is generated. For interrupt requests for peripheral modules, refer to the corresponding sections.

This flag becomes 0 under any of the following conditions:

- The IR flag becomes 0 when the interrupt request destination accepts an interrupt request.
- The IR flag becomes 0 by writing 0 to the IR flag. Note that when the interrupt request destination is the DTC or DMAC, do not write 0 to the IR flag.

(2) Level detection

This flag becomes 1 under any of the following conditions:

- The IR flag is 1 while an interrupt request for peripheral interrupts or external pin interrupts is generated. For interrupt requests for peripheral modules, refer to the corresponding sections.
- For group interrupts, the IR flag becomes 1 when the IS_j flag in the group interrupt request register (GRPIE0, GRPBE0, GRPBL0, GRPBL1, GRPBL2, GRPAL0, GRPAL1) is 1 (interrupt request is generated) while the EN_j bit in the group interrupt request enable register (GENIE0, GENBE0, GENBL0, GENBL1, GENBL2, GENAL0, GENAL1) is 1 (enabled) (j = 0 to 31).

This flag becomes 0 under any of the following conditions:

- The IR flag becomes 0 by clearing output of the peripheral module interrupt request. The IR flag does not become 0 when the interrupt request destination accepts the interrupt request. For interrupt requests for peripheral modules, refer to the corresponding sections.
- For group interrupts, the IR flag becomes 0 when the EN_j bit in the group interrupt request enable register is 0 (disabled) or when the IS_j flag in the group interrupt request register is 0 (interrupt request is not generated).

When level detection is selected for detecting external pin interrupts, set the input level of the IRQ_i pin to high (i = 0 to 15) to cancel the external pin interrupt that has occurred. When level detection is selected, do not write to the IR flag.

15.2.2 Interrupt Request Enable Register m (IERm) (m = 02h to 1Fh)

Address(es): ICU.IER02 0008 7202h to ICU.IER1F 0008 721Fh

b7	b6	b5	b4	b3	b2	b1	b0
IEN7	IEN6	IEN5	IEN4	IEN3	IEN2	IEN1	IEN0
Value after reset: 0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	IEN0	Interrupt Request Enable 0	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b1	IEN1	Interrupt Request Enable 1		R/W
b2	IEN2	Interrupt Request Enable 2		R/W
b3	IEN3	Interrupt Request Enable 3		R/W
b4	IEN4	Interrupt Request Enable 4		R/W
b5	IEN5	Interrupt Request Enable 5		R/W
b6	IEN6	Interrupt Request Enable 6		R/W
b7	IEN7	Interrupt Request Enable 7		R/W

Note: When the interrupt source of the interrupt vector number is reserved, set the corresponding bit to 0. The read value is 0.

The IERm register enables or disables output of the interrupt request to the interrupt request destination.

IENj Bit (Interrupt Request Enable j) (j = 0 to 7)

When the IENj bit is 1, an interrupt request is output to the destination. When the IENj bit is 0, an interrupt request is not output to the destination.

The IRn.IR flag (n = 016 to 255) is not affected by the IENj bit setting. Even when the IENj bit is 0, the IR flag changes according to the conditions described in section 15.2.1, Interrupt Request Register n (IRn) (n = 016 to 255).

The IERm.IENj bit is provided for each interrupt vector number.

Refer to Table 15.5, Interrupt Vector Table for the correspondence between interrupt sources and the IERm.IENj bit.

Note that m and j can be calculated by the following formula:

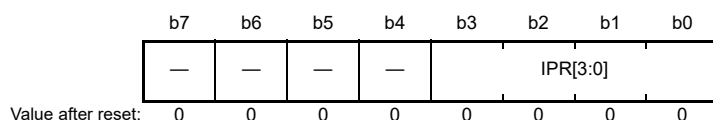
$m = \text{quotient when } n \text{ divided by } 8$

$j = \text{remainder when } n \text{ divided by } 8$

Refer to section 15.7.3.1, Interrupt Request Destination Setting Procedure for the procedure to set the IERm.IENj bit to select the interrupt request destination.

15.2.3 Interrupt Source Priority Register r (IPRr) (r = 000 to 255)

Address(es): ICU.IPR000 0008 7300h to ICU.IPR255 0008 73FFh



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	IPR[3:0]	Interrupt Priority Level Select	b3 b0 0 0 0 0: Level 0 (interrupt disabled) *1 0 0 0 1: Level 1 0 0 1 0: Level 2 0 0 1 1: Level 3 0 1 0 0: Level 4 0 1 0 1: Level 5 0 1 1 0: Level 6 0 1 1 1: Level 7 1 0 0 0: Level 8 1 0 0 1: Level 9 1 0 1 0: Level 10 1 0 1 1: Level 11 1 1 0 0: Level 12 1 1 0 1: Level 13 1 1 1 0: Level 14 1 1 1 1: Level 15 (highest)	R/W
b7 to b4	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. For an interrupt source of a fast interrupt, even when the IPR[3:0] bits are set to level 0, the priority level is level 15.

The IPRr register sets the interrupt priority level of an interrupt source that is assigned to the corresponding interrupt vector number.

IPR[3:0] Bits (Interrupt Priority Level Select)

The IPR[3:0] bits select the interrupt priority level of the corresponding interrupt source.

The priority level selected by the IPR[3:0] bits is used for only determining the priority level of interrupt requests to the CPU. It does not affect transfer requests to the DTC and DMAC.

The CPU accepts only interrupt requests that have the higher priority level than the processor interrupt priority level indicated by the PSW.IPL[3:0] bits.

When multiple interrupt requests are concurrently generated, the priority levels selected by the corresponding IPR[3:0] bits are compared. When multiple interrupt requests that have the same priority level are concurrently generated, the interrupt request that has the smallest interrupt vector number has priority.

Write to this register while the corresponding IERm.IENj bit is 0 (interrupt request is disabled) (m = 02h to 1Fh; j = 0 to 7).

Refer to Table 15.5, Interrupt Vector Table for the correspondence between interrupt vectors and the IPRr register.

Note that r matches the vector number when the interrupt vector number is 32 or greater.

15.2.4 Fast Interrupt Set Register (FIR)

Address(es): ICU.FIR 0008 72F0h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	FIEN	—	—	—	—	—	—	—	FVCT[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	FVCT[7:0]	Fast Interrupt Vector Number	Set the vector number of an interrupt source that is assigned to a fast interrupt.	R/W
b14 to b8	—	Reserved	The read value is 0. The write value should be 0.	R/W
b15	FIEN	Fast Interrupt Enable	0: Fast interrupt is disabled 1: Fast interrupt is enabled	R/W

The FIR register sets an interrupt source that is handled as the fast interrupt.

The fast interrupt is enabled only when the destination is the CPU. When the destination is the DTC or DMAC, the DTC or DMA transfer request is not affected by setting the interrupt vector number as the fast interrupt.

Write to this register while the corresponding IERm.IENj bit is 0 (m = 02h to 1Fh; j = 0 to 7).

Refer to section 15.9, Fast interrupt for details on the fast interrupt.

FVCT[7:0] Bits (Fast Interrupt Vector Number)

The FVCT[7:0] bits set the interrupt vector number of an interrupt source for the fast interrupt.

Refer to Table 15.5, Interrupt Vector Table for interrupt vector numbers that can be set in the FVCT[7:0] bits. Do not set an interrupt vector number that is reserved.

FIEN Bit (Fast Interrupt Enable)

The FIEN bit enables the fast interrupt to be used.

When the FIEN bit is 1, the interrupt source that is assigned to the interrupt vector number set by the FVCT[7:0] bits is handled as the fast interrupt.

When an interrupt request of the interrupt vector number set by the FVCT[7:0] bits is generated to the CPU while the FIEN bit is 1, an interrupt request is output to the CPU as the fast interrupt regardless of the IPRr register setting (r = 000 to 255). Note that the IPRr register setting is required when using the fast interrupt to exit software standby mode. Refer to section 15.10.3, Exiting Software Standby Mode for details.

15.2.5 Software Interrupt Generation Register (SWINTR)

Address(es): ICU.SWINTR 0008 72E0h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	SWINT
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SWINT	Software Interrupt Generation	The read value is 0. When writing 1 to this bit, a software interrupt request is generated. Writing 0 to this bit has no effect.	R/W
b7 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

The SWINTR register controls generation of a software interrupt request.

SWINT Bit (Software Interrupt Generation)

When the SWINT bit is set to 1, a software interrupt request (SWINT) is generated, and the IR027.IR flag becomes 1. A software interrupt request (SWINT) can be set as a DTC trigger, but it cannot be set as a DMAC trigger.

15.2.6 Software Interrupt 2 Generation Register (SWINT2R)

Address(es): ICU.SWINT2R 0008 72E1h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	SWINT 2
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SWINT2	Software Interrupt 2 Generation	The read value is 0. When writing 1 to this bit, a software interrupt request 2 is generated. Writing 0 to this bit has no effect.	R/W
b7 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

The SWINT2R register controls generation of software interrupt request 2.

SWINT2 Bit (Software Interrupt 2 Generation)

When the SWINT2 bit is set to 1, software interrupt request 2 (SWINT2) is generated, and the IR026.IR flag becomes 1. Software interrupt request 2 (SWINT2) can be set as the DTC trigger, but it cannot be set as the DMAC trigger.

15.2.7 DTC Transfer Request Enable Register n (DTCERn) (n = 026 to 255)

Address(es): ICU.DTCER026 0008 711Ah to ICU.DTCER255 0008 71FFh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	DTCE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	DTCE	DTC Transfer Request Enable	0: The corresponding interrupt source is selected as an interrupt request to the CPU or as the DMAC trigger. 1: The corresponding interrupt source is selected as the DTC trigger.	R/W
b7 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

The DTCERn register selects the interrupt source corresponding to interrupt vector number n as the DTC trigger. Do not set the same interrupt source for both the DTC trigger and DMAC trigger. Refer to **Table 15.5, Interrupt Vector Table** for the correspondence between interrupt sources and interrupt vector numbers and interrupt sources that can be used as the DTC trigger.

DTCE Bit (DTC Transfer Request Enable)

When the DTCE bit is set to 1, the corresponding interrupt source is selected as the DTC trigger.

This bit becomes 1 under the following condition:

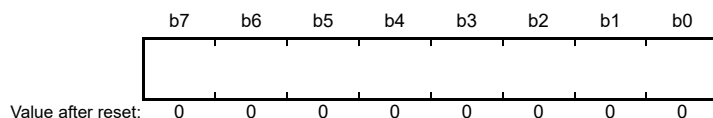
- 1 is written to the DTCE bit

This bit becomes 0 under any of the following conditions:

- The specified number of transfers is completed (for the chain transfer, the number of transfers for the last chain transfer is completed)
- 0 is written to the DTCE bit

15.2.8 DMAC Trigger Select Register m (DMRSRm) (m = DMAC channel number)

Address(es): ICU.DMRSR0 0008 7400h, ICU.DMRSR1 0008 7404h, ICU.DMRSR2 0008 7408h, ICU.DMRSR3 0008 740Ch,
ICU.DMRSR4 0008 7410h, ICU.DMRSR5 0008 7414h, ICU.DMRSR6 0008 7418h, ICU.DMRSR7 0008 741Ch



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	—	These bits set the interrupt vector number of the interrupt source as the DMAC trigger.	R/W

The DMRSRm register sets an interrupt source as the DMACm trigger.

Do not set the same vector number for multiple DMRSRm registers. Do not set the same interrupt source for both the DTC trigger and DMAC trigger. Otherwise, the operation is not guaranteed.

Set the interrupt vector number of an interrupt source used as the DMAC trigger in the DMRSRm register. Do not set vector numbers of interrupt sources that cannot be used as the DMAC trigger.

Refer to Table 15.5, Interrupt Vector Table for interrupt vector numbers of interrupt sources.

Write the DMRSRm register while the DMACm.DMCNT.DTE bit is 0.

15.2.9 IRQ Control Register i (IRQCRi) (i = 0 to 15)

Address(es): ICU.IRQCR0 0008 7500h to ICU.IRQCR15 0008 750Fh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	IRQMD[1:0]	—	—	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	—	Reserved	The read value is 0. The write value should be 0.	R/W
b3, b2	IRQMD[1:0]	IRQ Detection Select	b3 b2 0 0: Low level 0 1: Falling edge 1 0: Rising edge 1 1: Rising and falling edges	R/W
b7 to b4	—	Reserved	The read value is 0. The write value should be 0.	R/W

The IRQCRi register selects the detection method for external pin interrupts.

Write to this register while the corresponding IERm.IENj bit is 0 (m = 02h to 1Fh; j = 0 to 7). After writing to this register, set the IRn.IR flag to 0, and then set the IENj bit to 1 (n = 016 to 255). Note that setting the IR flag to 0 is not required when changing the detection method to level detection.

IRQMD[1:0] Bits (IRQ Detection Select)

The IRQMD[1:0] bits set the detection method for the IRQi pin interrupt (i = 0 to 15).

Refer to section 15.7.4, Setting the External Pin Interrupt for the procedure to set the external pin interrupts.

15.2.10 IRQ Pin Digital Filter Enable Register 0 (IRQFLTE0)

Address(es): ICU.IRQFLTE0 0008 7520h

b7	b6	b5	b4	b3	b2	b1	b0
FLTEN 7	FLTEN 6	FLTEN 5	FLTEN 4	FLTEN 3	FLTEN 2	FLTEN 1	FLTEN 0
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	FLTEN0	IRQ0 Digital Filter Enable	0: Digital filter is disabled. 1: Digital filter is enabled.	R/W
b1	FLTEN1	IRQ1 Digital Filter Enable		R/W
b2	FLTEN2	IRQ2 Digital Filter Enable		R/W
b3	FLTEN3	IRQ3 Digital Filter Enable		R/W
b4	FLTEN4	IRQ4 Digital Filter Enable		R/W
b5	FLTEN5	IRQ5 Digital Filter Enable		R/W
b6	FLTEN6	IRQ6 Digital Filter Enable		R/W
b7	FLTEN7	IRQ7 Digital Filter Enable		R/W

The IRQFLTE0 register enables and disables digital filters for pins IRQ0 to IRQ7.

FLTENi Bits (IRQi Digital Filter Enable) (i = 0 to 7)

When the FLTENi bit is 1, the digital filter for the IRQi pin is enabled. When the FLTENi bit is 0, the digital filter for the IRQi pin is disabled.

The signal input to the IRQi pin is sampled at the sampling clock set by the IRQFLTC0.FCLKSELi[1:0] bits. When the sampled signal level does not match three times in a row, the input signal is removed.

Refer to section 15.7.6, Digital Filter for details on the digital filter.

15.2.11 IRQ Pin Digital Filter Enable Register 1 (IRQFLTE1)

Address(es): ICU.IRQFLTE1 0008 7521h

b7	b6	b5	b4	b3	b2	b1	b0
FLTEN15	FLTEN14	FLTEN13	FLTEN12	FLTEN11	FLTEN10	FLTEN9	FLTEN8
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	FLTEN8	IRQ8 Digital Filter Enable	0: Digital filter is disabled. 1: Digital filter is enabled.	R/W
b1	FLTEN9	IRQ9 Digital Filter Enable		R/W
b2	FLTEN10	IRQ10 Digital Filter Enable		R/W
b3	FLTEN11	IRQ11 Digital Filter Enable		R/W
b4	FLTEN12	IRQ12 Digital Filter Enable		R/W
b5	FLTEN13	IRQ13 Digital Filter Enable		R/W
b6	FLTEN14	IRQ14 Digital Filter Enable		R/W
b7	FLTEN15	IRQ15 Digital Filter Enable		R/W

The IRQFLTE1 register enables or disables digital filters for pins IRQ8 to IRQ15.

FLTEN_i Bit (IRQ_i Digital Filter Enable) (i = 8 to 15)

When the FLTEN_i bit is 1, the digital filter for the IRQ_i pin is enabled. When the FLTEN_i bit is 0, the digital filter for the IRQ_i pin is disabled.

The signal input to the IRQ_i pin is sampled at the sampling clock set by the IRQFLTC1.FCLKSEL_i[1:0] bits. When the sampled signal level does not match three times in a row, the input signal is removed.

Refer to section 15.7.6, Digital Filter for details on the digital filter.

15.2.12 IRQ Pin Digital Filter Setting Register 0 (IRQFLTC0)

Address(es): ICU.IRQFLTC0 0008 7528h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
FCLKSEL7[1:0]		FCLKSEL6[1:0]		FCLKSEL5[1:0]		FCLKSEL4[1:0]		FCLKSEL3[1:0]		FCLKSEL2[1:0]		FCLKSEL1[1:0]		FCLKSEL0[1:0]	
0		0		0		0		0		0		0		0	

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	FCLKSEL0[1:0]	IRQ0 Digital Filter Sampling Clock	0 0: PCLKB	R/W
b3, b2	FCLKSEL1[1:0]	IRQ1 Digital Filter Sampling Clock	0 1: PCLKB/8	R/W
b5, b4	FCLKSEL2[1:0]	IRQ2 Digital Filter Sampling Clock	1 0: PCLKB/32	R/W
b7, b6	FCLKSEL3[1:0]	IRQ3 Digital Filter Sampling Clock	1 1: PCLKB/64	R/W
b9, b8	FCLKSEL4[1:0]	IRQ4 Digital Filter Sampling Clock		R/W
b11, b10	FCLKSEL5[1:0]	IRQ5 Digital Filter Sampling Clock		R/W
b13, b12	FCLKSEL6[1:0]	IRQ6 Digital Filter Sampling Clock		R/W
b15, b14	FCLKSEL7[1:0]	IRQ7 Digital Filter Sampling Clock		R/W

The IRQFLTC0 register sets the sampling clock of the digital filter for pins IRQ0 to IRQ7.

FCLKSELi[1:0] Bits (IRQi Digital Filter Sampling Clock) (i = 0 to 7)

The FCLKSELi[1:0] bits set the sampling clock of the digital filter for the IRQi pin.

Refer to section 15.7.6, Digital Filter for details on the digital filter.

15.2.13 IRQ Pin Digital Filter Setting Register 1 (IRQFLTC1)

Address(es): ICU.IRQFLTC1 0008 752Ah

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
FCLKSEL15[1:0]	FCLKSEL14[1:0]	FCLKSEL13[1:0]	FCLKSEL12[1:0]	FCLKSEL11[1:0]	FCLKSEL10[1:0]	FCLKSEL9[1:0]	FCLKSEL8[1:0]								
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	FCLKSEL8[1:0]	IRQ8 Digital Filter Sampling Clock	0 0: PCLKB	R/W
b3, b2	FCLKSEL9[1:0]	IRQ9 Digital Filter Sampling Clock	0 1: PCLKB/8	R/W
b5, b4	FCLKSEL10[1:0]	IRQ10 Digital Filter Sampling Clock	1 0: PCLKB/32	R/W
b7, b6	FCLKSEL11[1:0]	IRQ11 Digital Filter Sampling Clock	1 1: PCLKB/64	R/W
b9, b8	FCLKSEL12[1:0]	IRQ12 Digital Filter Sampling Clock		R/W
b11, b10	FCLKSEL13[1:0]	IRQ13 Digital Filter Sampling Clock		R/W
b13, b12	FCLKSEL14[1:0]	IRQ14 Digital Filter Sampling Clock		R/W
b15, b14	FCLKSEL15[1:0]	IRQ15 Digital Filter Sampling Clock		R/W

The IRQFLTC1 register sets the sampling clock of the digital filter for pins IRQ8 to IRQ15.

FCLKSELi[1:0] Bits (IRQi Digital Filter Sampling Clock) (i = 8 to 15)

The FCLKSELi[1:0] bits set the sampling clock of the digital filter for the IRQi pin.

Refer to section 15.7.6, Digital Filter for details on the digital filter.

15.2.14 Non-Maskable Interrupt Status Register (NMISR)

Address(es): ICU.NMISR 0008 7580h

b7	b6	b5	b4	b3	b2	b1	b0
—	EXNMIST	LVD2ST	LVD1ST	IWDTST	WDTST	OSTST	NMIST
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	NMIST	NMI Status Flag	0: NMI pin interrupt is not requested. 1: NMI pin interrupt is requested.	R
b1	OSTST	Oscillation Stop Detection Interrupt Status Flag	0: Oscillation stop detection interrupt is not requested. 1: Oscillation stop detection interrupt is requested.	R
b2	WDTST	WDT Underflow/Refresh Error Status Flag	0: WDT underflow/refresh error interrupt is not requested. 1: WDT underflow/refresh error interrupt is requested.	R
b3	IWDTST	IWDT Underflow/Refresh Error Status Flag	0: IWDT underflow/refresh error interrupt is not requested. 1: IWDT underflow/refresh error interrupt is requested.	R
b4	LVD1ST	Voltage Monitoring 1 Interrupt Status Flag	0: Voltage monitoring 1 interrupt is not requested. 1: Voltage monitoring 1 interrupt is requested.	R
b5	LVD2ST	Voltage Monitoring 2 Interrupt Status Flag	0: Voltage monitoring 2 interrupt is not requested. 1: Voltage monitoring 2 interrupt is requested.	R
b6	EXNMIST	Expanded Non-Maskable Interrupt Status Flag	0: Expanded non-maskable interrupt is not requested. 1: Expanded non-maskable interrupt is requested.	R
b7	—	Reserved	This bit is read as 0 and cannot be modified.	R

The NMISR register indicates whether a non-maskable interrupt request has occurred.

Each flag in the NMISR register is not affected by the setting of the corresponding bit in the NMICR register.

In the non-maskable interrupt handler, read the NMISR register to check if the other non-maskable interrupt has occurred. Confirm that all the status flags are 0 before exiting the interrupt handler.

NMIST Flag (NMI Status Flag)

The NMIST flag indicates whether an NMI pin interrupt request has been generated.

This flag is read-only. To set the NMIST flag to 0, set the NMICLR.NMICLR bit to 1.

This flag becomes 1 under the following condition:

- An edge set in the NMICR.NMIMD bit is input to the NMI pin

This flag becomes 0 under the following condition:

- 1 is written to the NMICLR.NMICLR bit

OSTST Flag (Oscillation Stop Detection Interrupt Status Flag)

The OSTST flag indicates whether an oscillation stop detection interrupt request has been generated.

The OSTST flag is read-only. To set the OSTST flag to 0, set the NMICLR.OSTCLR bit to 1.

This flag becomes 1 under the following condition:

- An oscillation stop detection interrupt occurs

This flag becomes 0 under the following condition:

- 1 is written to the NMICLR.OSTCLR bit

WDTST Flag (WDT Underflow/Refresh Error Status Flag)

The WDTST flag indicates whether an WDT underflow/refresh error interrupt request is generated.

The WDTST flag is read-only. To set the WDTST flag to 0, set the NMICLR.WDTCLR bit to 1.

This flag becomes 1 under the following condition:

- A WDT underflow/refresh error interrupt occurs while the WDTRCR.RSTIRQS bit is 0

This flag becomes 0 under the following condition:

- 1 is written to the NMICLR.WDTCLR bit

IWDTST Flag (IWDT Underflow/Refresh Error Status Flag)

The IWDTST flag indicates whether an IWDT underflow/refresh error interrupt request is generated.

The IWDTST flag is read-only. To set the IWDTST flag to 0, set the NMICLR.IWDTCLR bit to 1.

This flag becomes 1 under the following condition:

- An IWDT underflow/refresh error interrupt occurs while the IWDTRCR.RSTIRQS bit is 0

This flag becomes 0 under the following condition:

- 1 is written to the NMICLR.IWDTCLR bit

LVD1ST Flag (Voltage Monitoring 1 Interrupt Status Flag)

The LVD1ST flag indicates whether a voltage monitoring 1 interrupt request is generated.

The LVD1ST flag is read-only. To set the LVD1ST flag to 0, set the NMICLR.LVD1CLR bit to 1.

This flag becomes 1 under the following condition:

- A voltage monitoring 1 interrupt occurs while the LVD1CR1.LVD1IRQSEL bit is 0

This flag becomes 0 under the following condition:

- 1 is written to the NMICLR.LVD1CLR bit

LVD2ST Flag (Voltage Monitoring 2 Interrupt Status Flag)

The LVD2ST flag indicates whether a voltage monitoring 2 interrupt request is generated.

The LVD2ST flag is read-only. To set the LVD2ST flag to 0, set the NMICLR.LVD2CLR bit to 1.

This flag becomes 1 under the following condition:

- A voltage monitoring 2 interrupt occurs while the LVD2CR1.LVD2IRQSEL bit is 0

This flag becomes 0 under the following condition:

- 1 is written to the NMICLR.LVD2CLR bit

EXNMIST Flag (Expanded Non-Maskable Interrupt Status Flag)

The EXNMIST flag indicates whether an expanded non-maskable interrupt request is generated.

The EXNMIST flag is read-only. To set the EXNMIST flag to 0, clear all error status flags of the expanded non-maskable interrupt request sources.

This flag becomes 1 under the following condition:

- When any of the flag in the EXNMISR register is 1 and the corresponding bit in the EXNMIER register is 1

This flag becomes 0 under the following condition:

- When all sources which set the EXNMIST flag to 1 are cleared.

15.2.15 Non-Maskable Interrupt Enable Register (NMIER)

Address(es): ICU.NMIER 0008 7581h

b7	b6	b5	b4	b3	b2	b1	b0
—	EXNMI EN	LVD2E N	LVD1E N	IWDTE N	WDTE N	OSTEN	NMIEN
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	NMIEN	NMI Pin Interrupt Enable	0: NMI pin interrupt is disabled. 1: NMI pin interrupt is enabled.	R/(W) *1
b1	OSTEN	Oscillation Stop Detection Interrupt Enable	0: Oscillation stop detection interrupt is disabled. 1: Oscillation stop detection interrupt is enabled.	R/(W) *1
b2	WDTEN	WDT Underflow/Refresh Error Enable	0: WDT underflow/refresh error interrupt is disabled. 1: WDT underflow/refresh error interrupt is enabled.	R/(W) *1
b3	IWDTEN	IWDT Underflow/Refresh Error Enable	0: IWDT underflow/refresh error interrupt is disabled. 1: IWDT underflow/refresh error interrupt is enabled.	R/(W) *1
b4	LVD1EN	Voltage Monitoring 1 Interrupt Enable	0: Voltage monitoring 1 interrupt is disabled. 1: Voltage monitoring 1 interrupt is enabled.	R/(W) *1
b5	LVD2EN	Voltage Monitoring 2 Interrupt Enable	0: Voltage monitoring 2 interrupt is disabled. 1: Voltage monitoring 2 interrupt is enabled.	R/(W) *1
b6	EXNMIEN	Expanded Non-Maskable Interrupt Enable	0: Expanded non-maskable interrupt is disabled. 1: Expanded non-maskable interrupt is enabled.	R/(W) *1
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Note 1. Once this bit is set to 1, it cannot be set to 0 by software.

The NMIER register enables and disables generation of non-maskable interrupt requests. When each bit is 1, the corresponding interrupt source is used as a non-maskable interrupt.

NMIEN Bit (NMI Pin Interrupt Enable)

The NMIEN bit enables and disables using the NMI pin interrupt.

OSTEN Bit (Oscillation Stop Detection Interrupt Enable)

The OSTEN bit enables and disables generation of a non-maskable interrupt by the oscillation stop detection interrupt. When the oscillation stop detection interrupt is used as a maskable interrupt, leave this bit set to 0.

WDTEN Bit (WDT Underflow/Refresh Error Enable)

The WDTEN bit enables and disables generation of a non-maskable interrupt by the WDT underflow/refresh error interrupt.

When the WDT underflow/refresh error interrupt is used as a maskable interrupt, leave this bit set to 0.

IWDTEN Bit (IWDT Underflow/Refresh Error Enable)

The IWDTEN bit enables and disables generation of a non-maskable interrupt by the IWDT underflow/refresh error interrupt.

When the IWDT underflow/refresh error interrupt is used as a maskable interrupt, leave this bit set to 0.

LVD1EN Bit (Voltage Monitoring 1 Interrupt Enable)

The LVD1EN bit enables and disables generation of a non-maskable interrupt by the voltage monitoring 1 interrupt. When the voltage monitoring 1 interrupt is used as a maskable interrupt, leave this bit set to 0.

LVD2EN Bit (Voltage Monitoring 2 Interrupt Enable)

The LVD2EN bit enables and disables generation of a non-maskable interrupt by the voltage monitoring 2 interrupt. When the voltage monitoring 2 interrupt is used as a maskable interrupt, leave this bit set to 0.

EXNMIEN Bit (Expanded Non-Maskable Interrupt Enable)

The EXNMIEN bit enables and disables generation of a non-maskable interrupt by the expanded non-maskable interrupt. When the expanded non-maskable interrupt is used as a maskable interrupt, leave this bit set to 0.

15.2.16 Non-Maskable Interrupt Status Clear Register (NMICLR)

Address(es): ICU.NMICLR 0008 7582h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	LVD2CLR	LVD1CLR	IWDTCCLR	WDTCLR	OSTCLR	NMICLR
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	NMICLR	NMI Clear	The read value is 0. When 1 is written to this bit, the NMISR.NMIST flag becomes 0. Writing 0 to this bit has no effect.	R/(W)
b1	OSTCLR	OST Clear	The read value is 0. When 1 is written to this bit, the NMISR.OSTST flag becomes 0. Writing 0 to this bit has no effect.	R/(W)
b2	WDTCLR	WDT Clear	The read value is 0. When 1 is written to this bit, the NMISR.WDTST flag becomes 0. Writing 0 to this bit has no effect.	R/(W)
b3	IWDTCCLR	IWDT Clear	The read value is 0. When 1 is written to this bit, the NMISR.IWDTST flag becomes 0. Writing 0 to this bit has no effect.	R/(W)
b4	LVD1CLR	LVD1 Clear	The read value is 0. When 1 is written to this bit, the NMISR.LVD1ST flag becomes 0. Writing 0 to this bit has no effect.	R/(W)
b5	LVD2CLR	LVD2 Clear	The read value is 0. When 1 is written to this bit, the NMISR.LVD2ST flag becomes 0. Writing 0 to this bit has no effect.	R/(W)
b7-b6	—	Reserved	The read value is 0. The write value should be 0.	R/W

The NMICLR register clears each flag in the NMISR register.

When writing 1 to a bit in this register, the corresponding status flag becomes 0.

15.2.17 NMI Pin Interrupt Control Register (NMICR)

Address(es): ICU.NMICR 0008 7583h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	NMIMD	—	—	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	The read value is 0. The write value should be 0.	R/W
b3	NMIMD	NMI Detection Set	0: Falling edge 1: Rising edge	R/W
b7 to b4	—	Reserved	The read value is 0. The write value should be 0.	R/W

The NMICR register selects a detection method for the NMI pin interrupt.

Write to the NMICR register while the NMIER.NMIEN bit is 0.

15.2.18 Expanded Non-Maskable Interrupt Status Register (EXNMISR)

Address(es): ICU.EXNMISR 0008 7584h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	DPFPU ST	RAMST
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RAMST	RAM Error Interrupt Status Flag	0: RAM error interrupt is not requested. 1: RAM error interrupt is requested.	R
b1	DPFPUST	Double-Precision Floating-Point Exception Status Flag	0: Double-precision floating-point exception is not requested. 1: Double-precision floating-point exception is requested.	R
b7 to b2	—	Reserved	These bits are read as 0 and cannot be modified.	R

The EXNMISR register indicates whether a non-maskable interrupt request assigned to expanded non-maskable interrupt source has occurred. Each flag in the EXNMISR register is not affected by the setting of the corresponding bit in the EXNMIER register.

RAMST Flag (RAM Error Interrupt Status Flag)

The RAMST flag indicates whether an RAM error interrupt request is generated from the RAM (including the expanded RAM and ECCRAM).

The RAMST flag is read-only. To set the RAMST flag to 0, clear all of the error status flags of the RAM. Refer to section 60.3.3, RAM Error Interrupt Function for details.

This flag becomes 1 under the following condition:

- A parity check error interrupt occurs (When the RAM.RAMSTS.RAMERR or RAM.EXRAMSTS.EXRAMERR flag becomes 1)
- An ECC error interrupt occurs (When the RAM.ECCRAM1STS.ECC1ERR or RAM.ECCRAM2STS.ECC2ERR flag becomes 1)

This flag becomes 0 under the following condition:

- When all sources which set the RAMST flag to 1 are cleared.

DPFPUST Flag (Double-Precision Floating-Point Exception Status Flag)

The DPFPUST flag indicates whether an double-precision floating-point exception is generated by double-precision floating-point coprocessor.

This flag becomes 1 under the following condition:

- A double-precision floating-point exception occurs

This flag becomes 0 under the following condition:

- 1 is written to the EXNMICLR.DPFPUCLR bit

15.2.19 Expanded Non-Maskable Interrupt Enable Register (EXNMIER)

Address(es): ICU.EXNMIER 0008 7585h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	DPFPU EN	RAME N
Value after reset:	1	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b0	RAMEN	RAM Error Interrupt Enable	0: RAM error interrupt is disabled. 1: RAM error interrupt is enabled.	R/W
b1	DPFPUEN	Double-Precision Floating-Point Exception Enable	0: Double-precision floating-point exception is disabled. 1: Double-precision floating-point exception is enabled.	R/W
b6 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R
b7	—	Reserved	This bit is read as 1. The write value should be 1.	R/W

The EXNMIER register enables and disables generation of non-maskable interrupt requests assigned to expanded non-maskable interrupt source. When each bit is 1, the corresponding interrupt source is used as a non-maskable interrupt. Set this register before setting the NMIER.EXNMIEN bit to 1.

RAMEN Bit (RAM Error Interrupt Enable)

The RAMEN bit enables and disables generation of a non-maskable interrupt by the RAM error interrupt from the RAM (including the expanded RAM and ECCRAM).

When the RAM error interrupt is used as a maskable interrupt, set this bit to 0.

DPFPUEN Bit (Double-Precision Floating-Point Exception Enable)

The DPFPUEN bit enables and disables generation of a non-maskable interrupt by the double-precision floating-point exception from double-precision floating-point coprocessor.

When the double-precision floating-point exception is used as a maskable interrupt, leave this bit set to 0.

15.2.20 Expanded Non-Maskable Interrupt Status Clear Register (EXNMICLR)

Address(es): ICU.EXNMICLR 0008 7586h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	DPFPU CLR	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	This bit is read as 0. The write value should be 0.	R
b1	DPFPUCLR	Double-Precision Floating-Point Exception Clear	The read value is 0. When 1 is written to this bit, the EXNMISR.DPFPUS flag becomes 0. Writing 0 to this bit has no effect.	R/(W)
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R

The EXNMICLR register clears each flag in the EXNMISR register.

When writing 1 to a bit in this register, the corresponding status flag becomes 0.

15.2.21 NMI Pin Digital Filter Enable Register (NMIFLTE)

Address(es): ICU.NMIFLTE 0008 7590h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	NFLTE N
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	NFLTEN	NMI Digital Filter Enable	0: Digital filter is disabled. 1: Digital filter is enabled.	R/W
b7 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

The NMIFLTE register enables or disables the digital filter for the NMI pin.

NFLTEN Bit (NMI Digital Filter Enable)

When the NFLTEN bit is 1, the digital filter is enabled. When the NFLTEN bit is 0, the digital filter is disabled.

The signal input to the NMI pin is sampled at the sampling clock set by the NMIFLTC.NFCLKSEL[1:0] bits. When the sampled signal level does not match three times in a row, the input signal is removed.

Refer to section 15.7.6, Digital Filter for details on the digital filter.

15.2.22 NMI Pin Digital Filter Setting Register (NMIFLTC)

Address(es): ICU.NMIFLTC 0008 7594h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	NFCLKSEL[1:0]	
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	NFCLKSEL[1:0]	NMI Digital Filter Sampling Clock	b1 b0 0 0: PCLKB 0 1: PCLKB/8 1 0: PCLKB/32 1 1: PCLKB/64	R/W
b7 to b2	—	Reserved	The read value is 0. The write value should be 0.	R/W

The NMIFLTC register sets the sampling clock of the digital filter for the NMI pin.

NFCLKSEL[1:0] Bits (NMI Digital Filter Sampling Clock)

The NFCLKSEL[1:0] bits select the sampling clock of the digital filter for the NMI pin.

Refer to section 15.7.6, Digital Filter for details on the digital filter.

15.2.23 Group IE0 Interrupt Request Register (GRPIE0), Group BE0 Interrupt Request Register (GRPBE0), Group BL0/BL1/BL2 Interrupt Request Register (GRPBL0/GRPBL1/GRPBL2), Group AL0/AL1 Interrupt Request Register (GRPAL0/GRPAL1)

Address(es): ICU.GRPIE0 0008 75B0h, ICU.GRPBE0 0008 7600h, ICU.GRPBL0 0008 7630h, ICU.GRPBL1 0008 7634h, ICU.GRPBL2 0008 7638h, ICU.GRPAL0 0008 7830h, ICU.GRPAL1 0008 7834h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	IS31	IS30	IS29	IS28	IS27	IS26	IS25	IS24	IS23	IS22	IS21	IS20	IS19	IS18	IS17	IS16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	IS15	IS14	IS13	IS12	IS11	IS10	IS9	IS8	IS7	IS6	IS5	IS4	IS3	IS2	IS1	IS0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	IS0	Interrupt Status Flag 0	0: Interrupt is not requested. 1: Interrupt is requested.	R
b1	IS1	Interrupt Status Flag 1		R
b2	IS2	Interrupt Status Flag 2		R
b3	IS3	Interrupt Status Flag 3		R
b4	IS4	Interrupt Status Flag 4		R
b5	IS5	Interrupt Status Flag 5		R
b6	IS6	Interrupt Status Flag 6		R
b7	IS7	Interrupt Status Flag 7		R
b8	IS8	Interrupt Status Flag 8		R
b9	IS9	Interrupt Status Flag 9		R
b10	IS10	Interrupt Status Flag 10		R
b11	IS11	Interrupt Status Flag 11		R
b12	IS12	Interrupt Status Flag 12		R
b13	IS13	Interrupt Status Flag 13		R
b14	IS14	Interrupt Status Flag 14		R
b15	IS15	Interrupt Status Flag 15		R
b16	IS16	Interrupt Status Flag 16		R
b17	IS17	Interrupt Status Flag 17		R
b18	IS18	Interrupt Status Flag 18		R
b19	IS19	Interrupt Status Flag 19		R
b20	IS20	Interrupt Status Flag 20		R
b21	IS21	Interrupt Status Flag 21		R
b22	IS22	Interrupt Status Flag 22		R
b23	IS23	Interrupt Status Flag 23		R
b24	IS24	Interrupt Status Flag 24		R
b25	IS25	Interrupt Status Flag 25		R
b26	IS26	Interrupt Status Flag 26		R
b27	IS27	Interrupt Status Flag 27		R
b28	IS28	Interrupt Status Flag 28		R
b29	IS29	Interrupt Status Flag 29		R
b30	IS30	Interrupt Status Flag 30		R
b31	IS31	Interrupt Status Flag 31		R

These registers indicate the status of each interrupt request of group interrupt sources.

The GRPIE0 register contains the statuses of interrupt sources that are detected by edge detection and use ICLK as the operating clock.

The GRPBE0 register contains the statuses of interrupt sources that are detected by edge detection and use PCLKB as the operating clock.

Registers GRPBL0, GRPBL1, and GRPBL2 contain the statuses of interrupt sources that are detected by level detection and use PCLKB as the operating clock.

The GRPAL0 and GRPAL1 registers contain the statuses of interrupt sources that are detected by level detection and use PCLKA as the operating clock.

These registers are collectively referred to as the “group interrupt request register”.

Refer to section 15.4.4, Group Interrupts for details on group interrupts.

ISj Flag (Interrupt Status Flag j) (j = 0 to 31)

The ISj flag indicates the status of interrupt sources assigned to group interrupts.

The ISj flag becomes 1 only when the corresponding ENj bit in the group interrupt request enable register is 1. When any of the ISj flags becomes 1, the IRn.IR flag corresponding to the group interrupt becomes 1 (n = 016 to 255).

(1) Group IE0

This flag becomes 1 under the following condition:

- The GRPIE0.ISj flag becomes 1 when the corresponding peripheral module interrupt request is generated while the GENIE0.ENj bit is 1.

This flag becomes 0 under the following condition:

- The GRPIE0.ISj flag becomes 0 when the GCRIE0.CLRj bit is set to 1.

(2) Group BE0

This flag becomes 1 under the following condition:

- The GRPBE0.ISj flag becomes 1 when the corresponding peripheral module interrupt request is generated while the GENBE0.ENj bit is 1.

This flag becomes 0 under the following condition:

- The GRPBE0.ISj flag becomes 0 when the GCRBE0.CLRj bit is set to 1.

(3) Group BL0/BL1/BL2

This flag becomes 1 under the following condition:

- The GRPBL0/GRPBL1/GRPBL2.ISj flag becomes 1 while the corresponding peripheral module interrupt request is generated when the GENBL0/GENBL1/GENBL2.ENj bit is 1.

This flag becomes 0 under any of the following conditions:

- The GRPBL0/GRPBL1/GRPBL2.ISj flag becomes 0 when the corresponding peripheral module interrupt request is cleared.
- The GRPBL0/GRPBL1/GRPBL2.ISj flag becomes 0 when the GENBL0/GENBL1/GENBL2.ENj bit is set to 0.

(4) Group AL0/AL1

This flag becomes 1 under the following condition:

- The GRPAL0/GRPAL1.ISj flag becomes 1 while the corresponding peripheral module interrupt request is generated when the GENAL0/GENAL1.ENj bit is 1.

This flag becomes 0 under any of the following conditions:

- The GRPAL0/GRPAL1.ISj flag becomes 0 when the corresponding peripheral module interrupt request is cleared.
- The GRPAL0/GRPAL1.ISj flag becomes 0 when the GENAL0/GENAL1.ENj bit is set to 0.

15.2.24 Group IE0 Interrupt Request Enable Register (GENIE0), Group BE0 Interrupt Request Enable Register (GENBE0), Group BL0/BL1/BL2 Interrupt Request Enable Register (GENBL0/GENBL1/GENBL2), Group AL0/AL1 Interrupt Request Enable Register (GENAL0/GENAL1)

Address(es): ICU.GENIE0 0008 75B4h, ICU.GENBE0 0008 7640h, ICU.GENBL0 0008 7670h, ICU.GENBL1 0008 7674h, ICU.GENBL2 0008 7678h, ICU.GENAL0 0008 7870h, ICU.GENAL1 0008 7874h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
EN31	EN30	EN29	EN28	EN27	EN26	EN25	EN24	EN23	EN22	EN21	EN20	EN19	EN18	EN17	EN16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
EN15	EN14	EN13	EN12	EN11	EN10	EN9	EN8	EN7	EN6	EN5	EN4	EN3	EN2	EN1	EN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	EN0	Interrupt Request Enable 0	0: Interrupt request is disabled. 1: Interrupt request is enabled.	R/W
b1	EN1	Interrupt Request Enable 1		R/W
b2	EN2	Interrupt Request Enable 2		R/W
b3	EN3	Interrupt Request Enable 3		R/W
b4	EN4	Interrupt Request Enable 4		R/W
b5	EN5	Interrupt Request Enable 5		R/W
b6	EN6	Interrupt Request Enable 6		R/W
b7	EN7	Interrupt Request Enable 7		R/W
b8	EN8	Interrupt Request Enable 8		R/W
b9	EN9	Interrupt Request Enable 9		R/W
b10	EN10	Interrupt Request Enable 10		R/W
b11	EN11	Interrupt Request Enable 11		R/W
b12	EN12	Interrupt Request Enable 12		R/W
b13	EN13	Interrupt Request Enable 13		R/W
b14	EN14	Interrupt Request Enable 14		R/W
b15	EN15	Interrupt Request Enable 15		R/W
b16	EN16	Interrupt Request Enable 16		R/W
b17	EN17	Interrupt Request Enable 17		R/W
b18	EN18	Interrupt Request Enable 18		R/W
b19	EN19	Interrupt Request Enable 19		R/W
b20	EN20	Interrupt Request Enable 20		R/W
b21	EN21	Interrupt Request Enable 21		R/W
b22	EN22	Interrupt Request Enable 22		R/W
b23	EN23	Interrupt Request Enable 23		R/W
b24	EN24	Interrupt Request Enable 24		R/W
b25	EN25	Interrupt Request Enable 25		R/W
b26	EN26	Interrupt Request Enable 26		R/W
b27	EN27	Interrupt Request Enable 27		R/W
b28	EN28	Interrupt Request Enable 28		R/W
b29	EN29	Interrupt Request Enable 29		R/W
b30	EN30	Interrupt Request Enable 30		R/W
b31	EN31	Interrupt Request Enable 31		R/W

Note: When a bit has no corresponding interrupt source (bit is reserved), set the bit to 0.

These registers select whether the IS_j flag in the group interrupt request register is set to 1 when each interrupt request for group interrupt sources is generated. These registers are collectively referred to as the “group interrupt request enable register”.

Registers GENIE0, GENBE0, GENBL0/GENBL1/GENBL2, and GENAL0/GENAL1 control the IS_j flag in registers GRPIE0, GRPBE0, GRPBL0/GRPBL1/GRPBL2, and GRPAL0/GRPAL1, respectively.

Refer to section 15.4.4, Group Interrupts for details on group interrupts.

EN_j Bits (Interrupt Request Enable j) (j = 0 to 31)

The EN_j bits select whether the corresponding IS_j flag in the group interrupt request register is set to 1 when an interrupt request assigned to group interrupts is generated.

(1) Group IE0

When a peripheral module interrupt request is generated while the corresponding GENIE0.EN_j bit is 1, the GRPIE0.IS_j flag becomes 1. When the EN_j bit is 0, the IS_j flag does not become 1.

Even when the EN_j bit is set to 0, the IS_j flag does not change.

(2) Group BE0

When a peripheral module interrupt request is generated while the corresponding GENBE0.EN_j bit is 1, the GRPBE0.IS_j flag becomes 1. When the EN_j bit is 0, the IS_j flag does not become 1.

Even when the EN_j bit is set to 0, the IS_j flag does not change.

(3) Group BL0/BL1/BL2

When a peripheral module interrupt request is generated while the corresponding GENBL0/GENBL1/GENBL2.EN_j bit is 1, the GRPBL0/GRPBL1/GRPBL2.IS_j flag becomes 1. When the EN_j bit is 0, the IS_j flag does not become 1.

When the EN_j bit is set to 0, the IS_j flag becomes 0.

(4) Group AL0/AL1

When a peripheral module interrupt request is generated while the corresponding GENAL0/GENAL1.EN_j bit is 1, the GRPAL0/GRPAL1.IS_j flag becomes 1. When the EN_j bit is 0, the IS_j flag does not become 1.

When the EN_j bit is set to 0, the IS_j flag becomes 0.

15.2.25 Group IE0 Interrupt Clear Register (GCRIE0), Group BE0 Interrupt Clear Register (GCRBE0)

Address(es): ICU.GCRIE0 0008 75B8h, ICU.GCRBE0 0008 7680h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
CLR31	CLR30	CLR29	CLR28	CLR27	CLR26	CLR25	CLR24	CLR23	CLR22	CLR21	CLR20	CLR19	CLR18	CLR17	CLR16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
CLR15	CLR14	CLR13	CLR12	CLR11	CLR10	CLR9	CLR8	CLR7	CLR6	CLR5	CLR4	CLR3	CLR2	CLR1	CLR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CLR0	Interrupt Source Clear 0	The read value is 0. When the CLRj bit is set to 1, the corresponding interrupt status flag (GRPIE0/GRPBE0.ISj) becomes 0 (j = 0 to 31). Writing 0 has no effect.	R/(W)
b1	CLR1	Interrupt Source Clear 1		R/(W)
b2	CLR2	Interrupt Source Clear 2		R/(W)
b3	CLR3	Interrupt Source Clear 3		R/(W)
b4	CLR4	Interrupt Source Clear 4		R/(W)
b5	CLR5	Interrupt Source Clear 5		R/(W)
b6	CLR6	Interrupt Source Clear 6		R/(W)
b7	CLR7	Interrupt Source Clear 7		R/(W)
b8	CLR8	Interrupt Source Clear 8		R/(W)
b9	CLR9	Interrupt Source Clear 9		R/(W)
b10	CLR10	Interrupt Source Clear 10		R/(W)
b11	CLR11	Interrupt Source Clear 11		R/(W)
b12	CLR12	Interrupt Source Clear 12		R/(W)
b13	CLR13	Interrupt Source Clear 13		R/(W)
b14	CLR14	Interrupt Source Clear 14		R/(W)
b15	CLR15	Interrupt Source Clear 15		R/(W)
b16	CLR16	Interrupt Source Clear 16		R/(W)
b17	CLR17	Interrupt Source Clear 17		R/(W)
b18	CLR18	Interrupt Source Clear 18		R/(W)
b19	CLR19	Interrupt Source Clear 19		R/(W)
b20	CLR20	Interrupt Source Clear 20		R/(W)
b21	CLR21	Interrupt Source Clear 21		R/(W)
b22	CLR22	Interrupt Source Clear 22		R/(W)
b23	CLR23	Interrupt Source Clear 23		R/(W)
b24	CLR24	Interrupt Source Clear 24		R/(W)
b25	CLR25	Interrupt Source Clear 25		R/(W)
b26	CLR26	Interrupt Source Clear 26		R/(W)
b27	CLR27	Interrupt Source Clear 27		R/(W)
b28	CLR28	Interrupt Source Clear 28		R/(W)
b29	CLR29	Interrupt Source Clear 29		R/(W)
b30	CLR30	Interrupt Source Clear 30		R/(W)
b31	CLR31	Interrupt Source Clear 31		R/(W)

Note: Write 1 to only the bit corresponding to the flag that is cleared, and write 0 to the other bits.

The GCRIE0 and GCRBE0 registers clear the GRPIE0.ISj and GRPBE0.ISj flags, respectively (j = 0 to 31). Refer to section 15.4.4, Group Interrupts for details on group interrupts.

CLRj Bit (Interrupt Source Clear j) (j = 0 to 31)

When the GCRIE0/GCRBE0.CLRj bit is set to 1, the GRPIE0/GRPBE0.ISj flag becomes 0.

15.2.26 Software Configurable Interrupt B Request Register k (PIBRk) (k = 0h to Bh)

Address(es): ICU.PIBR0 0008 7700h, ICU.PIBR1 0008 7701h, ICU.PIBR2 0008 7702h, ICU.PIBR3 0008 7703h, ICU.PIBR4 0008 7704h, ICU.PIBR5 0008 7705h, ICU.PIBR6 0008 7706h, ICU.PIBR7 0008 7707h, ICU.PIBR8 0008 7708h, ICU.PIBR9 0008 7709h, ICU.PIBRA 0008 770Ah, ICU.PIBRB 0008 770Bh

b7	b6	b5	b4	b3	b2	b1	b0
PIR7	PIR6	PIR5	PIR4	PIR3	PIR2	PIR1	PIR0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	PIR0	Software Configurable Interrupt B Status Flag 0	When reading 0: Interrupt is not requested. 1: Interrupt is requested.	R/W
b1	PIR1	Software Configurable Interrupt B Status Flag 1		R/W
b2	PIR2	Software Configurable Interrupt B Status Flag 2	When writing *1 0: Ignored. 1: The software configurable interrupt B status flag is cleared.	R/W
b3	PIR3	Software Configurable Interrupt B Status Flag 3		R/W
b4	PIR4	Software Configurable Interrupt B Status Flag 4		R/W
b5	PIR5	Software Configurable Interrupt B Status Flag 5		R/W
b6	PIR6	Software Configurable Interrupt B Status Flag 6		R/W
b7	PIR7	Software Configurable Interrupt B Status Flag 7		R/W

Note 1. Do not use bit manipulation instructions. If a bit manipulation instruction is used, multiple status flags may be cleared. Write 1 only to the flag to be cleared and write 0 to the other flags (write to this register in 8-bit units).

The PIBRk register is used for polling of interrupt requests of interrupt sources that is assigned to software configurable interrupt B by software. For an interrupt request of software configurable interrupt B set in the SLIBXRn or SLIBRn register, use the corresponding IRn.IR flag for polling (n = 128 to 207).

Refer to Table 15.3, Interrupt Sources for Software Configurable Interrupt B for correspondence between interrupt source numbers and interrupt sources for software configurable interrupt B.

PIRj Flag (Software Configurable Interrupt B Status Flag j) (j = 0 to 7)

When an interrupt request of interrupt sources assigned to software configurable interrupt B is generated, the corresponding PIBRk.PIRj flag becomes 1 regardless of whether the interrupt source is selected in the SLIBXRn or SLIBRn register.

Although the PIRj flag does not become 0 after the interrupt request is accepted by the destination (CPU, DTC, DMAC), generation of interrupt requests is not affected.

When using the PIRj flag for polling, set the PIRj flag to 0 by writing 1 to the flag in advance.

This flag becomes 1 under the following condition:

- An interrupt request is generated

This flag becomes 0 under the following condition:

- 1 is written to the PIBRk.PIRj flag

15.2.27 Software Configurable Interrupt A Request Register k (PIAR_k) (k = 0h to Ah, Ch)

Address(es): ICU.PIAR0 0008 7900h, ICU.PIAR1 0008 7901h, ICU.PIAR2 0008 7902h, ICU.PIAR3 0008 7903h, ICU.PIAR4 0008 7904h, ICU.PIAR5 0008 7905h, ICU.PIAR6 0008 7906h, ICU.PIAR7 0008 7907h, ICU.PIAR8 0008 7908h, ICU.PIAR9 0008 7909h, ICU.PIARA 0008 790Ah, ICU.PIARC 0008 790Ch

b7	b6	b5	b4	b3	b2	b1	b0
PIR7	PIR6	PIR5	PIR4	PIR3	PIR2	PIR1	PIR0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	PIR0	Software Configurable Interrupt A Status Flag 0	When reading 0: Interrupt is not requested. 1: Interrupt is requested.	R/W
b1	PIR1	Software Configurable Interrupt A Status Flag 1		R/W
b2	PIR2	Software Configurable Interrupt A Status Flag 2	When writing *1 0: Ignored. 1: The Software Configurable interrupt A status flag is cleared.	R/W
b3	PIR3	Software Configurable Interrupt A Status Flag 3		R/W
b4	PIR4	Software Configurable Interrupt A Status Flag 4		R/W
b5	PIR5	Software Configurable Interrupt A Status Flag 5		R/W
b6	PIR6	Software Configurable Interrupt A Status Flag 6		R/W
b7	PIR7	Software Configurable Interrupt A Status Flag 7		R/W

Note 1. Do not use bit manipulation instructions. If a bit manipulation instruction is used, multiple status flags may be cleared. Write 1 only to the flag to be cleared and write 0 to the other flags (write to this register in 8-bit units).

The PIAR_k register is used for polling interrupt requests of interrupt sources that is assigned to software configurable interrupt A by software. For an interrupt request of software configurable interrupt A set in the SLIAR_n register, use the corresponding IR_n.IR flag for polling (n = 208 to 255).

Refer to Table 15.4, Interrupt Sources for Software Configurable Interrupt A for correspondence between interrupt source numbers and interrupt sources for software configurable interrupt A.

PIR_j Flag (Software Configurable Interrupt A Status Flag j) (j = 0 to 7)

When an interrupt request of interrupt sources assigned to software configurable interrupt A is generated, the corresponding PIAR_k.PIR_j flag becomes 1 regardless of whether the interrupt source is selected in the SLIAR_n register. Although the PIR_j flag does not become 0 after the interrupt request is accepted by the destination (CPU, DTC, DMAC), generation of interrupt requests is not affected.

When using the PIR_j flag for polling, set the PIR_j flag to 0 by writing 1 to the flag in advance.

This flag becomes 1 under the following condition:

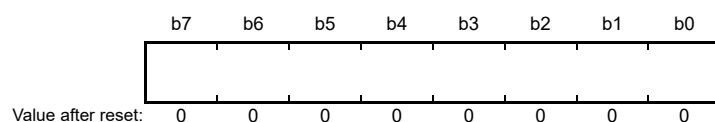
- An interrupt request is generated

This flag becomes 0 under the following condition:

- 1 is written to the PIAR_k.PIR_j flag

15.2.28 Software Configurable Interrupt B Source Select Register Xn (SLIBXRn) (n = 128 to 143)

Address(es): ICU.SLIBXR128 0008 7780h to ICU.SLIBXR143 0008 778Fh



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	—	00h: No interrupt source selected. 01h: Interrupt source number 1 : FEh: Interrupt source number 254 FFh: Interrupt source number 255	R/W*1

Note 1. Writing to these bits is ignored while the SLIPRCR.WPRC bit is set to 1.

The SLIBXRn register is used to assign interrupt vector numbers 128 to 143 to interrupt sources assigned to software configurable interrupt B.

Table 15.3, Interrupt Sources for Software Configurable Interrupt B lists interrupt sources assigned to software configurable interrupt B. Set the SLIBXRn register to an interrupt source number that is not reserved. When 00h or FFh are set, no interrupt source is assigned to interrupt vector number n.

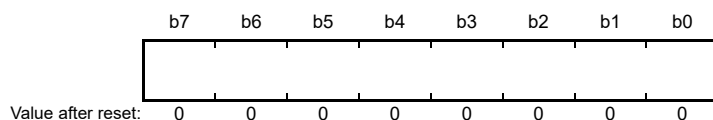
Do not assign the same interrupt source to multiple registers from registers SLIBXRn and SLIBRn.

Some interrupt sources can be used to start the DTC or DMAC. Refer to Table 15.3, Interrupt Sources for Software Configurable Interrupt B for details on which sources can be used to start the DTC or DMAC.

Refer to section 15.7.7, Setting Software Configurable Interrupts for the procedure to set software configurable interrupts.

15.2.29 Software Configurable Interrupt B Source Select Register n (SLIBRn) (n = 144 to 207)

Address(es): ICU.SLIBR144 0008 7790h to ICU.SLIBR207 0008 77CFh



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	—	00h: No interrupt source selected. 01h: Interrupt source number 1 : FEh: Interrupt source number 254 FFh: No interrupt source selected.	R/W*1

Note 1. Writing to these bits is ignored while the SLIPRCR.WPRC bit is set to 1.

The SLIBRn register is used to assign interrupt vector numbers 144 to 207 to interrupt sources for software configurable interrupt B.

Table 15.3, Interrupt Sources for Software Configurable Interrupt B lists interrupt sources for software configurable interrupt B. Set the SLIBRn register to an interrupt source number that is not reserved. When 00h or FFh are set, no interrupt source is assigned to interrupt vector number n.

Do not assign the same interrupt source to multiple registers from registers SLIBXRn and SLIBRn.

Some interrupt sources can be used to start the DTC or DMAC. Refer to Table 15.3, Interrupt Sources for Software Configurable Interrupt B for details on which sources can be used to start the DTC or DMAC.

Although interrupt sources of interrupt vector numbers 144 and 145 can be used to start EXDMAC0 and EXDMAC1, respectively, only interrupt source number 20, TPU1.TGI1A (TGRA input capture/compare match) can be set as a source to start the EXDMAC. Set only one of either the SLIBR144 register or SLIBR145 register to 20, and set the corresponding bit (SELEXD0 or SELEXD1 bit) in the SELEXDR register to 0.

Refer to section 15.7.7, Setting Software Configurable Interrupts for the procedure to set software configurable interrupts.

Table 15.3 Interrupt Sources for Software Configurable Interrupt B (1/3)

Interrupt Source No.	Category	Interrupt Request Source	Name	Start the DTC	Start the DMAC	Interrupt Status Flag
0	—	None	No interrupt selected (initial value)	N/A	N/A	PIBR0.PIR0
1	Edge	CMT2	CM12 (CMCOR compare match)	✓	✓	PIBR0.PIR1
2		CMT3	CM13 (CMCOR compare match)	✓	✓	PIBR0.PIR2
3		TMR0	CM1A0 (TCORA compare match)	✓	N/A	PIBR0.PIR3
4			CM1B0 (TCORB compare match)	✓	N/A	PIBR0.PIR4
5			OVI0 (TCNT overflow)	N/A	N/A	PIBR0.PIR5
6		TMR1	CM1A1 (TCORA compare match)	✓	N/A	PIBR0.PIR6
7			CM1B1 (TCORB compare match)	✓	N/A	PIBR0.PIR7
8			OVI1 (TCNT overflow)	N/A	N/A	PIBR1.PIR0
9		TMR2	CM1A2 (TCORA compare match)	✓	N/A	PIBR1.PIR1
10			CM1B2 (TCORB compare match)	✓	N/A	PIBR1.PIR2
11			OVI2 (TCNT overflow)	N/A	N/A	PIBR1.PIR3
12		TMR3	CM1A3 (TCORA compare match)	✓	N/A	PIBR1.PIR4
13			CM1B3 (TCORB compare match)	✓	N/A	PIBR1.PIR5
14			OVI3 (TCNT overflow)	N/A	N/A	PIBR1.PIR6
15		TPU0	TGI0A (TGRA input capture/compare match)	✓	✓	PIBR1.PIR7
16			TGI0B (TGRB input capture/compare match)	✓	N/A	PIBR2.PIR0
17			TGI0C (TGRC input capture/compare match)	✓	N/A	PIBR2.PIR1
18			TGI0D (TGRD input capture/compare match)	✓	N/A	PIBR2.PIR2
19			TCI0V (TCNT overflow)	N/A	N/A	PIBR2.PIR3
20*1		TPU1	TGI1A (TGRA input capture/compare match)	✓	✓	PIBR2.PIR4
21			TGI1B (TGRB input capture/compare match)	✓	N/A	PIBR2.PIR5
22			TCI1V (TCNT overflow)	N/A	N/A	PIBR2.PIR6
23			TCI1U (TCNT underflow)	N/A	N/A	PIBR2.PIR7
24		TPU2	TGI2A (TGRA input capture/compare match)	✓	✓	PIBR3.PIR0
25			TGI2B (TGRB input capture/compare match)	✓	N/A	PIBR3.PIR1
26			TCI2V (TCNT overflow)	N/A	N/A	PIBR3.PIR2
27			TCI2U (TCNT underflow)	N/A	N/A	PIBR3.PIR3
28		TPU3	TGI3A (TGRA input capture/compare match)	✓	✓	PIBR3.PIR4
29			TGI3B (TGRB input capture/compare match)	✓	N/A	PIBR3.PIR5
30			TGI3C (TGRC input capture/compare match)	✓	N/A	PIBR3.PIR6
31			TGI3D (TGRD input capture/compare match)	✓	N/A	PIBR3.PIR7
32			TCI3V (TCNT overflow)	N/A	N/A	PIBR4.PIR0
33		TPU4	TGI4A (TGRA input capture/compare match)	✓	✓	PIBR4.PIR1
34			TGI4B (TGRB input capture/compare match)	✓	N/A	PIBR4.PIR2
35			TCI4V (TCNT overflow)	N/A	N/A	PIBR4.PIR3
36			TCI4U (TCNT underflow)	N/A	N/A	PIBR4.PIR4
37		TPU5	TGI5A (TGRA input capture/compare match)	✓	✓	PIBR4.PIR5
38			TGI5B (TGRB input capture/compare match)	✓	N/A	PIBR4.PIR6
39			TCI5V (TCNT overflow)	N/A	N/A	PIBR4.PIR7
40			TCI5U (TCNT underflow)	N/A	N/A	PIBR5.PIR0

Table 15.3 Interrupt Sources for Software Configurable Interrupt B (2/3)

Interrupt Source No.	Category	Interrupt Request Source	Name	Start the DTC	Start the DMAC	Interrupt Status Flag
41	Edge	CMTW0	IC0I0 (input capture of the CMWICR0 register)	✓	✓	PIBR5.PIR1
42			IC1I0 (input capture of the CMWICR1 register)	✓	✓	PIBR5.PIR2
43			OC0I0 (output compare of the CMWOCR0 register)	✓	✓	PIBR5.PIR3
44			OC1I0 (output compare of the CMWOCR1 register)	✓	✓	PIBR5.PIR4
45		CMTW1	IC0I1 (input capture of the CMWICR0 register)	✓	✓	PIBR5.PIR5
46			IC1I1 (input capture of the CMWICR1 register)	✓	✓	PIBR5.PIR6
47			OC0I1 (output compare of the CMWOCR0 register)	✓	✓	PIBR5.PIR7
48			OC1I1 (output compare of the CMWOCR1 register)	✓	✓	PIBR6.PIR0
49		RTC	CUP (carry interrupt)	N/A	N/A	PIBR6.PIR1
50		CAN0	RXF0 (receive FIFO interrupt)	N/A	N/A	PIBR6.PIR2
51			TXF0 (transmit FIFO interrupt)	N/A	N/A	PIBR6.PIR3
52			RXM0 (mailboxes 0 to 31 message reception completed)	N/A	N/A	PIBR6.PIR4
53			TXM0 (mailboxes 0 to 31 message transmission completed)	N/A	N/A	PIBR6.PIR5
54		CAN1	RXF1 (receive FIFO interrupt)	N/A	N/A	PIBR6.PIR6
55			TXF1 (transmit FIFO interrupt)	N/A	N/A	PIBR6.PIR7
56			RXM1 (mailboxes 0 to 31 message reception completed)	N/A	N/A	PIBR7.PIR0
57			TXM1 (mailboxes 0 to 31 message transmission completed)	N/A	N/A	PIBR7.PIR1
58		CAN2	RXF2 (receive FIFO interrupt)	N/A	N/A	PIBR7.PIR2
59			TXF2 (transmit FIFO interrupt)	N/A	N/A	PIBR7.PIR3
60			RXM2 (mailboxes 0 to 31 message reception completed)	N/A	N/A	PIBR7.PIR4
61			TXM2 (mailboxes 0 to 31 message transmission completed)	N/A	N/A	PIBR7.PIR5
62		USB0	USBI0 (status interrupt from 15 sources)	N/A	N/A	PIBR7.PIR6
63		Reserved	—	N/A	N/A	PIBR7.PIR7
64		S12AD	S12ADI (A/D conversion end)	✓	✓	PIBR8.PIR0
65			S12GBADI (group B A/D conversion end interrupt)	✓	✓	PIBR8.PIR1
66			S12GCADI (group C A/D conversion end interrupt)	✓	✓	PIBR8.PIR2
67		Reserved	—	N/A	N/A	PIBR8.PIR3
68		S12AD1	S12ADI1 (A/D conversion end)	✓	✓	PIBR8.PIR4
69			S12GBADI1 (group B A/D conversion end interrupt)	✓	✓	PIBR8.PIR5
70			S12GCADI1 (group C A/D conversion end interrupt)	✓	✓	PIBR8.PIR6
71		Reserved	—	N/A	N/A	PIBR8.PIR7
72		Reserved	—	N/A	N/A	PIBR9.PIR0
73		Reserved	—	N/A	N/A	PIBR9.PIR1
74		Reserved	—	N/A	N/A	PIBR9.PIR2
75		Reserved	—	N/A	N/A	PIBR9.PIR3
76		Reserved	—	N/A	N/A	PIBR9.PIR4
77		Reserved	—	N/A	N/A	PIBR9.PIR5
78		Reserved	—	N/A	N/A	PIBR9.PIR6
79		ELC	ELSR18I (ELC interrupt)	✓	✓	PIBR9.PIR7
80			ELSR19I (ELC interrupt)	✓	✓	PIBRA.PIR0
81 to 84		Reserved	—	N/A	N/A	—

Table 15.3 Interrupt Sources for Software Configurable Interrupt B (3/3)

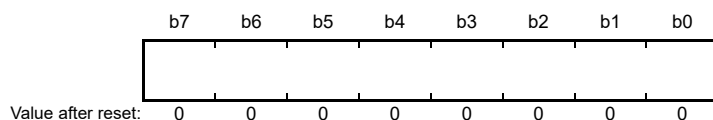
Interrupt Source No.	Category	Interrupt Request Source	Name	Start the DTC	Start the DMAC	Interrupt Status Flag
85	Edge	TSIP	PROC_BUSY (procedure completed)	N/A	N/A	PIBRA.PIR5
86			ROMOK (ROM falsification detected)	N/A	N/A	PIBRA.PIR6
87			LONG_PLG (operation completed)	N/A	N/A	PIBRA.PIR7
88			TEST_BUSY (test busy)	N/A	N/A	PIBRB.PIR0
89			WRRDY0 (write ready 0)	N/A	N/A	PIBRB.PIR1
90			WRRDY1 (write ready 1)	N/A	N/A	PIBRB.PIR2
91			WRRDY4 (write ready 4)	N/A	N/A	PIBRB.PIR3
92			RDRDY0 (read ready 0)	N/A	N/A	PIBRB.PIR4
93			RDRDY1 (read ready 1)	N/A	N/A	PIBRB.PIR5
94			INTEGRATE_WRRDY (integrate write ready)	N/A	N/A	PIBRB.PIR6
95			INTEGRATE_RDRDY (integrate read ready)	N/A	N/A	PIBRB.PIR7
96 to 254	—	Reserved	—	N/A	N/A	—
255	—	Reserved/ none *2	—/No interrupt selected	N/A	N/A	—

Note 1. No. 20 can be used as a trigger for EXDMAC0 or EXDMAC1.

Note 2. "Reserved" for the SLIBXRn register, and "none" for the SLIBRn register.

15.2.30 Software Configurable Interrupt A Source Select Register n (SLIARn) (n = 208 to 255)

Address(es): ICU.SLIAR208 0008 79D0h to ICU.SLIAR255 0008 79FFh



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	—	00h: No interrupt source selected. 01h: Interrupt source number 1 : FEh: Interrupt source number 254 FFh: No interrupt source selected.	R/(W) *1

Note 1. Writing to these bits is ignored while the SLIPRCR.WPRC bit is set to 1.

The SLIARn register is used to assign interrupt vector numbers 208 to 255 to interrupt sources for software configurable interrupt A.

Table 15.4, Interrupt Sources for Software Configurable Interrupt A lists interrupt sources for software configurable interrupt A. Set the SLIARn register to an interrupt source number that is not reserved. When 00h or FFh are set, no interrupt source is assigned to interrupt vector number n.

Do not assign the same interrupt source to multiple SLIARn registers.

Some interrupt sources can be used to start the DTC or DMAC. Refer to Table 15.4, Interrupt Sources for Software Configurable Interrupt A for details on which sources can be used to start the DTC or DMAC.

Although interrupt sources of interrupt vector numbers 208 and 209 can be used to start EXDMAC0 and EXDMAC1, respectively, only interrupt source number 8, MTU1.TGIA1 (TGRA input capture/compare match) can be set as a source to start the EXDMAC. Set only one of either the SLIAR208 register or SLIAR209 register to 8, and set the corresponding bit (SELEXD0 or SELEXD1) in the SELEXDR register to 1.

Refer to section 15.7.7, Setting Software Configurable Interrupts for the procedure to set software configurable interrupts.

Table 15.4 Interrupt Sources for Software Configurable Interrupt A (1/3)

Interrupt Source No.	Category	Interrupt Request Source	Name	Start the DTC	Start the DMAC	Interrupt Status Flag
0	—	None	No interrupt selected (initial value)	N/A	N/A	PIAR0.PIR0
1	Edge	MTU0	TGIA0 (TGRA input capture/compare match)	✓	✓	PIAR0.PIR1
2			TGIB0 (TGRB input capture/compare match)	✓	✓	PIAR0.PIR2
3			TGIC0 (TGRC input capture/compare match)	✓	✓	PIAR0.PIR3
4			TGID0 (TGRD input capture/compare match)	✓	✓	PIAR0.PIR4
5			TCIV0 (TCNT overflow)	N/A	N/A	PIAR0.PIR5
6			TGIE0 (TGRE compare match)	N/A	N/A	PIAR0.PIR6
7			TGIF0 (TGRF compare match)	N/A	N/A	PIAR0.PIR7
8*1		MTU1	TGIA1 (TGRA input capture/compare match)	✓	✓	PIAR1.PIR0
9			TGIB1 (TGRB input capture/compare match)	✓	✓	PIAR1.PIR1
10			TCIV1 (TCNT overflow)	N/A	N/A	PIAR1.PIR2
11			TCIU1 (TCNT underflow)	N/A	N/A	PIAR1.PIR3
12		MTU2	TGIA2 (TGRA input capture/compare match)	✓	✓	PIAR1.PIR4
13			TGIB2 (TGRB input capture/compare match)	✓	✓	PIAR1.PIR5
14			TCIV2 (TCNT overflow)	N/A	N/A	PIAR1.PIR6
15			TCIU2 (TCNT underflow)	N/A	N/A	PIAR1.PIR7
16		MTU3	TGIA3 (TGRA input capture/compare match)	✓	✓	PIAR2.PIR0
17			TGIB3 (TGRB input capture/compare match)	✓	✓	PIAR2.PIR1
18			TGIC3 (TGRC input capture/compare match)	✓	✓	PIAR2.PIR2
19			TGID3 (TGRD input capture/compare match)	✓	✓	PIAR2.PIR3
20			TCIV3 (TCNT overflow)	N/A	N/A	PIAR2.PIR4
21		MTU4	TGIA4 (TGRA input capture/compare match)	✓	✓	PIAR2.PIR5
22			TGIB4 (TGRB input capture/compare match)	✓	✓	PIAR2.PIR6
23			TGIC4 (TGRC input capture/compare match)	✓	✓	PIAR2.PIR7
24			TGID4 (TGRD input capture/compare match)	✓	✓	PIAR3.PIR0
25			TCIV4 (TCNT overflow/underflow (only for complementary PWM mode))	✓	✓	PIAR3.PIR1
26		Reserved	—	N/A	N/A	PIAR3.PIR2
27		MTU5	TGIU5 (TGRU input capture/compare match)	✓	✓	PIAR3.PIR3
28			TGIV5 (TGRV input capture/compare match)	✓	✓	PIAR3.PIR4
29			TGIW5 (TGRW input capture/compare match)	✓	✓	PIAR3.PIR5
30		MTU6	TGIA6 (TGRA input capture/compare match)	✓	✓	PIAR3.PIR6
31			TGIB6 (TGRB input capture/compare match)	✓	✓	PIAR3.PIR7
32			TGIC6 (TGRC input capture/compare match)	✓	✓	PIAR4.PIR0
33			TGID6 (TGRD input capture/compare match)	✓	✓	PIAR4.PIR1
34			TCIV6 (TCNT overflow)	N/A	N/A	PIAR4.PIR2
35		MTU7	TGIA7 (TGRA input capture/compare match)	✓	✓	PIAR4.PIR3
36			TGIB7 (TGRB input capture/compare match)	✓	✓	PIAR4.PIR4
37			TGIC7 (TGRC input capture/compare match)	✓	✓	PIAR4.PIR5
38			TGID7 (TGRD input capture/compare match)	✓	✓	PIAR4.PIR6
39			TCIV7 (TCNT overflow/underflow (only for complementary PWM mode))	✓	✓	PIAR4.PIR7
40		Reserved	—	N/A	N/A	PIAR5.PIR0

Table 15.4 Interrupt Sources for Software Configurable Interrupt A (2/3)

Interrupt Source No.	Category	Interrupt Request Source	Name	Start the DTC	Start the DMAC	Interrupt Status Flag
41	Edge	MTU8	TGIA8 (TGRA input capture/compare match)	✓	✓	PIAR5.PIR1
42			TGIB8 (TGRB input capture/compare match)	✓	✓	PIAR5.PIR2
43			TGIC8 (TGRC input capture/compare match)	✓	✓	PIAR5.PIR3
44			TGID8 (TGRD input capture/compare match)	✓	✓	PIAR5.PIR4
45			TCIV8 (TCNT overflow)	N/A	N/A	PIAR5.PIR5
46		Reserved	—	N/A	N/A	PIAR5.PIR6
47		GPTW0	GTCIA0 (GTCCRA input capture/compare match)	✓	✓	PIAR5.PIR7
48			GTCIB0 (GTCCRB input capture/compare match)	✓	✓	PIAR6.PIR0
49			GTCIC0 (GTCCRC compare match)	✓	✓	PIAR6.PIR1
50			GTCID0 (GTCCRD compare match)	✓	✓	PIAR6.PIR2
51			GDTE0 (dead time error)	N/A	N/A	PIAR6.PIR3
52			GTCIE0 (GTCCRE compare match)	✓	✓	PIAR6.PIR4
53			GTCIF0 (GTCCRF compare match)	✓	✓	PIAR6.PIR5
54			GTCIV0 (GTCNT overflow (GTPR compare match))	✓	✓	PIAR6.PIR6
55			GTCIU0 (GTCNT underflow)	✓	✓	PIAR6.PIR7
56		Reserved	—	N/A	N/A	PIAR7.PIR0
57		Reserved	—	N/A	N/A	PIAR7.PIR1
58		GPTW1	GTCIA1 (GTCCRA input capture/compare match)	✓	✓	PIAR7.PIR2
59			GTCIB1 (GTCCRB input capture/compare match)	✓	✓	PIAR7.PIR3
60			GTCIC1 (GTCCRC compare match)	✓	✓	PIAR7.PIR4
61			GTCID1 (GTCCRD compare match)	✓	✓	PIAR7.PIR5
62			GDTE1 (dead time error)	N/A	N/A	PIAR7.PIR6
63			GTCIE1 (GTCCRE compare match)	✓	✓	PIAR7.PIR7
64			GTCIF1 (GTCCRF compare match)	✓	✓	PIAR8.PIR0
65			GTCIV1 (GTCNT overflow (GTPR compare match))	✓	✓	PIAR8.PIR1
66			GTCIU1 (GTCNT underflow)	✓	✓	PIAR8.PIR2
67		GPTW2	GTCIA2 (GTCCRA input capture/compare match)	✓	✓	PIAR8.PIR3
68			GTCIB2 (GTCCRB input capture/compare match)	✓	✓	PIAR8.PIR4
69			GTCIC2 (GTCCRC compare match)	✓	✓	PIAR8.PIR5
70			GTCID2 (GTCCRD compare match)	✓	✓	PIAR8.PIR6
71			GDTE2 (dead time error)	N/A	N/A	PIAR8.PIR7
72			GTCIE2 (GTCCRE compare match)	✓	✓	PIAR9.PIR0
73			GTCIF2 (GTCCRF compare match)	✓	✓	PIAR9.PIR1
74			GTCIV2 (GTCNT overflow (GTPR compare match))	✓	✓	PIAR9.PIR2
75			GTCIU2 (GTCNT underflow)	✓	✓	PIAR9.PIR3
76		GPTW3	GTCIA3 (GTCCRA input capture/compare match)	✓	✓	PIAR9.PIR4
77			GTCIB3 (GTCCRB input capture/compare match)	✓	✓	PIAR9.PIR5
78			GTCIC3 (GTCCRC compare match)	✓	✓	PIAR9.PIR6
79			GTCID3 (GTCCRD compare match)	✓	✓	PIAR9.PIR7
80			GDTE3 (dead time error)	N/A	N/A	PIARA.PIR0
81			GTCIE3 (GTCCRE compare match)	✓	✓	PIARA.PIR1
82			GTCIF3 (GTCCRF compare match)	✓	✓	PIARA.PIR2
83			GTCIV3 (GTCNT overflow (GTPR compare match))	✓	✓	PIARA.PIR3
84			GTCIU3 (GTCNT underflow)	✓	✓	PIARA.PIR4

Table 15.4 Interrupt Sources for Software Configurable Interrupt A (3/3)

Interrupt Source No.	Category	Interrupt Request Source	Name	Start the DTC	Start the DMAC	Interrupt Status Flag
85	Edge	Reserved	—	N/A	N/A	PIARA.PIR5
86		EPTPC	IPLS (timer interrupt)	✓	✓	PIARA.PIR6
87 to 97		Reserved	—	N/A	N/A	—
98		PMGI0	PMGI0I (access completed interrupt)	N/A	N/A	PIARC.PIR2
99		PMGI1	PMGI1I (access completed interrupt)	N/A	N/A	PIARC.PIR3
100 to 254	—	Reserved	—	N/A	N/A	—
255	—	None	No interrupt selected	N/A	N/A	—

Note 1. No. 8 can be used as a trigger for EXDMAC0 or EXDMAC1.

15.2.31 EXDMAC Trigger Select Register (SELEXDR)

Address(es): ICU.SELEXDR 0008 7A01h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	SELEX D1	SELEX D0
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SELEXD0	EXDMAC0 Trigger Select	0: Interrupt source selected in the SLIBR144 register is used as the EXDMAC0 trigger. 1: Interrupt source selected in the SLIAR208 register is used as the EXDMAC0 trigger.	R/W*1
b1	SELEXD1	EXDMAC1 Trigger Select	0: Interrupt source selected in the SLIBR145 register is used as the EXDMAC1 trigger. 1: Interrupt source selected in the SLIAR209 register is used as the EXDMAC1 trigger.	R/W*1
b7 to b2	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. Writing to this bit is ignored while the SLIPRCR.WPRC bit is 1.

The SELEXDR register selects an interrupt source for software configurable interrupts as the EXDMAC trigger. Refer to section 15.4.5.3, EXDMAC Start Trigger by Software Configurable Interrupts for details on starting the EXDMAC by an interrupt request.

SELEXD0 Bit (EXDMAC0 Trigger Select)

The SELEXD0 bit selects an interrupt source set in the SLIBR144 register or SLIAR208 register as the EXDMAC0 trigger.

SELEXD1 Bit (EXDMAC1 Trigger Select)

The SELEXD1 bit selects an interrupt source set in the SLIBR145 register or SLIAR209 register as the EXDMAC1 trigger.

15.2.32 Software Configurable Interrupt Source Select Register Write Protect Register (SLIPRCR)

Address(es): ICU.SLIPRCR 0008 7A00h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	WPRC
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	WPRC	Software Configurable Interrupt Source Select Register Write Protect	0: Write enabled. 1: Write disabled.	R/(W) *1
b7 to b1	—	Reserved	The read value is 0. The write value should be 0.	R

Note 1. Once this bit is set to 1, it cannot be set to 0 by software.

The SLIPRCR register protects registers that control assignment of software configurable interrupts from being written to.

WPRC Bit (Software Configurable Interrupt Source Select Register Write Protect)

The WPRC bit disables writing to the SLIBXRn, SLIBRn, SLIARn, and SELEXDR registers.

Once this bit is set to 1, it cannot be set to 0 by software.

After assigning software configurable interrupts, confirm that the WPRC bit is 1 before the corresponding interrupt request is generated. Refer to section 15.7.7, Setting Software Configurable Interrupts for the procedure to set software configurable interrupts.

15.3 Vector Table

There are two types of exceptions detected by the ICU: maskable interrupts (hereinafter referred to as “interrupts”) and non-maskable interrupts.

When the CPU accepts an interrupt or non-maskable interrupt, it acquires a 4-byte vector address from the vector table.

15.3.1 Interrupt Vector Table

The vector table that is used for maskable interrupts is called the interrupt vector table.

The interrupt vector table is allocated to a 1024-byte area (4 bytes × 256 sources) beginning with the address set in the INTB register in the CPU. Set the INTB register before enabling interrupts. Set a multiple of 4 in the INTB register.

An unconditional trap is generated when the INT or BRK instruction is executed. Interrupt vectors for unconditional traps use the same area as the interrupt vector table. The BRK instruction is assigned to interrupt vector number 0. The INT instruction is assigned to the interrupt vector number corresponding to the value set as the operand (0 to 255).

Table 15.5 lists details of the interrupt vectors. Details of the headings in Table 15.5 are listed below.

Heading	Description
Interrupt request generated by	Name of the source that generates the interrupt request (module symbol)
Name	Name of the interrupt source (symbol)
Vector no.	Interrupt vector number
Vector address offset	Offset from the address set in the INTB register
Interrupt detection method	“Edge” indicates that the interrupt is detected by edge detection. “Level” indicates that the interrupt is detected by level detection.
CPU interrupt	Interrupt source indicated by “√” can be used as an interrupt source to the CPU.
Start the DTC	Interrupt source indicated by “√” can be used as the DTC trigger.
Start the DMAC	Interrupt source indicated by “√” can be used as the DMAC trigger.
Exit from SSBY	Interrupt source indicated by “√” can be used as a source to exit software standby mode.
Exit from ACS	Interrupt source indicated by “√” can be used as a source to exit all-module clock stop mode.
IER	Name of the bit in the IER register corresponding to the interrupt vector number
IPR	Name of the IPR register corresponding to the interrupt source
DTCER	Name of the DTCER register corresponding to the DTC trigger

Table 15.5 Interrupt Vector Table (1/6)

Interrupt Request is Generated By	Name	Vector No.	Vector Address Offset	Interrupt Detection Method	CPU Interrupt	Start the DTC	Start the DMAC	Exit from SSBY	Exit from ACS	IER	IPR	DTCER
—	For an unconditional trap	0	0000h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	1	0004h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	2	0008h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	3	000Ch	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	4	0010h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	5	0014h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	6	0018h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	7	001Ch	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	8	0020h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	9	0024h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	10	0028h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	11	002Ch	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	12	0030h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	13	0034h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	14	0038h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	For an unconditional trap	15	003Ch	—	N/A	N/A	N/A	N/A	N/A	—	—	—
BSC	BUSERR	16	0040h	Level	✓	N/A	N/A	N/A	N/A	IER02.IEN0	IPR000	—
ICU*1	GROUPIE0	17	0044h	Level	✓	N/A	N/A	N/A	N/A	IER02.IEN1	IPR000	—
RAM	RAMERR*2	18	0048h	Level	✓	N/A	N/A	N/A	N/A	IER02.IEN2	IPR000	—
—	Reserved	19	004Ch	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	Reserved	20	0050h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
FCU	FIFERR	21	0054h	Level	✓	N/A	N/A	N/A	N/A	IER02.IEN5	IPR001	—
—	Reserved	22	0058h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
FCU	FRDYI	23	005Ch	Edge	✓	N/A	N/A	N/A	N/A	IER02.IEN7	IPR002	—
—	Reserved	24	0060h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	Reserved	25	0064h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
ICU	SWINT2	26	0068h	Edge	✓	✓	N/A	N/A	N/A	IER03.IEN2	IPR003	DTCER026
	SWINT	27	006Ch	Edge	✓	✓	N/A	N/A	N/A	IER03.IEN3		DTCER027
CMT0	CMI0 (for OS)	28	0070h	Edge	✓	✓	✓	N/A	N/A	IER03.IEN4	IPR004	DTCER028
CMT1	CMI1	29	0074h	Edge	✓	✓	✓	N/A	N/A	IER03.IEN5	IPR005	DTCER029
CMTW0	CMWI0	30	0078h	Edge	✓	✓	✓	N/A	N/A	IER03.IEN6	IPR006	DTCER030
CMTW1	CMWI1	31	007Ch	Edge	✓	✓	✓	N/A	N/A	IER03.IEN7	IPR007	DTCER031
—	Reserved	32	0080h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	Reserved	33	0084h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
USB0	D0FIFO0	34	0088h	Edge	✓	✓	✓	N/A	N/A	IER04.IEN2	IPR034	DTCER034
	D1FIFO0	35	008Ch	Edge	✓	✓	✓	N/A	N/A	IER04.IEN3	IPR035	DTCER035
—	Reserved	36	0090h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	Reserved	37	0094h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
RSPi0	SPRI0	38	0098h	Edge	✓	✓	✓	N/A	N/A	IER04.IEN6	IPR038	DTCER038
	SPTI0	39	009Ch	Edge	✓	✓	✓	N/A	N/A	IER04.IEN7	IPR039	DTCER039
RSPi1	SPRI1	40	00A0h	Edge	✓	✓	✓	N/A	N/A	IER05.IEN0	IPR040	DTCER040
	SPTI1	41	00A4h	Edge	✓	✓	✓	N/A	N/A	IER05.IEN1	IPR041	DTCER041
QSPI	SPRI	42	00A8h	Edge	✓	✓	✓	N/A	N/A	IER05.IEN2	IPR042	DTCER042
	SPTI	43	00ACh	Edge	✓	✓	✓	N/A	N/A	IER05.IEN3	IPR043	DTCER043

Table 15.5 Interrupt Vector Table (2/6)

Interrupt Request is Generated By	Name	Vector No.	Vector Address Offset	Interrupt Detection Method	CPU Interrupt	Start the DTC	Start the DMAC	Exit from SSBY	Exit from ACS	IER	IPR	DTCER
SDHI	SBFAI	44	00B0h	Edge	✓	✓	✓	N/A	N/A	IER05.IEN4	IPR044	DTCER044
MMCIF	MBFAI	45	00B4h	Edge	✓	✓	✓	N/A	N/A	IER05.IEN5	IPR045	DTCER045
SSIE0	SSITX10	46	00B8h	Edge	✓	✓	✓	N/A	N/A	IER05.IEN6	IPR046	DTCER046
	SSIRX10	47	00BCh	Edge	✓	✓	✓	N/A	N/A	IER05.IEN7	IPR047	DTCER047
SSIE1	SSIRT11	48	00C0h	Edge	✓	✓	✓	N/A	N/A	IER06.IEN0	IPR048	DTCER048
—	Reserved	49	00C4h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
RIIC1	RX11	50	00C8h	Edge	✓	✓	✓	N/A	N/A	IER06.IEN2	IPR050	DTCER050
	TX11	51	00CCh	Edge	✓	✓	✓	N/A	N/A	IER06.IEN3	IPR051	DTCER051
RIIC0	RX10	52	00D0h	Edge	✓	✓	✓	N/A	N/A	IER06.IEN4	IPR052	DTCER052
	TX10	53	00D4h	Edge	✓	✓	✓	N/A	N/A	IER06.IEN5	IPR053	DTCER053
RIIC2	RX12	54	00D8h	Edge	✓	✓	✓	N/A	N/A	IER06.IEN6	IPR054	DTCER054
	TX12	55	00DCh	Edge	✓	✓	✓	N/A	N/A	IER06.IEN7	IPR055	DTCER055
—	Reserved	56	00E0h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	Reserved	57	00E4h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
SCI0	RX10	58	00E8h	Edge	✓	✓	✓	N/A	N/A	IER07.IEN2	IPR058	DTCER058
	TX10	59	00ECh	Edge	✓	✓	✓	N/A	N/A	IER07.IEN3	IPR059	DTCER059
SCI1	RX11	60	00F0h	Edge	✓	✓	✓	N/A	N/A	IER07.IEN4	IPR060	DTCER060
	TX11	61	00F4h	Edge	✓	✓	✓	N/A	N/A	IER07.IEN5	IPR061	DTCER061
SCI2	RX12	62	00F8h	Edge	✓	✓	✓	N/A	N/A	IER07.IEN6	IPR062	DTCER062
	TX12	63	00FCh	Edge	✓	✓	✓	N/A	N/A	IER07.IEN7	IPR063	DTCER063
ICU	IRQ0	64	0100h	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN0	IPR064	DTCER064
	IRQ1	65	0104h	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN1	IPR065	DTCER065
	IRQ2	66	0108h	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN2	IPR066	DTCER066
	IRQ3	67	010Ch	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN3	IPR067	DTCER067
	IRQ4	68	0110h	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN4	IPR068	DTCER068
	IRQ5	69	0114h	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN5	IPR069	DTCER069
	IRQ6	70	0118h	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN6	IPR070	DTCER070
	IRQ7	71	011Ch	Edge/Level	✓	✓	✓	✓	✓	IER08.IEN7	IPR071	DTCER071
	IRQ8	72	0120h	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN0	IPR072	DTCER072
	IRQ9	73	0124h	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN1	IPR073	DTCER073
	IRQ10	74	0128h	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN2	IPR074	DTCER074
	IRQ11	75	012Ch	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN3	IPR075	DTCER075
	IRQ12	76	0130h	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN4	IPR076	DTCER076
	IRQ13	77	0134h	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN5	IPR077	DTCER077
	IRQ14	78	0138h	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN6	IPR078	DTCER078
	IRQ15	79	013Ch	Edge/Level	✓	✓	✓	✓	✓	IER09.IEN7	IPR079	DTCER079

Table 15.5 Interrupt Vector Table (3/6)

Interrupt Request is Generated By	Name	Vector No.	Vector Address Offset	Interrupt Detection Method	CPU Interrupt	Start the DTC	Start the DMAC	Exit from SSBY	Exit from ACS	IER	IPR	DTCER
SCI3	RXI3	80	0140h	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN0	IPR080	DTCER080
	TXI3	81	0144h	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN1	IPR081	DTCER081
SCI4	RXI4	82	0148h	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN2	IPR082	DTCER082
	TXI4	83	014Ch	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN3	IPR083	DTCER083
SCI5	RXI5	84	0150h	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN4	IPR084	DTCER084
	TXI5	85	0154h	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN5	IPR085	DTCER085
SCI6	RXI6	86	0158h	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN6	IPR086	DTCER086
	TXI6	87	015Ch	Edge	✓	✓	✓	N/A	N/A	IER0A.IEN7	IPR087	DTCER087
LVD1	LVD1	88	0160h	Edge	✓	N/A	N/A	✓	✓	IER0B.IEN0	IPR088	—
LVD2	LVD2	89	0164h	Edge	✓	N/A	N/A	✓	✓	IER0B.IEN1	IPR089	—
USB0	USBR0	90	0168h	Level	✓	N/A	N/A	✓	✓	IER0B.IEN2	IPR090	—
—	Reserved	91	016Ch	—	N/A	N/A	N/A	N/A	N/A	—	—	—
RTC	ALM	92	0170h	Edge	✓	N/A	N/A	✓	✓	IER0B.IEN4	IPR092	—
	PRD	93	0174h	Edge	✓	N/A	N/A	✓	✓	IER0B.IEN5	IPR093	—
—	Reserved	94	0178h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
IWDT	IWUNI*2	95	017Ch	Edge	✓	N/A	N/A	✓	✓	IER0B.IEN7	IPR095	—
WDT	WUNI*2	96	0180h	Edge	✓	N/A	N/A	N/A	N/A	IER0C.IEN0	IPR096	—
PDC	PCDFI	97	0184h	Edge	✓	✓	✓	N/A	N/A	IER0C.IEN1	IPR097	DTCER097
SCI7	RXI7	98	0188h	Edge	✓	✓	✓	N/A	N/A	IER0C.IEN2	IPR098	DTCER098
	TXI7	99	018Ch	Edge	✓	✓	✓	N/A	N/A	IER0C.IEN3	IPR099	DTCER099
SCI8	RXI8	100	0190h	Edge	✓	✓	✓	N/A	N/A	IER0C.IEN4	IPR100	DTCER100
	TXI8	101	0194h	Edge	✓	✓	✓	N/A	N/A	IER0C.IEN5	IPR101	DTCER101
SCI9	RXI9	102	0198h	Edge	✓	✓	✓	N/A	N/A	IER0C.IEN6	IPR102	DTCER102
	TXI9	103	019Ch	Edge	✓	✓	✓	N/A	N/A	IER0C.IEN7	IPR103	DTCER103
SCI10	RXI10	104	01A0h	Edge	✓	✓	✓	N/A	N/A	IER0D.IEN0	IPR104	DTCER104
	TXI10	105	01A4h	Edge	✓	✓	✓	N/A	N/A	IER0D.IEN1	IPR105	DTCER105
ICU*1	GROUPBE0	106	01A8h	Level	✓	N/A	N/A	N/A	N/A	IER0D.IEN2	IPR106	—
	GROUPBL2	107	01ACh	Level	✓	N/A	N/A	N/A	N/A	IER0D.IEN3	IPR107	—
RSP12	SPRI2	108	01B0h	Edge	✓	✓	✓	N/A	N/A	IER0D.IEN4	IPR108	DTCER108
	SPTI2	109	01B4h	Edge	✓	✓	✓	N/A	N/A	IER0D.IEN5	IPR109	DTCER109
ICU*1	GROUPBL0	110	01B8h	Level	✓	N/A	N/A	N/A	N/A	IER0D.IEN6	IPR110	—
	GROUPBL1	111	01BCh	Level	✓	N/A	N/A	N/A	N/A	IER0D.IEN7	IPR111	—
	GROUPAL0	112	01C0h	Level	✓	N/A	N/A	N/A	N/A	IER0E.IEN0	IPR112	—
	GROUPAL1	113	01C4h	Level	✓	N/A	N/A	N/A	N/A	IER0E.IEN1	IPR113	—
SCI11	RXI11	114	01C8h	Edge	✓	✓	✓	N/A	N/A	IER0E.IEN2	IPR114	DTCER114
	TXI11	115	01CCh	Edge	✓	✓	✓	N/A	N/A	IER0E.IEN3	IPR115	DTCER115
SCI12	RXI12	116	01D0h	Edge	✓	✓	✓	N/A	N/A	IER0E.IEN4	IPR116	DTCER116
	TXI12	117	01D4h	Edge	✓	✓	✓	N/A	N/A	IER0E.IEN5	IPR117	DTCER117
—	Reserved	118	01D8h	—	N/A	N/A	N/A	N/A	N/A	—	—	—
—	Reserved	119	01DCh	—	N/A	N/A	N/A	N/A	N/A	—	—	—
DMAC	DMAC0I	120	01E0h	Edge	✓	✓	N/A	N/A	N/A	IER0F.IEN0	IPR120	DTCER120
	DMAC1I	121	01E4h	Edge	✓	✓	N/A	N/A	N/A	IER0F.IEN1	IPR121	DTCER121
	DMAC2I	122	01E8h	Edge	✓	✓	N/A	N/A	N/A	IER0F.IEN2	IPR122	DTCER122
	DMAC3I	123	01ECh	Edge	✓	✓	N/A	N/A	N/A	IER0F.IEN3	IPR123	DTCER123
	DMAC74I	124	01F0h	Level	✓	N/A	N/A	N/A	N/A	IER0F.IEN4	IPR124	—

Table 15.5 Interrupt Vector Table (4/6)

Interrupt Request is Generated By	Name	Vector No.	Vector Address Offset	Interrupt Detection Method	CPU Interrupt	Start the DTC	Start the DMAC	Exit from SSBY	Exit from ACS	IER	IPR	DTCER
OST	OSTDI*2	125	01F4h	Edge	✓	N/A	N/A	N/A	N/A	IER0F.IEN5	IPR125	—
EXDMAC	EXDMAC0I	126	01F8h	Edge	✓	✓	N/A	N/A	N/A	IER0F.IEN6	IPR126	DTCER126
	EXDMAC1I	127	01FCh	Edge	✓	✓	N/A	N/A	N/A	IER0F.IEN7	IPR127	DTCER127
PERIB (software configurable interrupt B *3)	INTB128	128	0200h	Edge	✓	✓	✓	N/A	N/A	IER10.IEN0	IPR128	DTCER128
	INTB129	129	0204h	Edge	✓	✓	✓	N/A	N/A	IER10.IEN1	IPR129	DTCER129
	INTB130	130	0208h	Edge	✓	✓	✓	N/A	N/A	IER10.IEN2	IPR130	DTCER130
	INTB131	131	020Ch	Edge	✓	✓	✓	N/A	N/A	IER10.IEN3	IPR131	DTCER131
	INTB132	132	0210h	Edge	✓	✓	✓	N/A	N/A	IER10.IEN4	IPR132	DTCER132
	INTB133	133	0214h	Edge	✓	✓	✓	N/A	N/A	IER10.IEN5	IPR133	DTCER133
	INTB134	134	0218h	Edge	✓	✓	✓	N/A	N/A	IER10.IEN6	IPR134	DTCER134
	INTB135	135	021Ch	Edge	✓	✓	✓	N/A	N/A	IER10.IEN7	IPR135	DTCER135
	INTB136	136	0220h	Edge	✓	✓	✓	N/A	N/A	IER11.IEN0	IPR136	DTCER136
	INTB137	137	0224h	Edge	✓	✓	✓	N/A	N/A	IER11.IEN1	IPR137	DTCER137
	INTB138	138	0228h	Edge	✓	✓	✓	N/A	N/A	IER11.IEN2	IPR138	DTCER138
	INTB139	139	022Ch	Edge	✓	✓	✓	N/A	N/A	IER11.IEN3	IPR139	DTCER139
	INTB140	140	0230h	Edge	✓	✓	✓	N/A	N/A	IER11.IEN4	IPR140	DTCER140
	INTB141	141	0234h	Edge	✓	✓	✓	N/A	N/A	IER11.IEN5	IPR141	DTCER141
	INTB142	142	0238h	Edge	✓	✓	✓	N/A	N/A	IER11.IEN6	IPR142	DTCER142
	INTB143	143	023Ch	Edge	✓	✓	✓	N/A	N/A	IER11.IEN7	IPR143	DTCER143
	INTB144	144*4	0240h	Edge	✓	✓	✓	N/A	N/A	IER12.IEN0	IPR144	DTCER144
	INTB145	145*5	0244h	Edge	✓	✓	✓	N/A	N/A	IER12.IEN1	IPR145	DTCER145
	INTB146	146	0248h	Edge	✓	✓	✓	N/A	✓	IER12.IEN2	IPR146	DTCER146
	INTB147	147	024Ch	Edge	✓	✓	✓	N/A	✓	IER12.IEN3	IPR147	DTCER147
	INTB148	148	0250h	Edge	✓	✓	✓	N/A	✓	IER12.IEN4	IPR148	DTCER148
	INTB149	149	0254h	Edge	✓	✓	✓	N/A	✓	IER12.IEN5	IPR149	DTCER149
	INTB150	150	0258h	Edge	✓	✓	✓	N/A	✓	IER12.IEN6	IPR150	DTCER150
	INTB151	151	025Ch	Edge	✓	✓	✓	N/A	✓	IER12.IEN7	IPR151	DTCER151
	INTB152	152	0260h	Edge	✓	✓	✓	N/A	✓	IER13.IEN0	IPR152	DTCER152
	INTB153	153	0264h	Edge	✓	✓	✓	N/A	✓	IER13.IEN1	IPR153	DTCER153
	INTB154	154	0268h	Edge	✓	✓	✓	N/A	✓	IER13.IEN2	IPR154	DTCER154
	INTB155	155	026Ch	Edge	✓	✓	✓	N/A	✓	IER13.IEN3	IPR155	DTCER155
	INTB156	156	0270h	Edge	✓	✓	✓	N/A	✓	IER13.IEN4	IPR156	DTCER156
	INTB157	157	0274h	Edge	✓	✓	✓	N/A	✓	IER13.IEN5	IPR157	DTCER157
	INTB158	158	0278h	Edge	✓	✓	✓	N/A	N/A	IER13.IEN6	IPR158	DTCER158
	INTB159	159	027Ch	Edge	✓	✓	✓	N/A	N/A	IER13.IEN7	IPR159	DTCER159
	INTB160	160	0280h	Edge	✓	✓	✓	N/A	N/A	IER14.IEN0	IPR160	DTCER160
	INTB161	161	0284h	Edge	✓	✓	✓	N/A	N/A	IER14.IEN1	IPR161	DTCER161
	INTB162	162	0288h	Edge	✓	✓	✓	N/A	N/A	IER14.IEN2	IPR162	DTCER162
	INTB163	163	028Ch	Edge	✓	✓	✓	N/A	N/A	IER14.IEN3	IPR163	DTCER163
	INTB164	164	0290h	Edge	✓	✓	✓	N/A	N/A	IER14.IEN4	IPR164	DTCER164
	INTB165	165	0294h	Edge	✓	✓	✓	N/A	N/A	IER14.IEN5	IPR165	DTCER165
	INTB166	166	0298h	Edge	✓	✓	✓	N/A	N/A	IER14.IEN6	IPR166	DTCER166
	INTB167	167	029Ch	Edge	✓	✓	✓	N/A	N/A	IER14.IEN7	IPR167	DTCER167
	INTB168	168	02A0h	Edge	✓	✓	✓	N/A	N/A	IER15.IEN0	IPR168	DTCER168
	INTB169	169	02A4h	Edge	✓	✓	✓	N/A	N/A	IER15.IEN1	IPR169	DTCER169

Table 15.5 Interrupt Vector Table (5/6)

Interrupt Request is Generated By	Name	Vector No.	Vector Address Offset	Interrupt Detection Method	CPU Interrupt	Start the DTC	Start the DMAC	Exit from SSBY	Exit from ACS	IER	IPR	DTCER
PERIB (software configurable interrupt B *3)	INTB170	170	02A8h	Edge	✓	✓	✓	N/A	N/A	IER15.IEN2	IPR170	DTCER170
	INTB171	171	02ACh	Edge	✓	✓	✓	N/A	N/A	IER15.IEN3	IPR171	DTCER171
	INTB172	172	02B0h	Edge	✓	✓	✓	N/A	N/A	IER15.IEN4	IPR172	DTCER172
	INTB173	173	02B4h	Edge	✓	✓	✓	N/A	N/A	IER15.IEN5	IPR173	DTCER173
	INTB174	174	02B8h	Edge	✓	✓	✓	N/A	N/A	IER15.IEN6	IPR174	DTCER174
	INTB175	175	02BCh	Edge	✓	✓	✓	N/A	N/A	IER15.IEN7	IPR175	DTCER175
	INTB176	176	02C0h	Edge	✓	✓	✓	N/A	N/A	IER16.IEN0	IPR176	DTCER176
	INTB177	177	02C4h	Edge	✓	✓	✓	N/A	N/A	IER16.IEN1	IPR177	DTCER177
	INTB178	178	02C8h	Edge	✓	✓	✓	N/A	N/A	IER16.IEN2	IPR178	DTCER178
	INTB179	179	02CCh	Edge	✓	✓	✓	N/A	N/A	IER16.IEN3	IPR179	DTCER179
	INTB180	180	02D0h	Edge	✓	✓	✓	N/A	N/A	IER16.IEN4	IPR180	DTCER180
	INTB181	181	02D4h	Edge	✓	✓	✓	N/A	N/A	IER16.IEN5	IPR181	DTCER181
	INTB182	182	02D8h	Edge	✓	✓	✓	N/A	N/A	IER16.IEN6	IPR182	DTCER182
	INTB183	183	02DCh	Edge	✓	✓	✓	N/A	N/A	IER16.IEN7	IPR183	DTCER183
	INTB184	184	02E0h	Edge	✓	✓	✓	N/A	N/A	IER17.IEN0	IPR184	DTCER184
	INTB185	185	02E4h	Edge	✓	✓	✓	N/A	N/A	IER17.IEN1	IPR185	DTCER185
	INTB186	186	02E8h	Edge	✓	✓	✓	N/A	N/A	IER17.IEN2	IPR186	DTCER186
	INTB187	187	02ECh	Edge	✓	✓	✓	N/A	N/A	IER17.IEN3	IPR187	DTCER187
	INTB188	188	02F0h	Edge	✓	✓	✓	N/A	N/A	IER17.IEN4	IPR188	DTCER188
	INTB189	189	02F4h	Edge	✓	✓	✓	N/A	N/A	IER17.IEN5	IPR189	DTCER189
	INTB190	190	02F8h	Edge	✓	✓	✓	N/A	N/A	IER17.IEN6	IPR190	DTCER190
	INTB191	191	02FCh	Edge	✓	✓	✓	N/A	N/A	IER17.IEN7	IPR191	DTCER191
	INTB192	192	0300h	Edge	✓	✓	✓	N/A	N/A	IER18.IEN0	IPR192	DTCER192
	INTB193	193	0304h	Edge	✓	✓	✓	N/A	N/A	IER18.IEN1	IPR193	DTCER193
	INTB194	194	0308h	Edge	✓	✓	✓	N/A	N/A	IER18.IEN2	IPR194	DTCER194
	INTB195	195	030Ch	Edge	✓	✓	✓	N/A	N/A	IER18.IEN3	IPR195	DTCER195
	INTB196	196	0310h	Edge	✓	✓	✓	N/A	N/A	IER18.IEN4	IPR196	DTCER196
	INTB197	197	0314h	Edge	✓	✓	✓	N/A	N/A	IER18.IEN5	IPR197	DTCER197
	INTB198	198	0318h	Edge	✓	✓	✓	N/A	N/A	IER18.IEN6	IPR198	DTCER198
	INTB199	199	031Ch	Edge	✓	✓	✓	N/A	N/A	IER18.IEN7	IPR199	DTCER199
	INTB200	200	0320h	Edge	✓	✓	✓	N/A	N/A	IER19.IEN0	IPR200	DTCER200
	INTB201	201	0324h	Edge	✓	✓	✓	N/A	N/A	IER19.IEN1	IPR201	DTCER201
	INTB202	202	0328h	Edge	✓	✓	✓	N/A	N/A	IER19.IEN2	IPR202	DTCER202
	INTB203	203	032Ch	Edge	✓	✓	✓	N/A	N/A	IER19.IEN3	IPR203	DTCER203
	INTB204	204	0330h	Edge	✓	✓	✓	N/A	N/A	IER19.IEN4	IPR204	DTCER204
	INTB205	205	0334h	Edge	✓	✓	✓	N/A	N/A	IER19.IEN5	IPR205	DTCER205
	INTB206	206	0338h	Edge	✓	✓	✓	N/A	N/A	IER19.IEN6	IPR206	DTCER206
	INTB207	207	033Ch	Edge	✓	✓	✓	N/A	N/A	IER19.IEN7	IPR207	DTCER207
PERIA (software configurable interrupt A *6)	INTA208	208*4	0340h	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN0	IPR208	DTCER208
	INTA209	209*5	0344h	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN1	IPR209	DTCER209
	INTA210	210	0348h	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN2	IPR210	DTCER210
	INTA211	211	034Ch	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN3	IPR211	DTCER211
	INTA212	212	0350h	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN4	IPR212	DTCER212
	INTA213	213	0354h	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN5	IPR213	DTCER213
	INTA214	214	0358h	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN6	IPR214	DTCER214

Table 15.5 Interrupt Vector Table (6/6)

Interrupt Request is Generated By	Name	Vector No.	Vector Address Offset	Interrupt Detection Method	CPU Interrupt	Start the DTC	Start the DMAC	Exit from SSBY	Exit from ACS	IER	IPR	DTCER
PERIA (software configurable interrupt A *6)	INTA215	215	035Ch	Edge	✓	✓	✓	N/A	N/A	IER1A.IEN7	IPR215	DTCER215
	INTA216	216	0360h	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN0	IPR216	DTCER216
	INTA217	217	0364h	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN1	IPR217	DTCER217
	INTA218	218	0368h	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN2	IPR218	DTCER218
	INTA219	219	036Ch	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN3	IPR219	DTCER219
	INTA220	220	0370h	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN4	IPR220	DTCER220
	INTA221	221	0374h	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN5	IPR221	DTCER221
	INTA222	222	0378h	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN6	IPR222	DTCER222
	INTA223	223	037Ch	Edge	✓	✓	✓	N/A	N/A	IER1B.IEN7	IPR223	DTCER223
	INTA224	224	0380h	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN0	IPR224	DTCER224
	INTA225	225	0384h	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN1	IPR225	DTCER225
	INTA226	226	0388h	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN2	IPR226	DTCER226
	INTA227	227	038Ch	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN3	IPR227	DTCER227
	INTA228	228	0390h	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN4	IPR228	DTCER228
	INTA229	229	0394h	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN5	IPR229	DTCER229
	INTA230	230	0398h	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN6	IPR230	DTCER230
	INTA231	231	039Ch	Edge	✓	✓	✓	N/A	N/A	IER1C.IEN7	IPR231	DTCER231
	INTA232	232	03A0h	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN0	IPR232	DTCER232
	INTA233	233	03A4h	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN1	IPR233	DTCER233
	INTA234	234	03A8h	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN2	IPR234	DTCER234
	INTA235	235	03ACh	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN3	IPR235	DTCER235
	INTA236	236	03B0h	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN4	IPR236	DTCER236
	INTA237	237	03B4h	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN5	IPR237	DTCER237
	INTA238	238	03B8h	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN6	IPR238	DTCER238
	INTA239	239	03BCh	Edge	✓	✓	✓	N/A	N/A	IER1D.IEN7	IPR239	DTCER239
	INTA240	240	03C0h	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN0	IPR240	DTCER240
	INTA241	241	03C4h	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN1	IPR241	DTCER241
	INTA242	242	03C8h	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN2	IPR242	DTCER242
	INTA243	243	03CCh	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN3	IPR243	DTCER243
	INTA244	244	03D0h	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN4	IPR244	DTCER244
	INTA245	245	03D4h	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN5	IPR245	DTCER245
	INTA246	246	03D8h	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN6	IPR246	DTCER246
	INTA247	247	03DCh	Edge	✓	✓	✓	N/A	N/A	IER1E.IEN7	IPR247	DTCER247
	INTA248	248	03E0h	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN0	IPR248	DTCER248
	INTA249	249	03E4h	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN1	IPR249	DTCER249
	INTA250	250	03E8h	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN2	IPR250	DTCER250
	INTA251	251	03ECh	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN3	IPR251	DTCER251
	INTA252	252	03F0h	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN4	IPR252	DTCER252
	INTA253	253	03F4h	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN5	IPR253	DTCER253
	INTA254	254	03F8h	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN6	IPR254	DTCER254
	INTA255	255	03FCh	Edge	✓	✓	✓	N/A	N/A	IER1F.IEN7	IPR255	DTCER255

Note: This table lists the interrupt vectors for the maximum specification. The interrupt vectors for individual products correspond to the functions listed in Table 1.2. For details, refer to Table 1.2, Comparison of Functions for Different Packages.

Note 1. For the group interrupt sources, refer to Table 15.7, Group Interrupt Requests.

Note 2. This is the case where the corresponding non-maskable interrupt enable bit is set to 0 (disabled).

Note 3. For the software configurable interrupt B sources, refer to Table 15.3, Interrupt Sources for Software Configurable Interrupt B.

Note that some interrupt sources cannot start the DTC, DMAC, or EXDMAC.

- Note 4. This can be used to start EXDMAC0. Set the SELEXDR.SELEXD0 bit to 0 when using the interrupt source assigned to interrupt vector number 144 as a trigger. Set the SELEXDR.SELEXD0 bit to 1 when using the interrupt source assigned to interrupt vector number 208 as a trigger. Refer to section 15.2.29, Software Configurable Interrupt B Source Select Register n (SLIBRn) (n = 144 to 207), section 15.2.30, Software Configurable Interrupt A Source Select Register n (SLIARn) (n = 208 to 255), and section 15.2.31, EXDMAC Trigger Select Register (SELEXDR).
- Note 5. This can be used to start EXDMAC1. Set the SELEXDR.SELEXD1 bit to 0 when using the interrupt source assigned to interrupt vector number 145 as a trigger. Set the SELEXDR.SELEXD1 bit to 1 when using the interrupt source assigned to interrupt vector number 209 as a trigger.
- Note 6. For the software configurable interrupt A sources, refer to Table 15.4, Interrupt Sources for Software Configurable Interrupt A. Note that some interrupt sources cannot start the DTC, DMAC, or EXDMAC.

15.3.2 Fast Interrupt Vector Area

The interrupt set as the fast interrupt uses the FINTV register in the CPU. Set the FINTV register before enabling the fast interrupt.

15.3.3 Non-maskable Interrupt Vector Area

Non-maskable interrupts use the vector area in the exception vector table.

The exception vector table is allocated to the 128-byte area (4 bytes × 32 sources) beginning with the address set in the EXTB register in the CPU. Set the EXTB register before enabling non-maskable interrupts. Set a multiple of 4 in the EXTB register.

15.4 Types of Interrupts

Interrupts are divided into maskable interrupts and non-maskable interrupts. Maskable interrupts can be masked by the PSW.I bit or IPL[3:0] bits of the processor status word in the CPU. Non-maskable interrupts can be accepted by the CPU regardless of those bits. While interrupt sources assigned to vector numbers 0 to 127 are fixed, an interrupt source assigned to each vector number from 128 to 255 (software configurable interrupt) can be selected from multiple sources. Note that maskable interrupts are referred to as interrupts in this chapter.

Figure 15.2 shows types of interrupts.

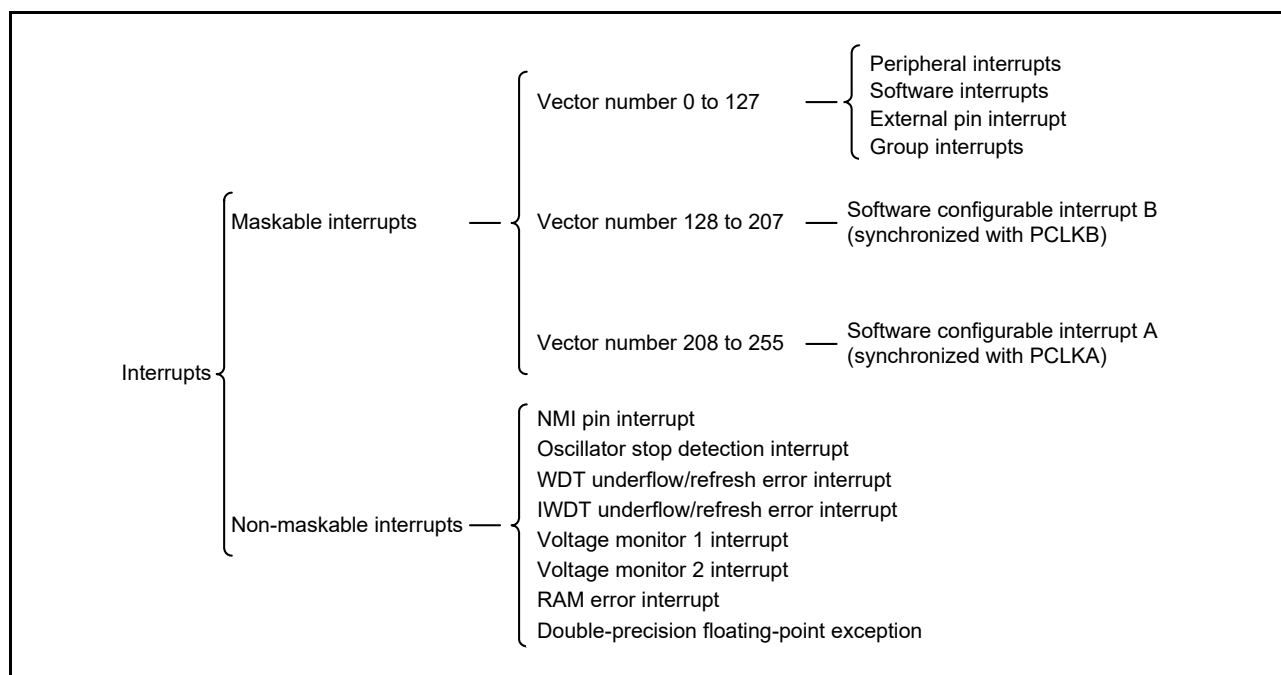


Figure 15.2 Types of Interrupts

15.4.1 Peripheral Interrupts

Peripheral interrupt is generated by peripherals. Peripheral interrupts sources assigned to vector numbers 0 to 127 cannot be assigned to the software configurable interrupts. Refer to section 15.4.5, Software Configurable Interrupts for details on software configurable interrupts.

15.4.2 Software Interrupts

When the SWINTR.SWINT bit and SWINT2R.SWINT2 bit are set to 1, the SWINT interrupt and SWINT2 interrupt occur, respectively.

15.4.3 External Pin Interrupt

An external pin interrupt is generated by signals input to the IRQ_i pin ($i = 0$ to 15). Refer to section 15.7.4, Setting the External Pin Interrupt for the procedure to set the external pin interrupt.

15.4.4 Group Interrupts

Multiple peripheral interrupt requests (up to 32 requests) are grouped together as one interrupt request. Interrupts are grouped depending on the peripheral operating clock (ICLK, PCLKB, or PCLKA) and method to detect interrupt requests (edge detection or level detection).

(1) Types of Group Interrupts

Table 15.6 lists types of group interrupts.

Table 15.6 Types of Group Interrupts

Interrupt Vector Number	Interrupt Name	Group Interrupt Source	
		Peripheral operating clock	Interrupt detection method
17	GROUPIE0	ICLK	Edge detection
106	GROUPBE0		
110	GROUPBL0	PCLKB	Level detection
111	GROUPBL1		
107	GROUPBL2		
112	GROUPAL0	PCLKA	
113	GROUPAL1		

(2) Configuration of Group Interrupts

When an interrupt request is generated while the corresponding EN_j bit in the group interrupt request enable register (GENIE0, GENBE0, GENBL0, GENBL1, GENBL2, GENAL0, GENAL1*1) is 1, the IS_j flag in the group interrupt request register (GRPIE0, GRPBE0, GRPBL0, GRPBL1, GRPBL2, GRPAL0, GRPAL1*1) becomes 1 (j = 0 to 31). Figure 15.3 shows the configuration of group interrupts.

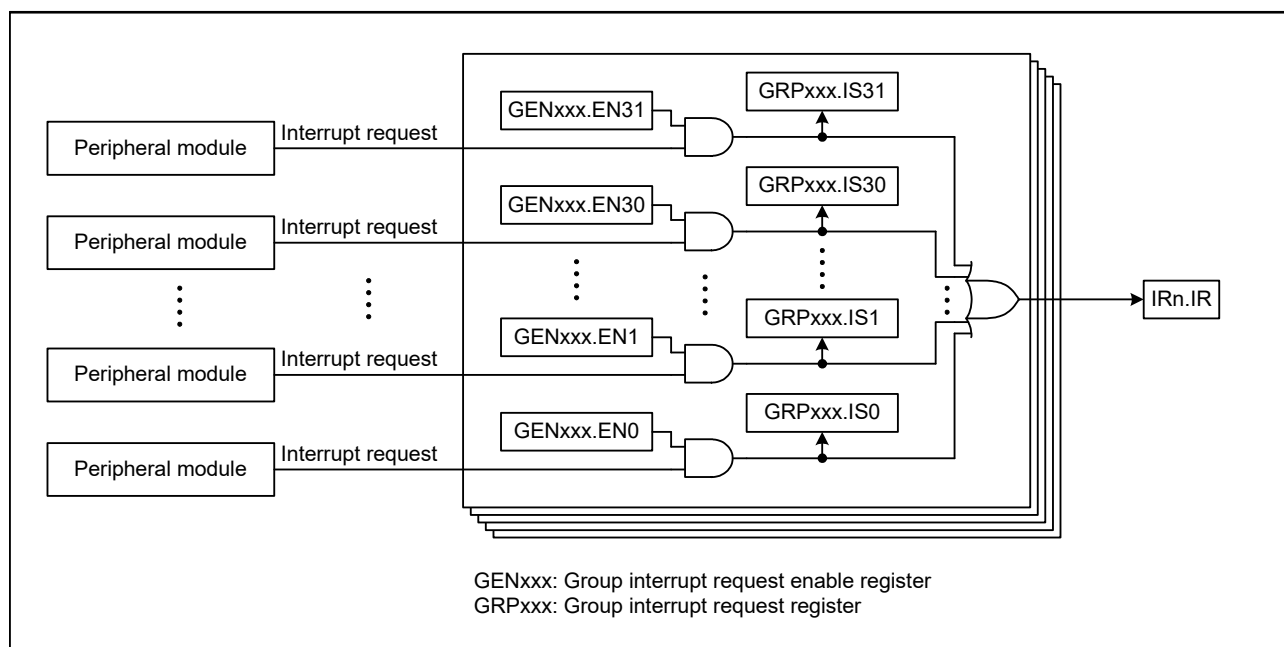


Figure 15.3 Group Interrupt Configuration (n = 17, 106, 107, 110 to 113)

Note 1. There is no register in the group to which no interrupt source is assigned.

(3) Group Interrupt Sources

Table 15.7 lists peripheral interrupt sources that are assigned to group interrupts.

Table 15.7 Group Interrupt Requests (1/3)

Group	No.	Interrupt Request Source	Name	Interrupt Request Enable Bit	Interrupt Status Flag	Interrupt Source Clear Bit	Vector No. (IRn.IR)
IE0	0	DPFPU	DPFPUX (double-precision floating-point exception)	GENIE0.EN0	GRPIE0.IS0	GCRIE0.CLR0	17
	1 to 31	Reserved	—	—	—	—	
BE0	0	CAN0	ERS0 (error interrupt)	GENBE0.EN0	GRPBE0.IS0	GCRBE0.CLR0	106
	1	CAN1	ERS1 (error interrupt)	GENBE0.EN1	GRPBE0.IS1	GCRBE0.CLR1	
	2	CAN2	ERS2 (error interrupt)	GENBE0.EN2	GRPBE0.IS2	GCRBE0.CLR2	
	3 to 31	Reserved	—	—	—	—	
BL0	0	SCI0	TEI0 (transmission end)	GENBL0.EN0	GRPBL0.IS0	—	110
	1		ERI0 (receive error)	GENBL0.EN1	GRPBL0.IS1	—	
	2	SCI1	TEI1 (transmission end)	GENBL0.EN2	GRPBL0.IS2	—	
	3		ERI1 (receive error)	GENBL0.EN3	GRPBL0.IS3	—	
	4	SCI2	TEI2 (transmission end)	GENBL0.EN4	GRPBL0.IS4	—	
	5		ERI2 (receive error)	GENBL0.EN5	GRPBL0.IS5	—	
	6	SCI3	TEI3 (transmission end)	GENBL0.EN6	GRPBL0.IS6	—	
	7		ERI3 (receive error)	GENBL0.EN7	GRPBL0.IS7	—	
	8	SCI4	TEI4 (transmission end)	GENBL0.EN8	GRPBL0.IS8	—	
	9		ERI4 (receive error)	GENBL0.EN9	GRPBL0.IS9	—	
	10	SCI5	TEI5 (transmission end)	GENBL0.EN10	GRPBL0.IS10	—	
	11		ERI5 (receive error)	GENBL0.EN11	GRPBL0.IS11	—	
	12	SCI6	TEI6 (transmission end)	GENBL0.EN12	GRPBL0.IS12	—	
	13		ERI6 (receive error)	GENBL0.EN13	GRPBL0.IS13	—	
	14, 15	Reserved	—	—	—	—	
	16	SCI12	TEI12 (transmission end)	GENBL0.EN16	GRPBL0.IS16	—	
	17		ERI12 (receive error)	GENBL0.EN17	GRPBL0.IS17	—	
	18		SCIX0 (Break Field Low width detection)	GENBL0.EN18	GRPBL0.IS18	—	
	19		SCIX1 (Control Field 0 match) (Control Field 1 match) (priority interrupt bit detection)	GENBL0.EN19	GRPBL0.IS19	—	
	20	Reserved	SCIX2 (bus collision detection)	GENBL0.EN20	GRPBL0.IS20	—	
	21		SCIX3 (valid edge detection)	GENBL0.EN21	GRPBL0.IS21	—	
	22, 23		—	—	—	—	
	24	QSPI	QSPSSLI (QSSL negation)	GENBL0.EN24	GRPBL0.IS24	—	
	25	Reserved	—	—	—	—	
	26	CAC	FERRI (frequency error)	GENBL0.EN26	GRPBL0.IS26	—	
	27		MENDI (measurement end)	GENBL0.EN27	GRPBL0.IS27	—	
	28		OVFI (overflow interrupt)	GENBL0.EN28	GRPBL0.IS28	—	
	29	DOC	DOPCI (data operation circuit interrupt)	GENBL0.EN29	GRPBL0.IS29	—	
	30	PDC	PCFEI (frame end interrupt)	GENBL0.EN30	GRPBL0.IS30	—	
	31		PCERI (error interrupt)	GENBL0.EN31	GRPBL0.IS31	—	

Table 15.7 Group Interrupt Requests (2/3)

Group	No.	Interrupt Request Source	Name	Interrupt Request Enable Bit	Interrupt Status Flag	Interrupt Source Clear Bit	Vector No. (IRn.IR)
BL1	0 to 2	Reserved	—	—	—	—	111
	3	SDHI	CDETI (card detection interrupt)	GENBL1.EN3	GRPBL1.IS3	—	
	4		CACI (card access interrupt)	GENBL1.EN4	GRPBL1.IS4	—	
	5		SDACI (SDIO access interrupt)	GENBL1.EN5	GRPBL1.IS5	—	
	6	MMCIF	CDETIO (card detection interrupt)	GENBL1.EN6	GRPBL1.IS6	—	
	7		ERRIO (error interrupt)	GENBL1.EN7	GRPBL1.IS7	—	
	8		ACCIO (access interrupt)	GENBL1.EN8	GRPBL1.IS8	—	
	9	POE3	OEI1 (output enable interrupt 1)	GENBL1.EN9	GRPBL1.IS9	—	
	10		OEI2 (output enable interrupt 2)	GENBL1.EN10	GRPBL1.IS10	—	
	11		OEI3 (output enable interrupt 3)	GENBL1.EN11	GRPBL1.IS11	—	
	12		OEI4 (output enable interrupt 4)	GENBL1.EN12	GRPBL1.IS12	—	
	13	RIIC0	TEI0 (transmission end)	GENBL1.EN13	GRPBL1.IS13	—	
	14		EEI0 (communication error/communication event)	GENBL1.EN14	GRPBL1.IS14	—	
	15	RIIC2	TEI2 (transmission end)	GENBL1.EN15	GRPBL1.IS15	—	
	16		EEI2 (communication error/communication event)	GENBL1.EN16	GRPBL1.IS16	—	
	17	SSIE0	SSIF0 (status interrupt)	GENBL1.EN17	GRPBL1.IS17	—	
	18	SSIE1	SSIF1 (status interrupt)	GENBL1.EN18	GRPBL1.IS18	—	
	19	Reserved	—	—	—	—	
	20	S12AD	S12CMPAI (compare interrupt)	GENBL1.EN20	GRPBL1.IS20	—	
	21		S12CMPBI (compare interrupt)	GENBL1.EN21	GRPBL1.IS21	—	
	22	S12AD1	S12CMPAI1 (compare interrupt)	GENBL1.EN22	GRPBL1.IS22	—	
	23		S12CMPBI1 (compare interrupt)	GENBL1.EN23	GRPBL1.IS23	—	
	24 to 27	Reserved	—	—	—	—	
	28	RIIC1	TEI1 (transmission end)	GENBL1.EN28	GRPBL1.IS28	—	
	29		EEI1 (communication error/communication event)	GENBL1.EN29	GRPBL1.IS29	—	
	30, 31	Reserved	—	—	—	—	
BL2	0 to 6	Reserved	—	—	—	—	107
	7	POEG	POEGGAI (group A interrupt)	GENBL2.EN7	GRPBL2.IS7	—	
	8		POEGGBI (group B interrupt)	GENBL2.EN8	GRPBL2.IS8	—	
	9		POEGGCI (group C interrupt)	GENBL2.EN9	GRPBL2.IS9	—	
	10		POEGGDI (group D interrupt)	GENBL2.EN10	GRPBL2.IS10	—	
	11 to 31	Reserved	—	—	—	—	

Table 15.7 Group Interrupt Requests (3/3)

Group	No.	Interrupt Request Source	Name	Interrupt Request Enable Bit	Interrupt Status Flag	Interrupt Source Clear Bit	Vector No. (IRn.IR)
AL0	0	SCI8	TEI8 (transmission end)	GENAL0.EN0	GRPAL0.IS0	—	112
	1		ERI8 (receive error)	GENAL0.EN1	GRPAL0.IS1	—	
	2, 3	Reserved	—	—	—	—	
	4	SCI9	TEI9 (transmission end)	GENAL0.EN4	GRPAL0.IS4	—	
	5		ERI9 (receive error)	GENAL0.EN5	GRPAL0.IS5	—	
	6, 7	Reserved	—	—	—	—	
	8	SCI10	TEI10 (transmission end)	GENAL0.EN8	GRPAL0.IS8	—	
	9		ERI10 (receive error)	GENAL0.EN9	GRPAL0.IS9	—	
	10, 11	Reserved	—	—	—	—	
	12	SCI11	TEI11 (transmission end)	GENAL0.EN12	GRPAL0.IS12	—	
	13		ERI11 (receive error)	GENAL0.EN13	GRPAL0.IS13	—	
	14, 15	Reserved	—	—	—	—	
	16	RSPI0	SPII0 (idle interrupt)	GENAL0.EN16	GRPAL0.IS16	—	
	17		SPEI0 (error interrupt)	GENAL0.EN17	GRPAL0.IS17	—	
	18	RSPI1	SPII1 (idle interrupt)	GENAL0.EN18	GRPAL0.IS18	—	
	19		SPEI1 (error interrupt)	GENAL0.EN19	GRPAL0.IS19	—	
	20	RSPI2	SPII2 (idle interrupt)	GENAL0.EN20	GRPAL0.IS20	—	
	21		SPEI2 (error interrupt)	GENAL0.EN21	GRPAL0.IS21	—	
	22	SCI7	TEI7 (transmission end)	GENAL0.EN22	GRPAL0.IS22	—	
	23		ERI7 (receive error)	GENAL0.EN23	GRPAL0.IS23	—	
	24 to 31	Reserved	—	—	—	—	
AL1	0	EPTPC	MINT (status interrupt)	GENAL1.EN0	GRPAL1.IS0	—	113
	1	PTPEDMAC	PINT (status interrupt)	GENAL1.EN1	GRPAL1.IS1	—	
	2	Reserved	—	—	—	—	
	3	Reserved	—	—	—	—	
	4	EDMAC0	EINT0 (status interrupt)	GENAL1.EN4	GRPAL1.IS4	—	
	5	EDMAC1	EINT1 (status interrupt)	GENAL1.EN5	GRPAL1.IS5	—	
	6, 7	Reserved	—	—	—	—	
	8	GLCDC	VPOS (line detection)	GENAL1.EN8	GRPAL1.IS8	—	
	9		GR1UF (graphic 1 underflow detection)	GENAL1.EN9	GRPAL1.IS9	—	
	10		GR2UF (graphic 2 underflow detection)	GENAL1.EN10	GRPAL1.IS10	—	
	11	DRW2D	DRW_IRQ (combined interrupt)	GENAL1.EN11	GRPAL1.IS11	—	
	12 to 31	Reserved	—	—	—	—	

15.4.5 Software Configurable Interrupts

An interrupt source assigned to each interrupt vector number from 128 to 255 can be selected from multiple sources. They are divided into software configurable interrupt B and software configurable interrupt A depending on the peripheral operating clock. Figure 15.4 shows the software configurable interrupt configuration.

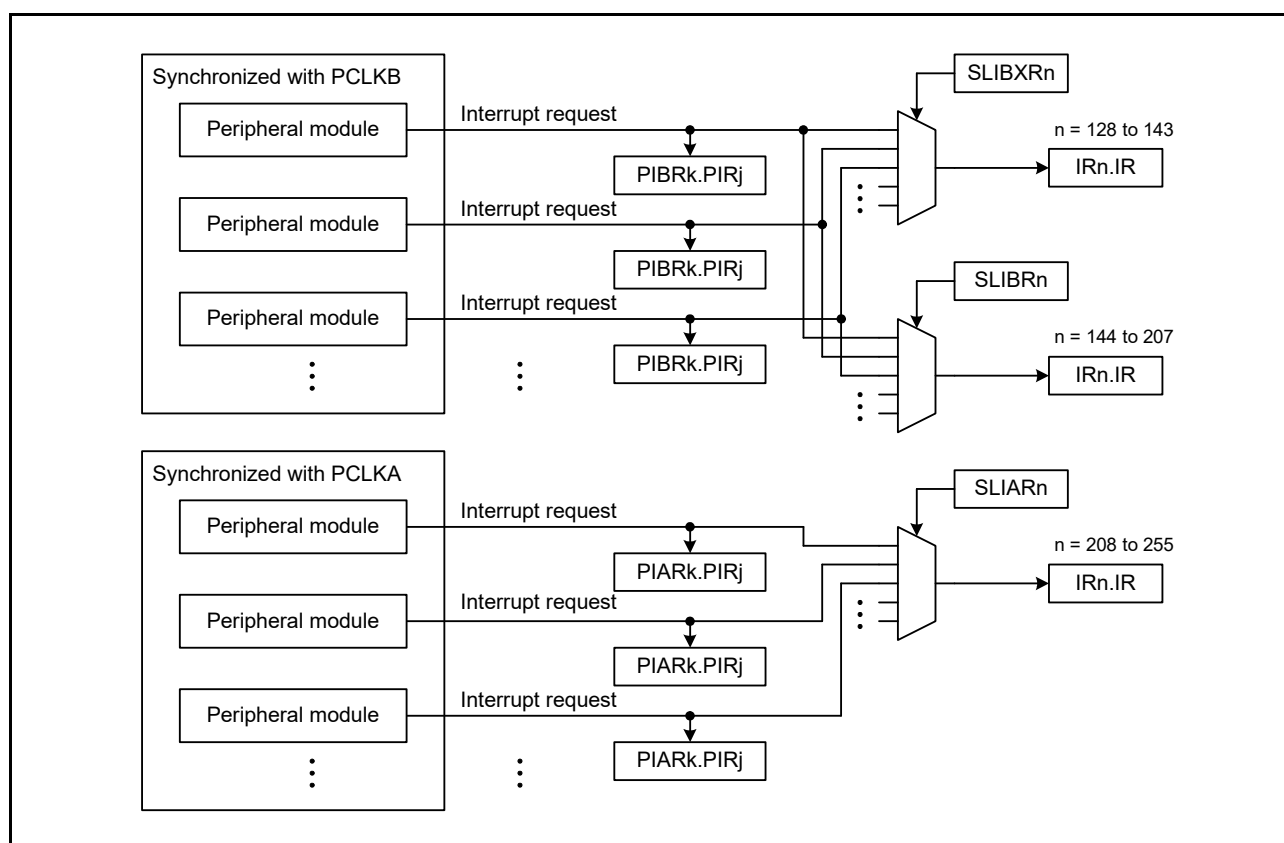


Figure 15.4 Software Configurable Interrupt Configuration

15.4.5.1 Software Configurable Interrupt B

For interrupt sources assigned to software configurable interrupts, interrupt sources of peripherals that operates in synchronization with PCLKB can be assigned to interrupt number from 128 to 207. The abbreviation for software configurable interrupt B is PERIB. Interrupt names are indicated by INTB128 to INTB207.

Refer to Table 15.3, Interrupt Sources for Software Configurable Interrupt B for interrupt sources that can be assigned to software configurable interrupt B.

15.4.5.2 Software Configurable Interrupt A

For interrupt sources assigned to software configurable interrupts, interrupt sources of peripherals that operates in synchronization with PCLKA can be assigned to interrupt number from 208 to 255. The abbreviation for software configurable interrupt A is PERIA. Interrupt names are indicated by INTA208 to INTA255.

Refer to Table 15.4, Interrupt Sources for Software Configurable Interrupt A for interrupt sources that can be assigned for software configurable interrupt A.

15.4.5.3 EXDMAC Start Trigger by Software Configurable Interrupts

Some interrupt sources assigned to software configurable interrupt B and software configurable interrupt A can be used as a trigger for EXDMAC0 or EXDMAC1.

To use an interrupt source as the EXDMAC0 trigger, assign the interrupt source to interrupt vector number 144 (INTB144) or 208 (INTA208). To use an interrupt source as the EXDMAC1 trigger, assign the interrupt source to interrupt vector number 145 (INTB145) or 209 (INTA209).

Only the TPU1.TGI1A interrupt of software configurable interrupt B and the MTU1.TGIA1 interrupt of software configurable interrupt A can be used as the EXDMAC trigger.

Triggers for EXDMAC0 and EXDMAC1 can be selected by setting bits SELEXDR.SELEXD0 and SELEXD1, respectively. Table 15.8 lists correspondences between the SELEXDR register settings and interrupt sources to start the EXDMAC.

Table 15.8 EXDMAC Start Triggers by Software Configurable Interrupts

SELEXDR Register	Bit Value	Interrupt Vector Number	Interrupt Request
SELEXD0 bit (EXDMAC0)	0	144 (INTB144)	TPU1.TGI1A (input capture/compare match)
	1	208 (INTA208)	MTU1.TGIA1 (input capture/compare match)
SELEXD1 bit (EXDMAC1)	0	145 (INTB145)	TPU1.TGI1A (input capture/compare match)
	1	209 (INTA209)	MTU1.TGIA1 (input capture/compare match)

15.4.6 Non-Maskable Interrupts

Non-maskable interrupts include the NMI pin interrupt, oscillation stop detection interrupt, WDT underflow/refresh error interrupt, IWDT underflow/refresh error interrupt, voltage monitoring 1 interrupt, voltage monitoring 2 interrupt, RAM error interrupt, and double-precision floating-point exception.

Non-maskable interrupts have the highest priority in all interrupts, including the fast interrupt, so are accepted regardless of the settings of the PSW.I bit (interrupt enable) and IPL[3:0] bits (processor interrupt priority level) in the CPU.

The NMISR register can be used to confirm whether a non-maskable interrupt is generated.

Only the CPU can be selected as the interrupt request destination of non-maskable interrupt. The DTC and DMAC cannot be selected.

15.5 Interrupt Detection

Level detection or edge detection can be used for detecting an interrupt request.

For interrupt requests from the peripherals, edge detection or level detection is fixed for each interrupt source. For interrupt requests of the external pin interrupt, edge detection or level detection can be selected with the `IRQCRi.IRQMD[1:0]` bits ($i = 0$ to 15).

Refer to Table 15.5, Interrupt Vector Table for the method to detect each interrupt request.

For group interrupts, interrupt sources are grouped depending on the method to detect interrupt requests.

Interrupt requests by interrupt sources assigned to groups IE0 and BE0 are detected by edge detection. Interrupt requests by interrupt sources assigned to groups BL0, BL1, BL2, AL0, and AL1 are detected by level detection. Note that group interrupts (GROUPIE0, GROUPBE0, GROUPBL0, GROUPBL1, GROUPBL2, GROUPAL0, GROUPAL1) are detected by level detection.

Refer to section 15.4.4, Group Interrupts for group interrupts. Refer to section 15.5.3, Group Interrupts Using Edge Detection and section 15.5.4, Group Interrupts Using Level Detection for interrupt requests of group interrupts.

15.5.1 Edge Detection

Figure 15.5 shows the operation of the `IRn.IR` flag at edge detection ($n = 023$ to 255).

The `IRn.IR` flag becomes 1 when the rising edge of the interrupt request signal is detected. Then, the `IRn.IR` flag does not become 0 by disabling the interrupt request of the peripheral module. When the CPU accepts the interrupt request or the DTC/DMAC accepts the transfer request, the `IRn.IR` flag automatically becomes 0. It is not required to set the `IRn.IR` flag to 0 by software. Refer to Table 15.9, Operations When Starting the DTC/DMAC for details on clearing the `IRn.IR` flag by DTC/DMAC.

For the external pin interrupts of interrupt vector number 64 to 79 and interrupt sources of interrupt vector number 88 to 95, the timing when the `IRn.IR` flag becomes 1 after an interrupt signal occurs is different from the other interrupts. For the external pin interrupts, the timing is delayed for internal delay plus two cycles of `PCLKB` after a signal is input to the `IRQ` pin ($i = 0$ to 15). For interrupts of interrupt vector number 88 to 95, the timing is delayed for two cycles of `PCLKB`.

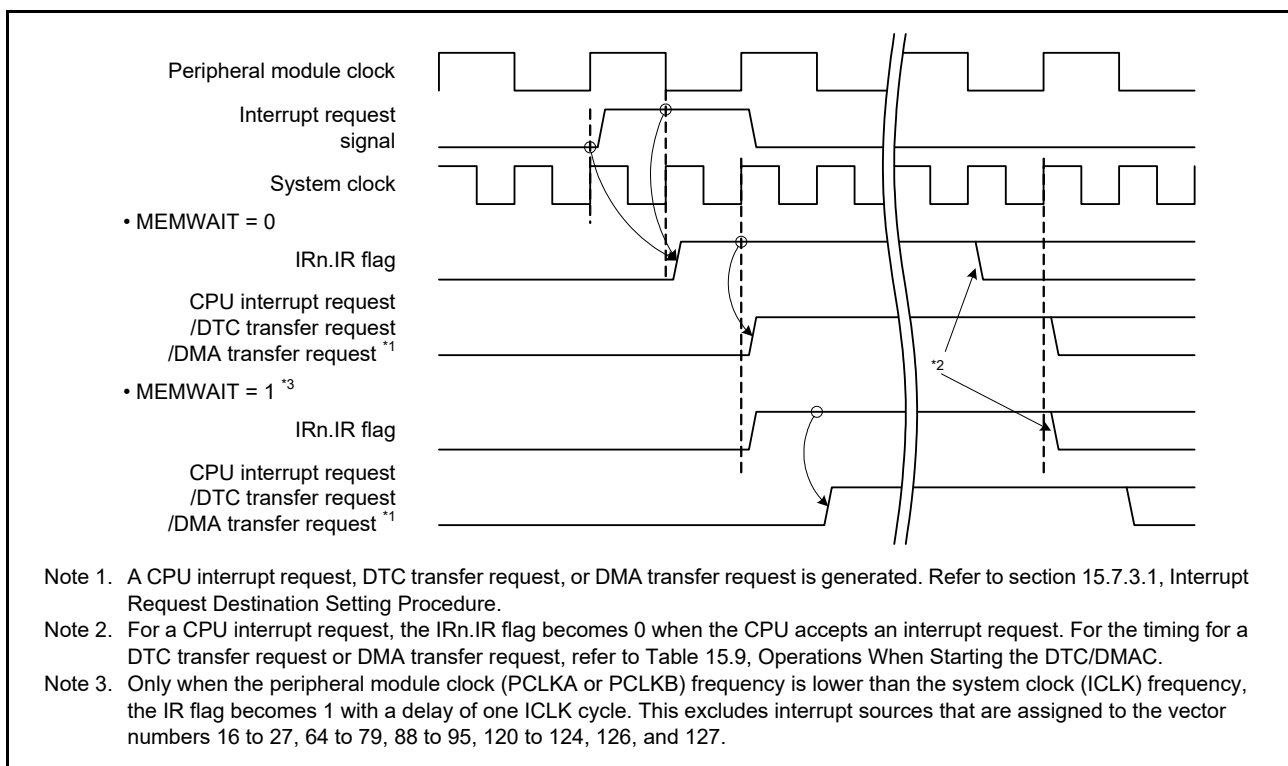


Figure 15.5 `IRn.IR` Flag Operation for Interrupts Detected by Edge Detection

(1) Detecting Consecutive Interrupt Request Signals

When interrupt request signals occur every cycle, the latter interrupt signal cannot be detected. To accept consecutive interrupt request signals, the interval of at least two cycles of the system clock or peripheral module clock, whichever is lower, is required between interrupt signals. Figure 15.6 shows the interval for accepting consecutive interrupt request signals.

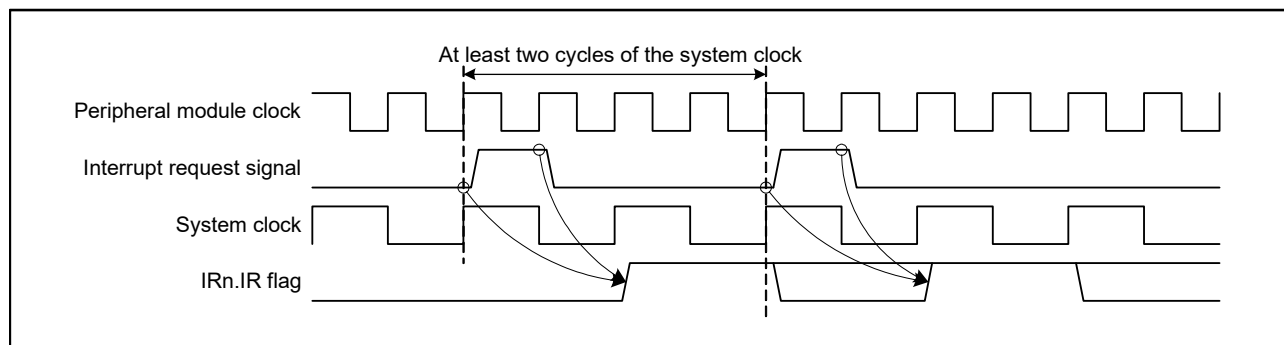


Figure 15.6 Accepting Consecutive Interrupt Signals
(when the system clock frequency is lower than the peripheral clock frequency)

When an interrupt request is generated again while the IRn.IR flag is 1, the interrupt request is ignored ($n = 023$ to 255). However, for transmit interrupt requests, receive interrupt requests, and buffer access interrupt requests of the SCI, RIIC, RSPI, QSPI, SSIE, SDHI, MMCIF, and PDC, when an interrupt request occurs while the IRn.IR flag is 1, the interrupt request is retained in the module. After the IRn.IR flag becomes 0, the IRn.IR flag is set to 1 again by the retained request. Refer to the descriptions for interrupts in each chapter of peripheral modules for details.

Figure 15.7 shows the timing when the IRn.IR flag is set again.

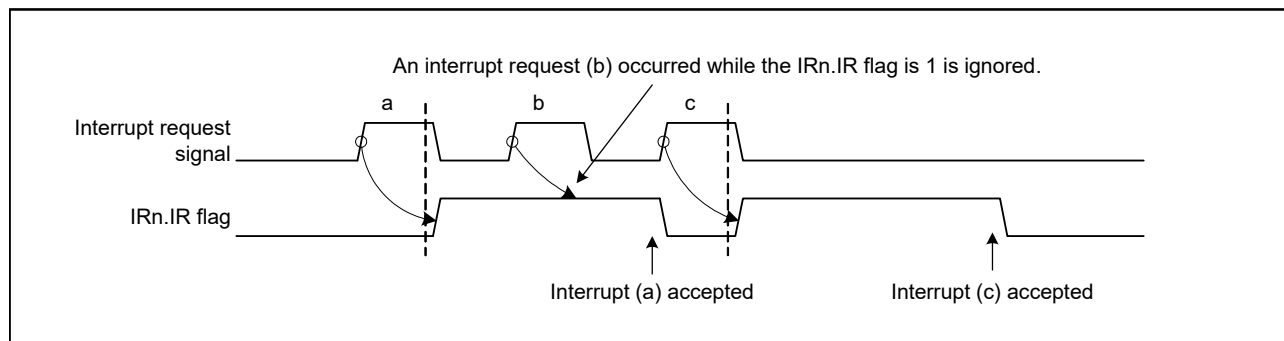


Figure 15.7 Timing to Set the IRn.IR Flag Again

(2) Relation between the IRn.IR flag and Interrupt Request Enable Bits

After the IRn.IR flag becomes 1, the IRn.IR flag does not become 0 even when an interrupt request enable bit in the corresponding peripheral module is set to 0.

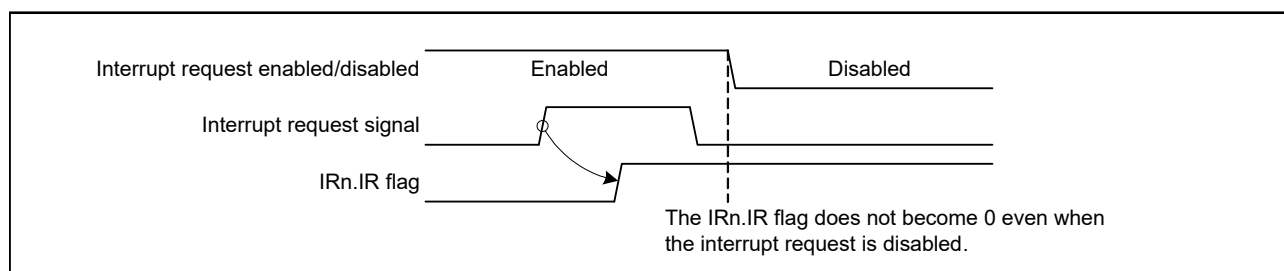


Figure 15.8 Relation Between Disabling the Interrupt Request and the IRn.IR Flag

15.5.2 Level Detection

Figure 15.9 shows operation of the interrupt request signal and the IRn.IR flag for level detection ($n = 016$ to 124). The IRn.IR flag is 1 while the interrupt request signal is 1. To set the IRn.IR flag to 0, set the corresponding interrupt request signal of the peripheral module to 0. Set the corresponding interrupt status flag of the peripheral module to 0 and wait for the time until the value is reflected in the IRn.IR flag before exiting the interrupt handler. Refer to (2) Notes on Writing to I/O Registers in section 5, I/O Registers for details on waiting for the reflection.

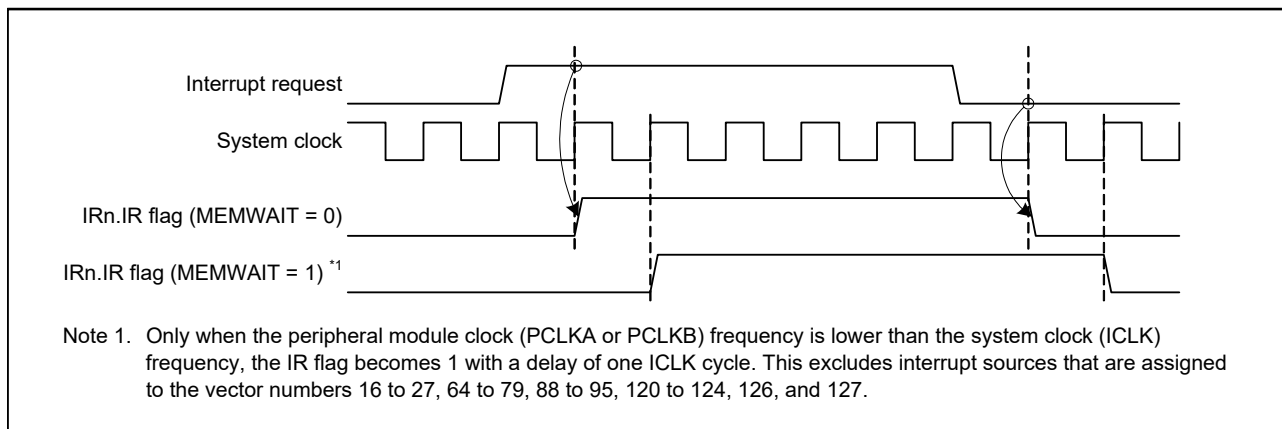


Figure 15.9 IRn.IR Flag Operation for Level Detection

Figure 15.10 shows an example of the procedure to handle interrupts for level detection.

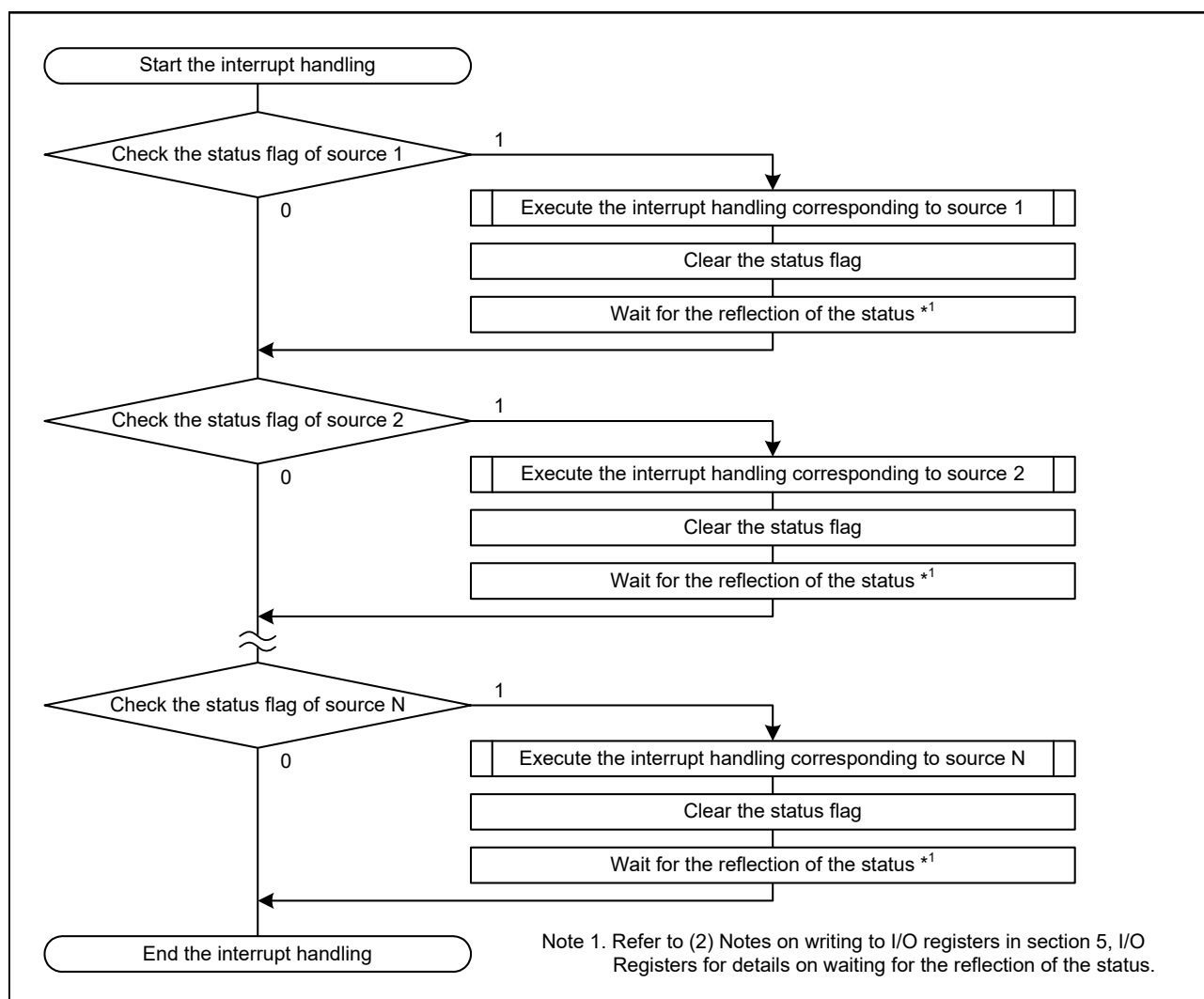


Figure 15.10 Example of Level Detection Interrupt Handling Procedure (N indicates the number of status flags)

15.5.3 Group Interrupts Using Edge Detection

Groups IE0 and BE0 of group interrupts include interrupt sources that are detected by edge detection.

While the IR017.IR flag corresponding to the GROUPIE0 interrupt and the IR106.IR flag corresponding to the GROUPBE0 interrupt become 1 under the same conditions as edge detection, the IR017.IR and IR106.IR flags become 0 under the same conditions as level detection.

When the rising edge of an interrupt request signal is detected while the corresponding GENIE0/GENBE0.ENj bit is 1, both the GRPIE0/GRPBE0.ISj flag and IR017/IR106.IR flag become 1 ($j = 0$ to 31). Then, GRPIE0/GRPBE0.ISj flag and IR017/IR106.IR flag do not become 0 by disabling the interrupt request of the peripheral module or setting the GENIE0/GENBE0.ENj bit to 0.

When the GCRIE0/GCRBE0.CLRj bit is set to 1, the GRPIE0/GRPBE0.ISj flag becomes 0, and consequently the IR017/IR106.IR flag becomes 0.

Figure 15.11 and Figure 15.12 show operation examples of group interrupts using edge detection. Figure 15.13 shows an example of operation when interrupt requests are generated by multiple interrupt sources in the same group.

Note: • There is no register in the group to which no interrupt source is assigned.

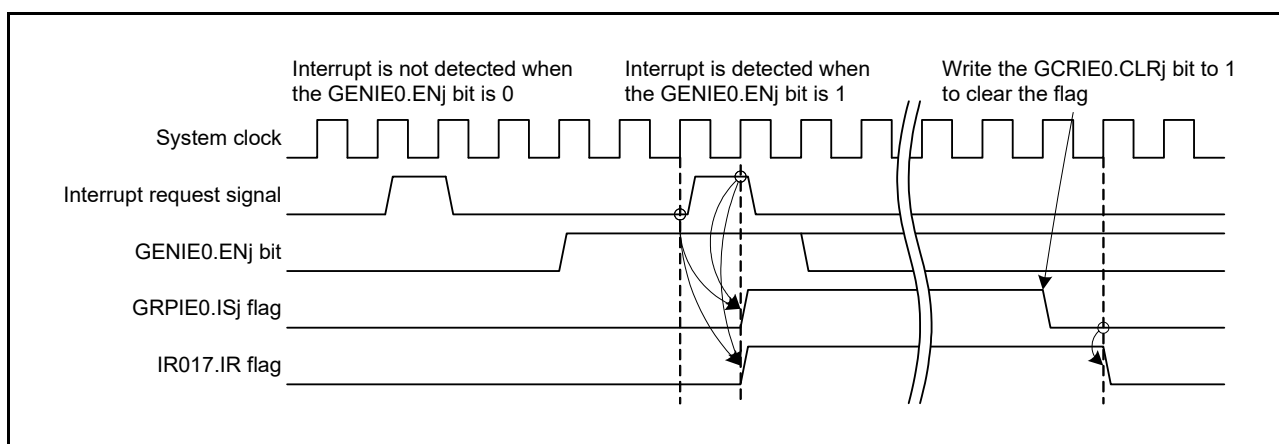


Figure 15.11 Example of Interrupt Request for Group Interrupt Using Edge Detection (Group IE0)

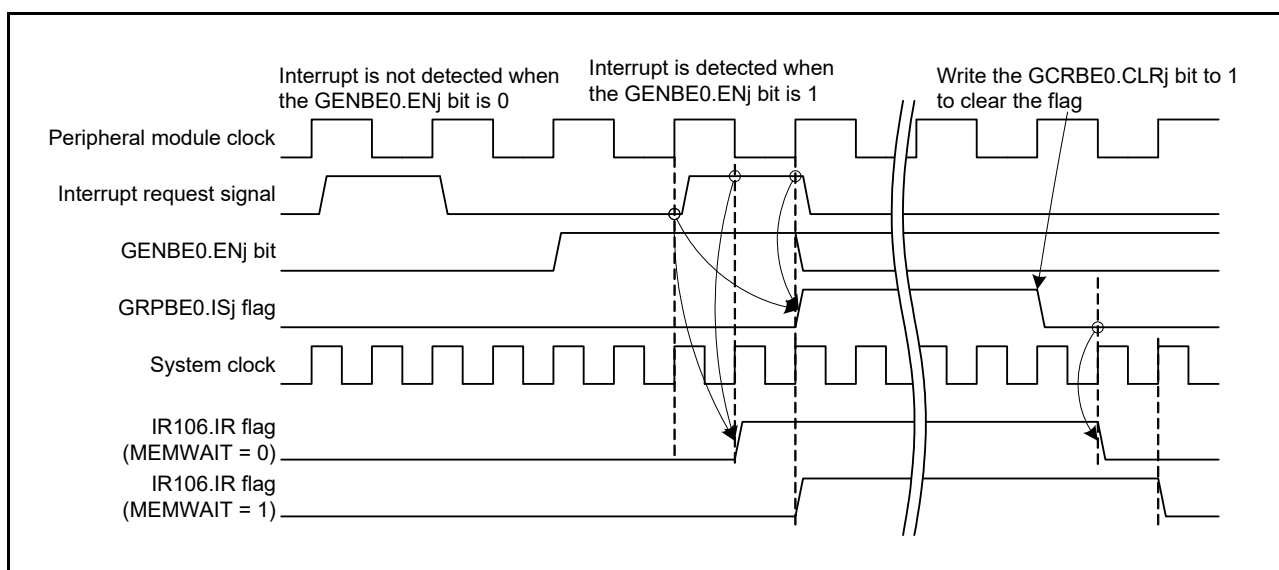


Figure 15.12 Example of Interrupt Request for Group Interrupt Using Edge Detection (Group BE0)

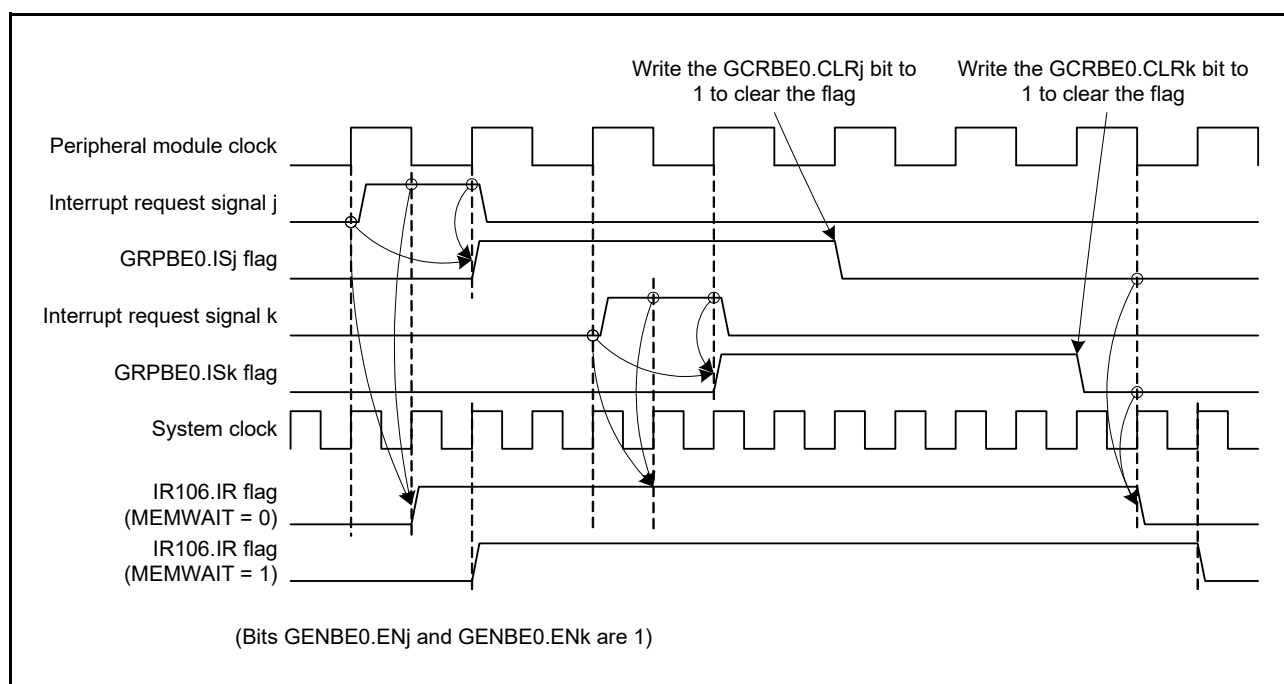


Figure 15.13 Example of Operation When Multiple Edge Detection Interrupt Requests Are Generated in the Same Group (Group BE0)

Figure 15.14 and Figure 15.15 show examples of the procedure to handle group interrupts using edge detection.

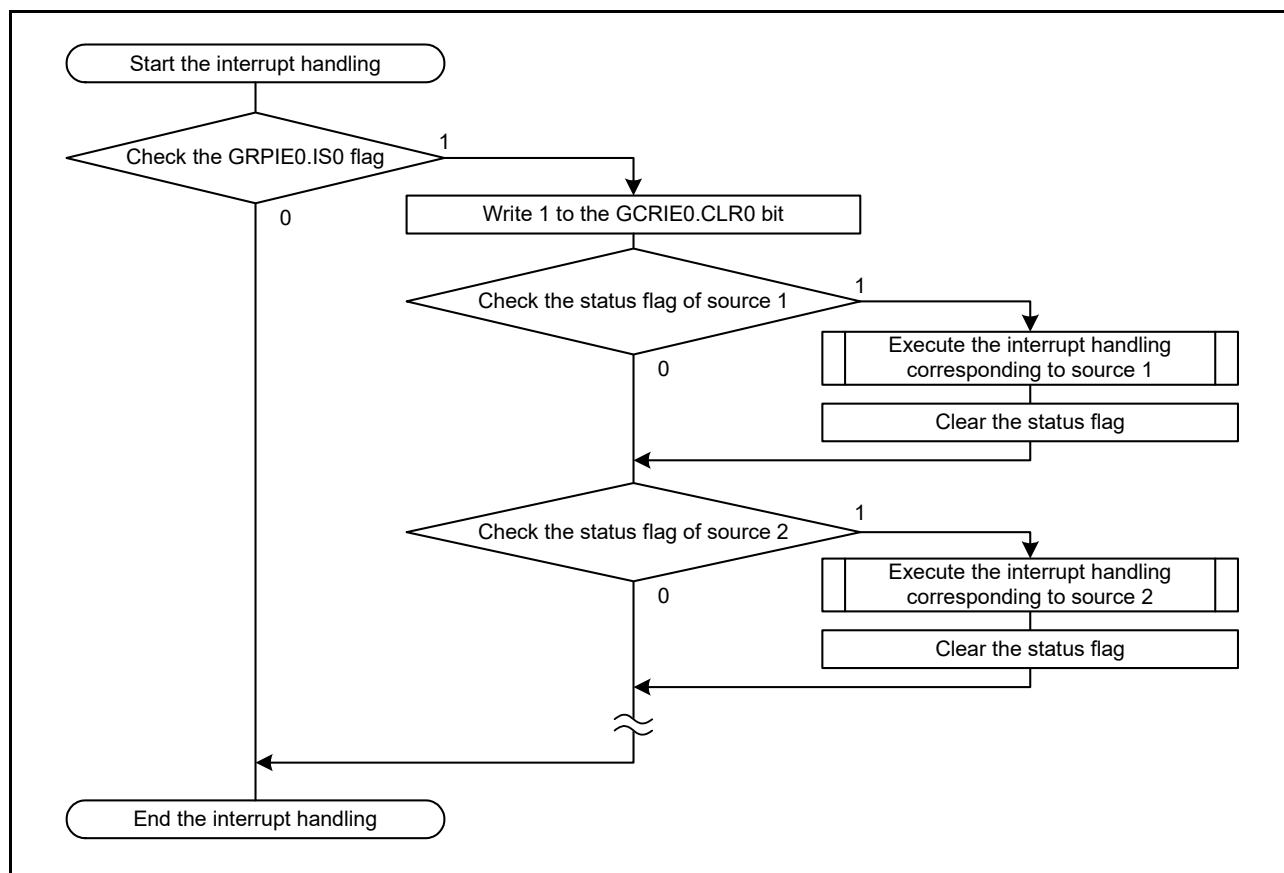


Figure 15.14 Example of Procedure to Handle Group Interrupts Using Edge Detection (Group IE0)

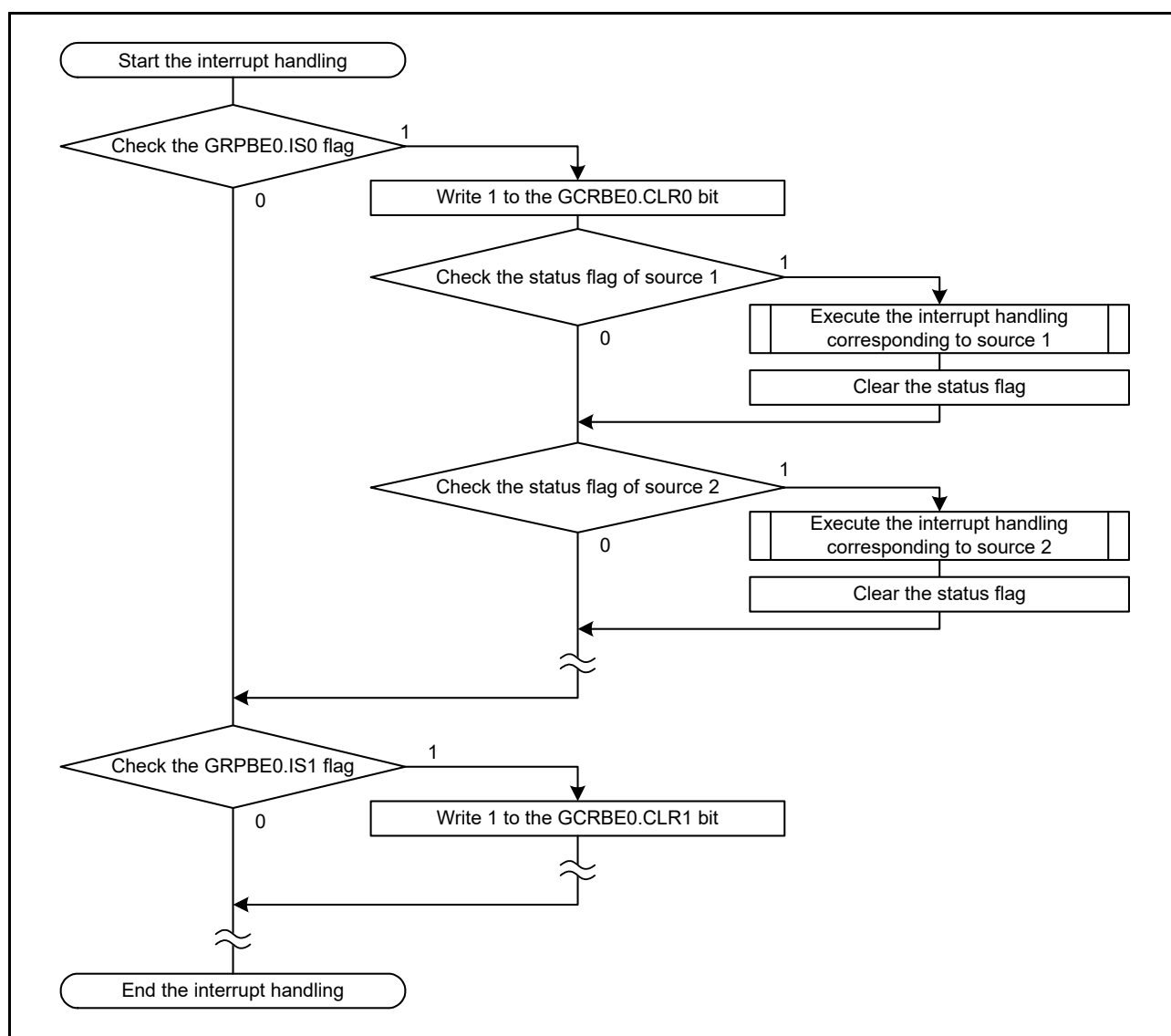


Figure 15.15 Example of Procedure to Handle Group Interrupts Using Edge Detection (Group BE0)

15.5.4 Group Interrupts Using Level Detection

Groups BL0, BL1, BL2, AL0, and AL1 of group interrupts includes interrupt sources that are detected by level detection. The IR110.IR flag corresponding to the GROUPBL0 interrupt, the IR111.IR flag corresponding to the GROUPBL1 interrupt, the IR107.IR flag corresponding to the GROUPBL2 interrupt, the IR112.IR flag corresponding to the GROUPAL0 interrupt, and the IR113.IR flag corresponding to the GROUPAL1 interrupt change under the same conditions as level detection.

When an interrupt signal becomes 1 while the corresponding GENBL0/GENBL1/GENBL2/GENAL0/GENAL1.ENj bit is 1, the GRPBL0/GRPBL1/GRPBL2/GRPAL0/GRPAL1.ISj flag and the IRn.IR flag become 1 ($j = 0$ to 31). Then, the GRPBL0/GRPBL1/GRPBL2/GRPAL0/GRPAL1.ISj flag and IRn.IR flag becomes 0 when the corresponding interrupt request signal becomes 0. Also, when the GENBL0/GENBL1/GENBL2/GENAL0/GENAL1.ENj bit is set to 0, the corresponding GRPBL0/GRPBL1/GRPBL2/GRPAL0/GRPAL1.ISj flag and IRn.IR flag become 0.

Figure 15.16 shows an operation example of group interrupts using edge detection. Figure 15.17 shows an example of operation when interrupt requests are generated by multiple interrupt sources in the same group.

Note: • There is no register in the group to which no interrupt source is assigned.

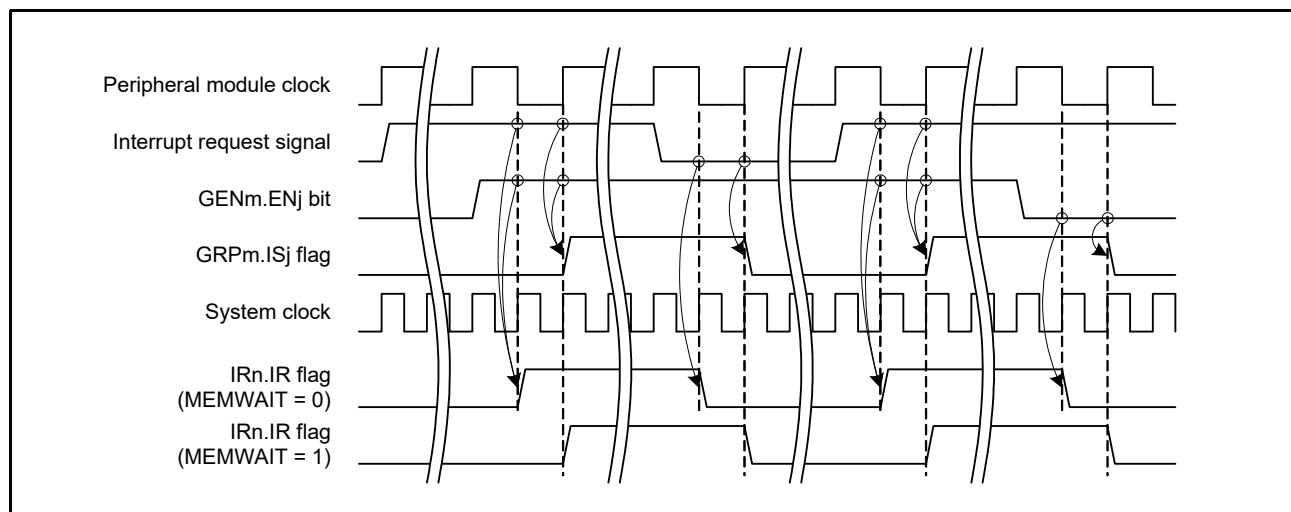


Figure 15.16 Operation Example of Group Interrupt Using Level Detection ($m = \text{BL0, BL1, BL2, AL0, AL1}$)

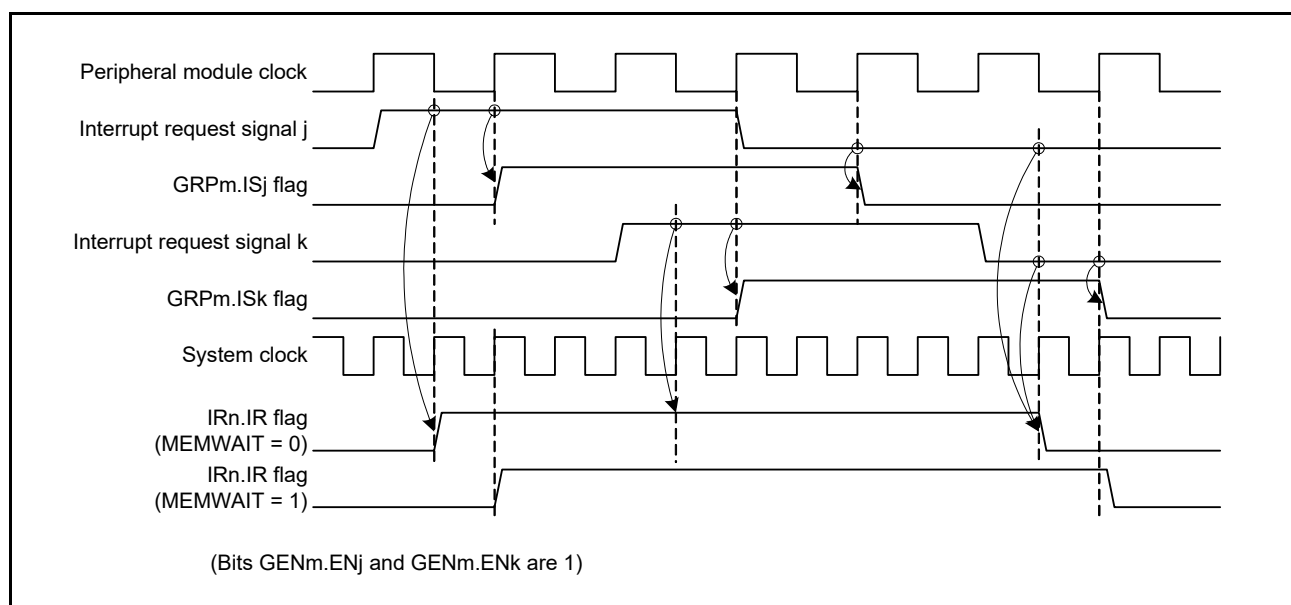


Figure 15.17 Operation Example When Multiple Interrupt Requests are Generated in the Same Group ($m = \text{BL0, BL1, BL2, AL0, AL1}$)

Figure 15.18 shows the procedure to handle group interrupts for level detection.

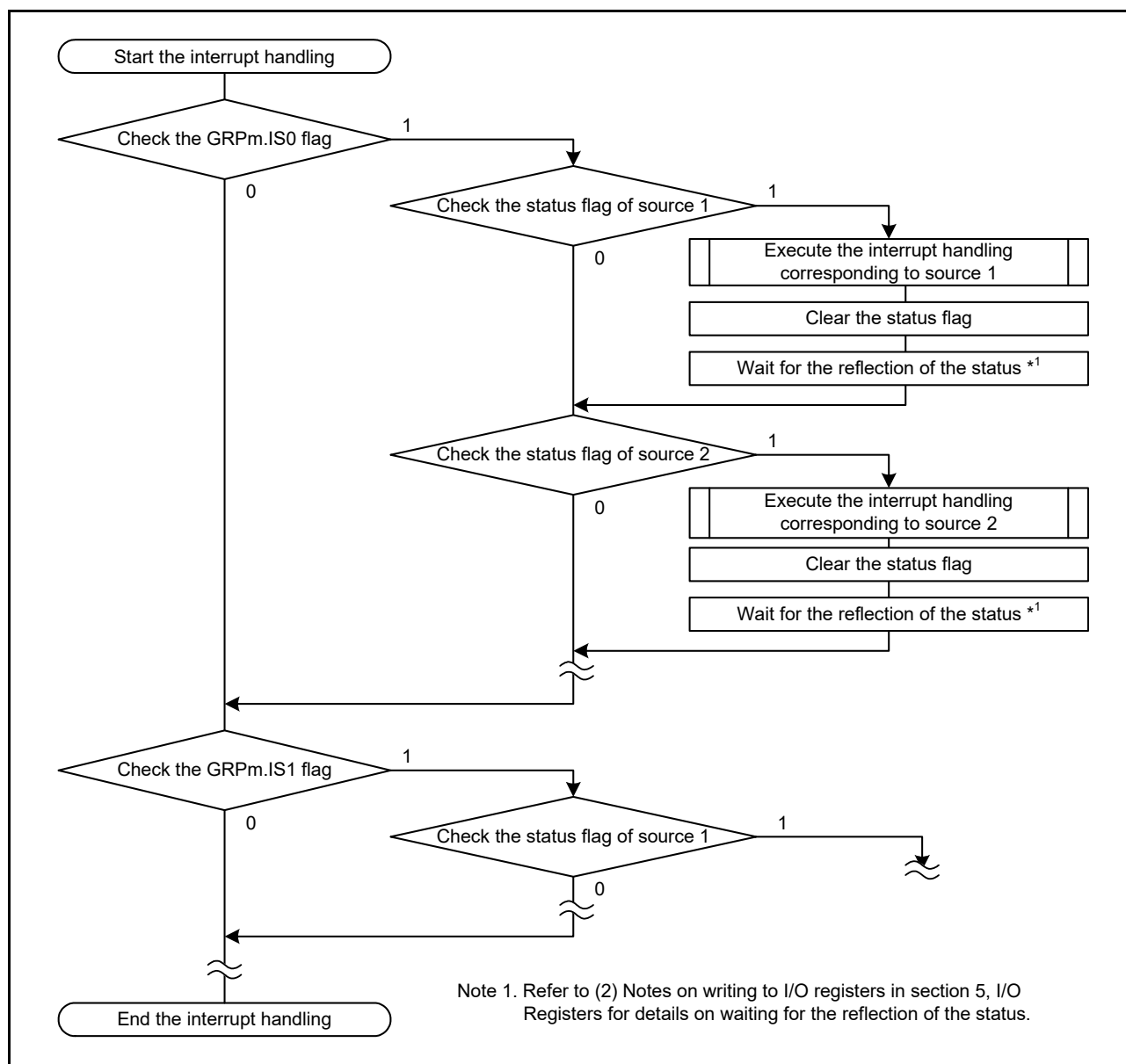


Figure 15.18 Example of Group Interrupt Handling Procedure for Level Detection (m = BL0, BL1, BL2, AL0, AL1)

15.5.5 Software Configurable Interrupts

Interrupt sources and interrupt requests for software configurable interrupts are detected by edge detection.

Figure 15.19 shows an operation example of the interrupt request and interrupt status flag for software configurable interrupts.

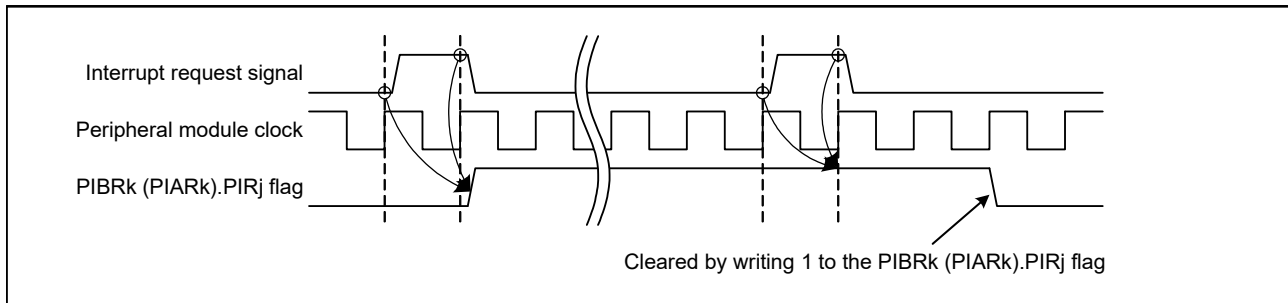


Figure 15.19 Operation Example of the Interrupt Request and Interrupt Status Flag for Software Configurable Interrupts

15.6 Determining Priority of Interrupt Requests

The ICU determines the priority for each interrupt request destination. The priority for each interrupt request destination is determined as follows.

(1) Determining Priority when the CPU is the Interrupt Request Destination

A source selected for the fast interrupt has the highest priority. Next to the fast interrupt, the priority is determined by the $IPRr.IPR[3:0]$ bit value, and an interrupt source with a larger value has priority ($r = 000$ to 255). If multiple sources have the same $IPRr.IPR[3:0]$ bit value, the priority is determined by the interrupt vector number, and the source with the smaller number has priority.

(2) Determining Priority when the DTC is the Interrupt Request Destination

The $IPRr.IPR[3:0]$ bits have no effect ($r = 000$ to 255). The priority is determined by only the interrupt vector number, and an interrupt source with a smaller number has priority.

(3) Determining Priority when the DMAC is the Interrupt Request Destination

The $IPRr.IPR[3:0]$ bits have no effect. The priority is determined by the DMAC channel number. Refer to section 18, DMA Controller (DMACa) for details on the DMAC channel priority.

15.7 Interrupt Setting Procedure

15.7.1 Enabling Interrupt Requests

The following describes the procedure to enable interrupt requests.

- (1) Set the interrupt request enable bit of the peripheral modules to enable output of the interrupt request.
- (2) For group interrupts, set the corresponding ENj bit in the group interrupt request enable register to 1 to enable output of the interrupt request to the ISj flag in the group interrupt request register (j = 0 to 31).
- (3) Set the corresponding IERm.IENj bit to 1 to enable output of the interrupt request to the interrupt request destination (m = 02h to 1Fh; j = 0 to 7).

After the above procedure is completed, when a peripheral interrupt occurs, the IRn.IR flag corresponding to the interrupt source becomes 1 (n = 016 to 255).

For group interrupts, the ISj flag in the group interrupt request register becomes 1, the IRn.IR flag corresponding to the group becomes 1, and an interrupt request is output to the interrupt request destination.

When the IERm.IENj bit is 0, the interrupt request corresponding to the interrupt source is not output to the interrupt request destination.

15.7.2 Disabling Interrupt Requests

The following describes the procedure to disable interrupt requests.

- (1) Set the corresponding IERm.IENj bit to 0 (m = 02h to 1Fh; j = 0 to 7).
 - (2) For group interrupts, set the corresponding ENj bit in the group interrupt request enable register to 0 to disable output of the interrupt request to the ISj flag in the group interrupt request register (j = 0 to 31).
 - (3) Set the interrupt request enable bit of the peripheral modules to disable output of the interrupt request. Read the register that has been set to confirm that the value is reflected.
 - (4) As needed, read the IRn.IR flag or set the IR flag to 0.*1
- For group interrupts, confirm that the ISj flag in the group interrupt request register is 0 or set the ISj flag to 0.

Note 1. When disabling the transmit interrupt request, receive interrupt request, or buffer access interrupt requests of the SCI, RIIC, RSPI, QSPI, SSIE, SDHI, MMCIF, or PDC, set the IRn.IR flag to 0 according to the above procedure. Refer to the description of interrupts in the corresponding section of peripheral modules for details.

15.7.3 Selecting Interrupt Request Destination

15.7.3.1 Interrupt Request Destination Setting Procedure

The destination of an interrupt request can be selected from the CPU, DTC, or DMAC for each interrupt source.

Destinations that can be selected differ depending on the interrupt source. Refer to Table 15.5, Interrupt Vector Table for details on the destinations. Do not select a destination that is not indicated as “✓” in Table 15.5.

When set the external pin interrupt as the DTC or DMAC trigger, set the IRQCRi.IRQMD[1:0] bits to select edge detection (i = 0 to 15).

The following describes the procedure to select a destination of an interrupt request.

(1) Setting Interrupt Sources as the DMAC trigger

Perform the following settings while the IERm.IENj bit is 0 of the interrupt source that is selected as the DMAC trigger (m = 02h to 1Fh; j = 0 to 7).

- (1) Set the interrupt vector number of the interrupt source used as the DMAC trigger in the DMRSRm register corresponding to the DMAC channel (m = DMAC channel number).^{*1}
- (2) Set the DMTMD.DCTG[1:0] bits corresponding to the DMAC channel to 01b in order to select the peripheral interrupt or external pin interrupt as the DMAC trigger.
- (3) Set the DMCNT.DTE bit corresponding to the DMAC channel to 1.

After the above settings are completed, set the corresponding IERm.IENj bit to 1.

Also, set the DMAST.DMST bit to 1 before or after the above settings.

Refer to section 18.3.7, Activating the DMAC in section 18, DMA Controller (DMACAA) for the procedure to set the DMAC.

(2) Setting Interrupt Sources as the DTC trigger

Perform the following setting while the IERm.IENj bit of the interrupt source that is selected as the DTC trigger is 0.

- (1) Set the DTCERn.DTCE bit corresponding to the interrupt vector number n used for the DTC trigger to 1 (n = 026 to 255).^{*1}

After the above setting is completed, set the IERm.IENj bit to 1.

Also set the DTCST.DTCST bit to 1 before or after the above settings.

Refer to section 20.5, DTC Setting Procedure in section 20, Data Transfer Controller (DTCb) for the procedure to set the DTC.

Note 1. Do not set the same interrupt source as DTC and DMAC triggers. Also, do not set the same interrupt source as triggers of multiple DMAC channels.

(3) Setting Interrupt Sources for the CPU

When an interrupt source is not selected as the DTC or DMAC trigger, the interrupt request is output to the CPU.

Set the IERm.IENj bit to 1 while the interrupt source is not selected as a DTC or DMAC trigger.

15.7.3.2 Operations When the DTC/DMAC Selected

Table 15.9 lists operations when the DTC or DMAC is set as an interrupt request destination.

Table 15.9 Operations When Starting the DTC/DMAC

Interrupt Request Destination	DISEL *1	Number of Remaining Transfers	Operation per Request	IR Flag Clear Timing *2	Interrupt Request Destination after Transfer
DTC *3	1	≠ 0	DTC transfer → CPU interrupt	Cleared when the CPU accepts an interrupt request.	DTC
		= 0	DTC transfer → CPU interrupt	Cleared when the CPU accepts an interrupt request.	CPU (DTCERn.DTCE bit becomes 0)
	0	≠ 0	DTC transfer	Cleared when the DTC starts data transfer.	DTC
		= 0	DTC transfer → CPU interrupt *4	Cleared when the CPU accepts an interrupt request. *4	CPU (DTCERn.DTCE bit becomes 0)
DMAC	1	≠ 0	DMA transfer → CPU interrupt	Cleared when the CPU accepts an interrupt request.	DMAC
		= 0	DMA transfer → CPU interrupt	Cleared when the CPU accepts an interrupt request.	CPU (DMACm.DMCNT.DTE bit becomes 0)
	0	≠ 0	DMA transfer	Cleared when the DMAC starts data transfer.	DMAC
		= 0	DMA transfer *4	Cleared when the DMAC starts data transfer. *4	CPU (DMACm.DMCNT.DTE bit becomes 0)

Note 1. For the DTC, set the DTC.MRB.DISEL bit; For the DMAC, set the DMACm.DMCSL.DISEL bit.

Note 2. When the IRn.IR flag is 1, an interrupt request (DTC or DMA transfer request) that is generated again is ignored.

Note 3. For chain transfer, the DTC transfer continues until the chain transfer ends. After chain transfer ends, whether a CPU interrupt occurs, the IRn.IR flag clear timing, and the interrupt request destination differ depending on the DISEL bit value the number of remaining transfers. For the chain transfer, refer to Table 20.4, Chain Transfer Conditions in section 20, Data Transfer Controller (DTCb).

Note 4. When the DISEL bit is 0 and the number of remaining transfers is 0, operations in the DTC and DMAC are different.

15.7.3.3 Changing the Interrupt Request Destination

Set the IERm.IENj bit to 0 before changing the interrupt request destination (m = 02h to 1Fh; j = 0 to 7).

(1) When the current interrupt request destination is the DMAC

To change interrupt request destinations or change the DMAC trigger to another interrupt source while the DMA transfer is not completed (DMCNT.DTE bit is not cleared) after the procedure described in (1) Setting Interrupt Sources as the DMAC trigger of section 15.7.3.1, Interrupt Request Destination Setting Procedure, follow the procedure below.

- (1) Set the IERm.IENj bit of both the current and new triggers to 0.
- (2) Check the DMAC transfer status. If transfer is not completed, wait until the completion of transfer.
- (3) Perform the procedure described in 15.7.3.1 Interrupt Request Destination Setting Procedure.

(2) When the current interrupt request destination is the DTC

To change interrupt request destinations or change the DTC transfer information while the DTC transfer is not completed (the DTCERn.DTCE bit is not cleared) after the procedure described in (2) Setting Interrupt Sources as the DTC trigger of section 15.7.3.1, Interrupt Request Destination Setting Procedure, follow the procedure below (n = 026 to 255).

- (1) Set the IERm.IENj bit of both the current and new triggers to 0 (m = 02h to 1Fh; j = 0 to 7).
- (2) Check the DTC transfer status. If transfer is not completed, wait until the completion of transfer.
- (3) Perform the procedure described in 15.7.3.1 Interrupt Request Destination Setting Procedure.

15.7.4 Setting the External Pin Interrupt

The following describes the procedure to use the external pin interrupt.

- (1) Set the IERm.IENj bit corresponding to the IRQi pin to 0 (interrupt request is disabled) (m = 02h to 1Fh; j = 0 to 7; i = 0 to 15).
- (2) Set the IRQFLTE0.FLTENi or IRQFLTE1.FLTENi bit to 0 (digital filter is disabled).
- (3) Set the IRQFLTC0.FCLKSELi[1:0] or IRQFLTC1.FCLKSELi[1:0] bits to select the sampling clock of the digital filter.
- (4) Set the I/O port and confirm the setting.
- (5) Set the IRQCRi.IRQMD[1:0] bits to select the detection method.
- (6) When edge detection is selected, set the corresponding IRn.IR flag to 0 (n = 016 to 255).
- (7) Set the IRQFLTE0.FLTENi or IRQFLTE1.FLTENi bit to 1 (digital filter is enabled).
- (8) To select the DTC as the interrupt request destination, set the DTCERn.DTCE bit. To select the DMAC as the interrupt request destination, set the DMRSRm register. When neither the DTCERn.DTCE bit nor the DMRSRm register is set, the interrupt request is sent to the CPU (m = DMAC channel number; n = 026 to 255).
- (9) Set the corresponding IERm.IENj bit to 1 (interrupt request is enabled).

15.7.5 Setting Non-Maskable Interrupts

After reset, non-maskable interrupts are disabled. To use non-maskable interrupts, follow the procedure below.

- (1) Set the stack pointer (SP).
- (2) When using the NMI pin, set the NMIFLTE.NFLTEN bit to 0 (digital filter is disabled).
- (3) When using the NMI pin, set the NMIFLTC.NFCLKSEL[1:0] bits to select the sampling clock of the digital filter.
- (4) When using the NMI pin, set the NMICR.NMIMD bit to select the edge for detection.
- (5) When using the NMI pin, write 1 to the NMICLR.NMICLR bit to set the NMISR.NMIST flag to 0.
- (6) When using the NMI pin, set the NMIFLTE.NFLTEN bit to 1 (digital filter is enabled).
- (7) To enable generation of the non-maskable interrupt, set the bit in the NMIER register corresponding to the used interrupt source to 1.

Once a bit in the NMIER register is set to 1 (enabled), the bit cannot be rewritten so it cannot be set to 0 (disabled). To disable a non-maskable interrupt that has been enabled, reset the MCU.

Refer to section 14, Exception Handling for details on the flow of non-maskable interrupt handling.

Excluding the EXNMIST flag, each flag in the NMISR register becomes 0 by writing 1 to the corresponding bit in the NMICLR register. To set the EXNMIST flag to 0, set each flag in the EXNMISR register that has become 1 to 0.

Confirm that all flags in the NMISR register are 0 before exiting the interrupt handler of non-maskable interrupts.

Non-maskable interrupts, excluding the NMI pin interrupt, can be used as a maskable interrupt. When using as a maskable interrupt, do not change the NMIER and EXNMIER register values from the value after reset. In addition, set the LVD1CR1.LVD1IRQSEL bit and LVD2CR1.LVD2IRQSEL bit to 1 when using voltage monitoring 1 interrupt and voltage monitoring 2 interrupt as a maskable interrupt.

15.7.6 Digital Filter

Noise included in signals input to pins IRQ_i and NMI can be reduced by enabling the digital filter ($i = 0$ to 15).

The digital filter samples signals input to pins using the sampling clock (PCLKB, PCLKB/8, PCLKB/32, PCLKB/64) for the digital filter, and passes the input signal only when three consecutive sampled signals are the same level.

When using the digital filter for the IRQ_i pin, refer to section 15.7.4, **Setting the External Pin Interrupt** to set the associated registers. When using the digital filter for the NMI pin, refer to section 15.7.5, **Setting Non-Maskable Interrupts** to set the associated registers.

When the external pin interrupt or the NMI pin interrupt is used as a source to exit software standby mode, the digital filter cannot be used. Set the IRQFLTE0.FLTEN_i bit, IRQFLTE1.FLTEN_i bit or NMIFLTE.NFLTEN bit to 0 before entering software standby mode. To enable the digital filter again, set the IRQFLTE0.FLTEN_i bit, IRQFLTE1.FLTEN_i bit, or NMIFLTE.NFLTEN bit to 1.

Figure 15.20 shows an example of digital filter operation.

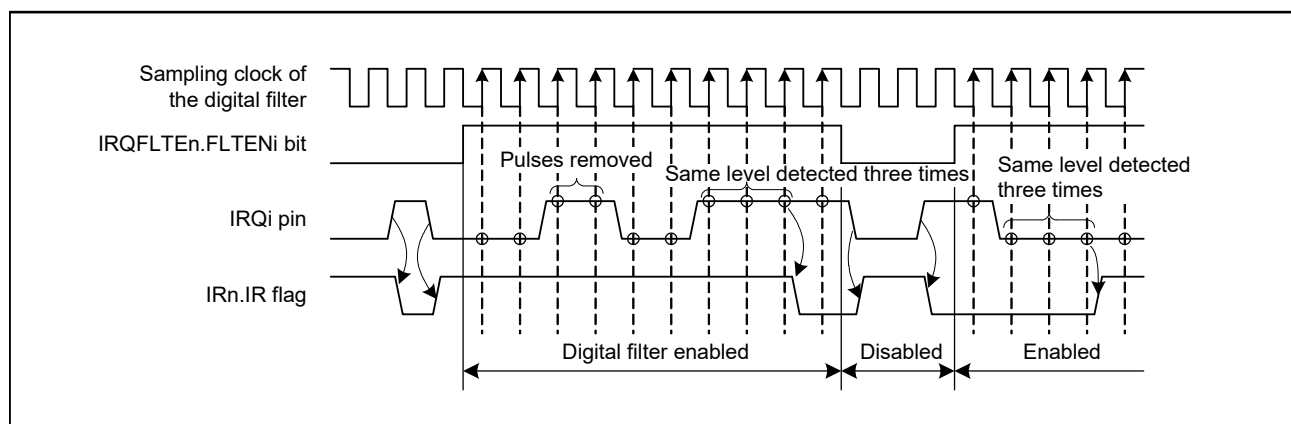


Figure 15.20 Digital Filter Operation Example (when the IRQCRi.IRQMD[1:0] bits are 00b (low level))

15.7.7 Setting Software Configurable Interrupts

The following describes the procedure to assign interrupt sources to software configurable interrupts.

- (1) Set the IER_m.IEN_j bit to 0 ($m = 02h$ to $1Fh$; $j = 0$ to 7). This setting is not required when the value does not change from the value after reset.
- (2) For software configurable interrupt B, set the interrupt source number in registers SLIBXR_n ($n = 128$ to 143) and SLIBR_n ($n = 144$ to 207). Refer to Table 15.3, **Interrupt Sources for Software Configurable Interrupt B** for details on interrupt source numbers that are assigned to software configurable interrupt B.
- (3) For software configurable interrupt A, set the interrupt source number in the SLIAR_n ($n = 208$ to 255) register. Refer to Table 15.4, **Interrupt Sources for Software Configurable Interrupt A** for details on interrupt source numbers that are assigned to software configurable interrupt A.
- (4) When starting the EXDMAC by a software configurable interrupt, set each bit in the SELEXDR register.
- (5) Set the SLIPRCR.WPRC bit to 1.
- (6) Confirm that the SLIPRCR.WPRC bit is 1.
- (7) Select the interrupt request destination from the CPU, DTC, or DMAC. Refer to section 15.7.3.1, **Interrupt Request Destination Setting Procedure** for details on the setting procedure.
- (8) Write 0 to the IR_n.IR flag only when edge detection is selected ($n = 128$ to 255).
- (9) Set the IER_m.IEN_j bit to 1.

15.7.7.1 Polling for Software Configurable Interrupts

When polling an interrupt request by reading the PIBRk.PIRj (k = 0h to Bh) or PIARk.PIRj (k = 0h to Ah, Ch), follow the procedure below (j = 0 to 7).

- (1) Set the peripheral interrupt used.
- (2) Clear the PIBRk.PIRj or PIARk.PIRj flag for polling by writing 1 to the flag.*1
- (3) Enable output of the peripheral interrupt request.
- (4) As needed, read the PIBRk.PIRj or PIARk.PIRj flag to check the value.
- (5) When clearing the PIBRk.PIRj or PIARk.PIRj flag, write 1 to the targeted flag.*1
- (6) As needed, repeat step (4) and (5).

Note 1. Do not use bit manipulation instructions. Multiple status flags may be cleared if a bit manipulation instruction is used. To clear a flag, write the PIBRk or PIARk register in 8-bit units as follows: set the flag that is to be cleared to 1 and set the other flags to 0.

15.8 Multiple Interrupt

To enable another interrupt while processing an interrupt (multiple interrupt), set the PSW.I bit to 1 (interrupt enabled) in the interrupt handler of an accepted interrupt.

The PSW.IPL[3:0] bits in the interrupt handler are the same value as the priority level of the accepted interrupt request. In this case, when an interrupt request with the higher priority level than the PSW.IPL[3:0] bit value is generated, the interrupt request is accepted.

The PSW.I bit can be rewritten only in supervisor mode. Since the PSW.PM bit becomes 0 (supervisor mode is selected) when an interrupt is accepted, the PSW.I bit can be rewritten in the interrupt handler.

15.9 Fast interrupt

The fast interrupt is an interrupt that the CPU can respond to fast. Only one interrupt source can be assigned to the fast interrupt.

The priority level of the fast interrupt is 15 (highest) regardless of the IPRr.IPR[3:0] bit setting (r = 000 to 255). Also, the fast interrupt has higher priority than the other interrupt sources of which the priority level is 15. Note that the fast interrupt cannot be accepted when the PSW.IPL[3:0] bits are 1111b (priority level 15).

To assign an interrupt source to the fast interrupt, set the FIR.FVCT[7:0] bits to select the vector number of the interrupt source, and set the FIR.FIEN bit to 1 (fast interrupt is enabled).

The fast interrupt is enabled only when the CPU is selected as the interrupt request destination. When the DTC or DMAC is selected as the destination, the fast interrupt is disabled.

Refer to section 2, CPU and section 14, Exception Handling for details on the fast interrupt.

15.10 Exiting Low Power Consumption State

Interrupts can be used for exiting sleep mode, all-module clock stop mode, and software standby mode.

Refer to section 11, Low Power Consumption for details. This section describes the procedure to set an interrupt source for exiting each low power consumption mode.

Refer to section 11.6.4, Deep Software Standby Mode for details on exiting deep software standby mode.

15.10.1 Exiting Sleep Mode

Non-maskable interrupts and all interrupt sources can be used for exiting sleep mode. The following conditions must be satisfied.

(1) Non-maskable interrupts

- Generation of the interrupt that is used for exiting is enabled in the NMIER register.

(2) Interrupts

- The CPU is selected as the interrupt request destination.
- The interrupt request is enabled by the IERm.IENj bit (m = 02h to 1Fh; j = 0 to 7).
- The priority level is higher than the PSW.IPL[3:0] bit value in the CPU.
- For group interrupts, the interrupt request is enabled by the corresponding ENj bit in the group interrupt request enable register (GENIE0, GENBE0, GENBL0, GENBL1, GENBL2, GENAL0, GENAL1) (j = 0 to 31).

15.10.2 Exiting All-Module Clock Stop Mode

Non-maskable interrupts and interrupt sources that have a “✓” in the Exit from ACS column in Table 15.5, Interrupt Vector Table can be used for exiting all-module clock stop mode. The conditions below must be satisfied.

(1) Non-maskable interrupts

- Generation of the interrupt that is used for exiting is enabled in the NMIER register.

(2) Interrupts

- The interrupt source can be used for exiting all-module clock stop mode.
- The CPU is selected as the interrupt request destination.
- The interrupt request is enabled by the IERm.IENj bit (m = 02h to 1Fh; j = 0 to 7).
- The priority level is higher than the PSW.IPL[3:0] bit value in the CPU.

15.10.3 Exiting Software Standby Mode

Non-maskable interrupts (excluding oscillation stop detection interrupt) and interrupt sources that have a “✓” in the Exit from SSBY column in Table 15.5, Interrupt Vector Table can be used for exiting software standby mode. The conditions below must be satisfied.

(1) Non-maskable interrupts

- Generation of the interrupt that is used for exiting is enabled in the NMIER register.
- When using the NMI pin interrupt, the digital filter is disabled.

(2) Interrupts

- The interrupt source can be used for exiting software standby mode.
- The CPU is selected as the interrupt request destination.
- The interrupt request is enabled by the IERm.IENj bit (m = 02h to 1Fh; j = 0 to 7).
- The priority level is higher than the PSW.IPL[3:0] bit value in the CPU.
When using the fast interrupt, set not only the FIR register but also the corresponding IPRr.IPR[3:0] bits (r = 000 to 255). Set the IPRr.IPR[3:0] bits to a higher level than the PSW.IPL[3:0] bit value in the CPU.
- When using the external pin interrupt, the digital filter of the IRQi pin used is disabled.

Refer to section 15.7.6, Digital Filter for details on the procedure to set the digital filter.

15.11 Usage Notes

15.11.1 Notes on the WAIT instruction When Using the Non-Maskable Interrupt

Confirm that all status flags in the NMISR register are 0 before executing the WAIT instruction.

15.11.2 Software Configurable Interrupts in All-Module Clock Stop Mode

When using an interrupt source assigned to a software configurable interrupt for exiting all-module clock stop mode, assign the interrupt source to software configurable interrupt B (INTB146 to INTB157) of interrupt vector numbers 146 to 157.

15.11.3 Interrupt Requests in Software Standby Mode

When an interrupt request occurs in software standby mode but the interrupt source is not set as a source for exiting software standby mode, the request is held in the ICU. The request is handled after exiting by another interrupt source. Note that the interrupt request for the external pin interrupt is not held.